UNIVERSITÉ DE GENÈVE

FACULTY OF SCIENCE
Department of Informatics

SIP stochastic information processing

Computational imaging

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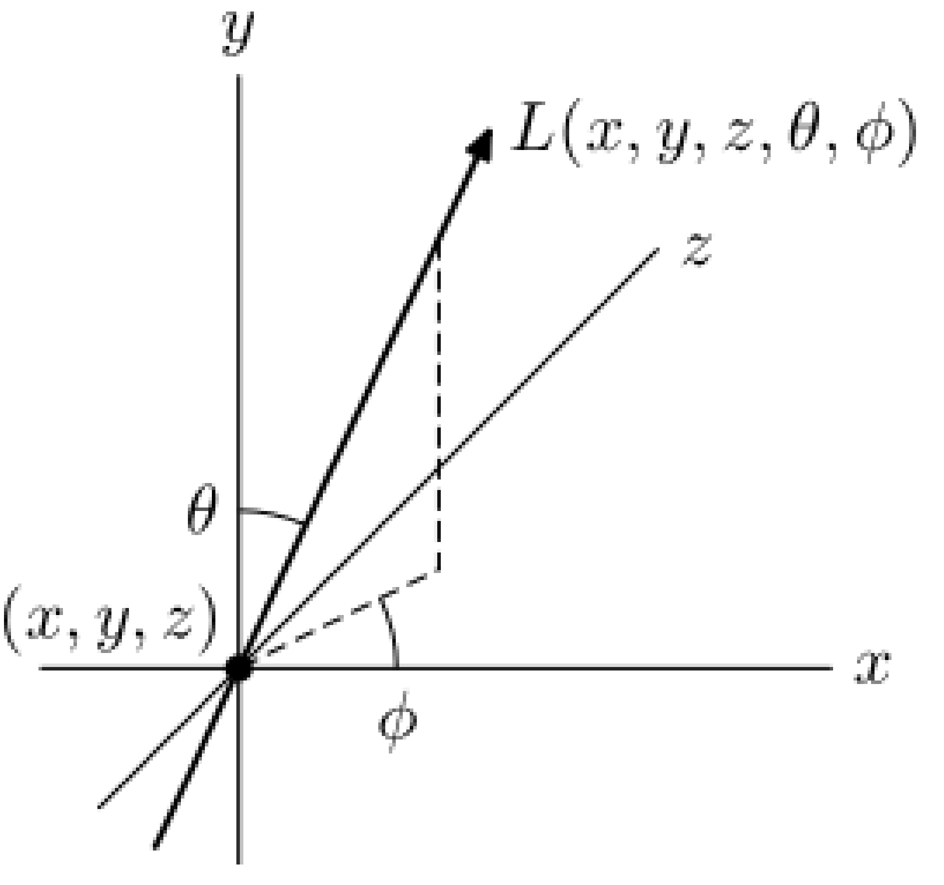
14X062

Content

- Computational photography
- Light field photography
- Conventional camera with coded aperture

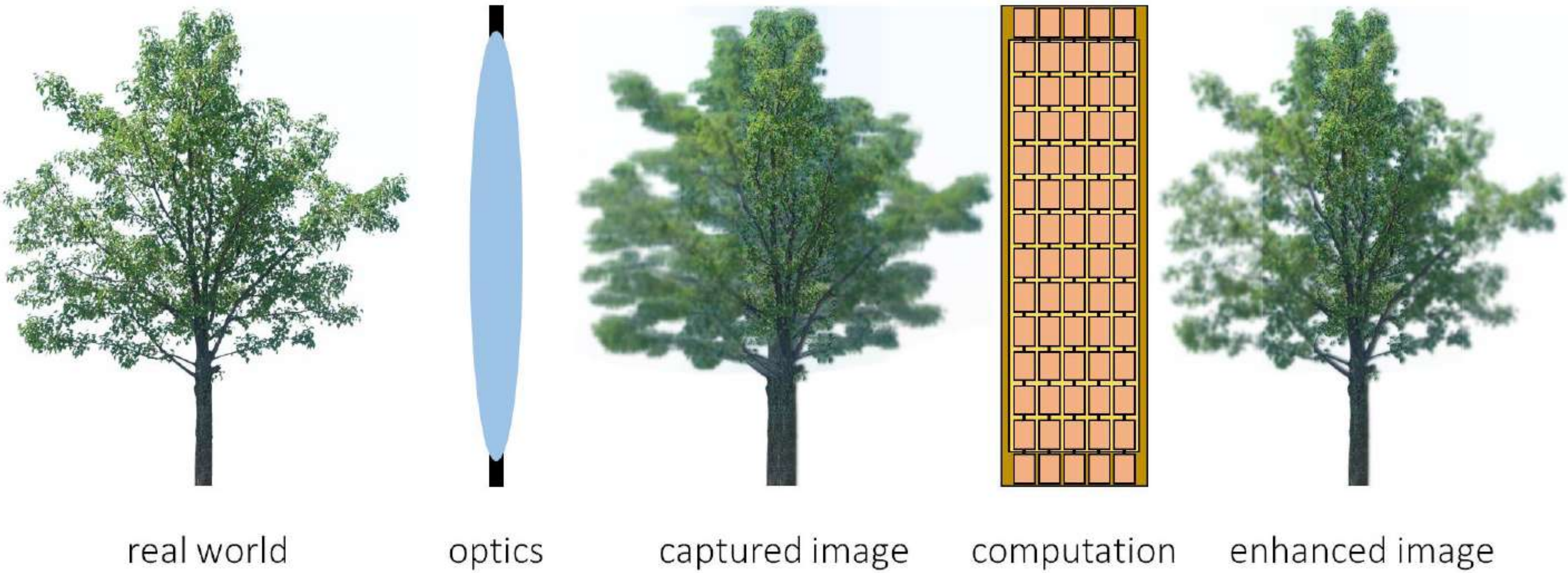
Computational photography

Imaging model



Computational photography

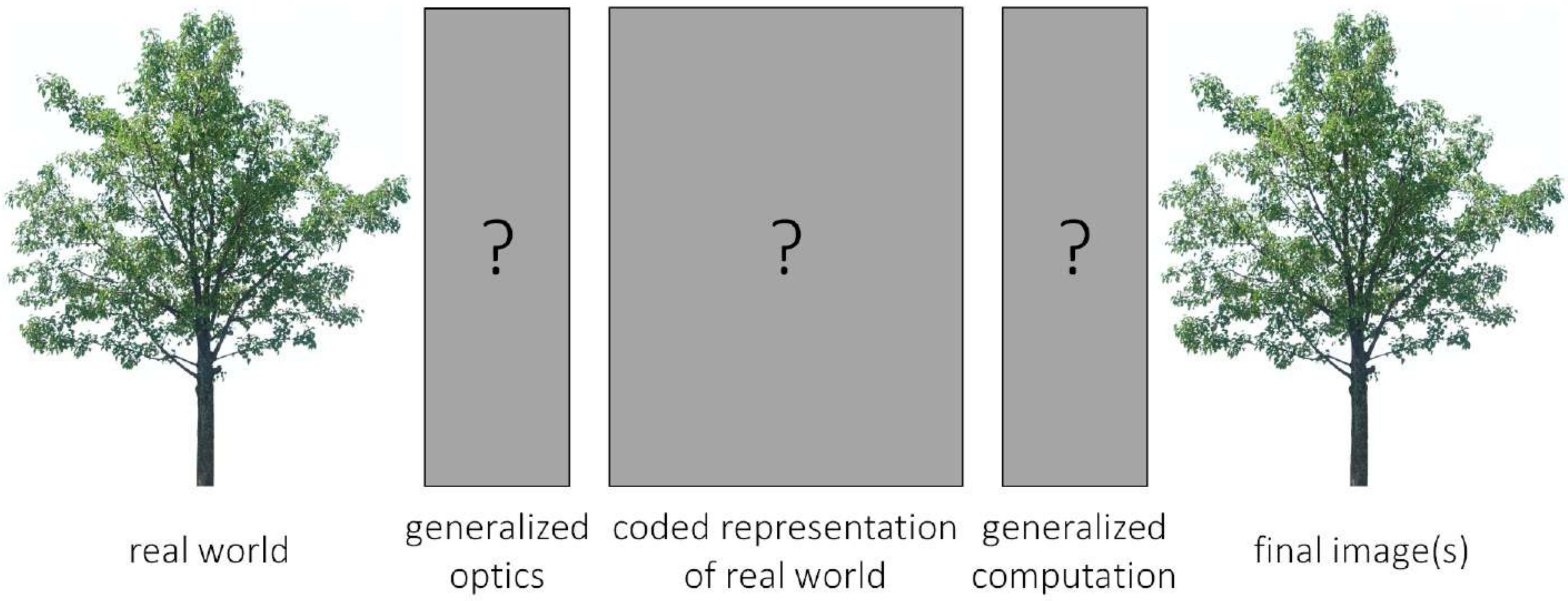
Conventional photography



- Optics capture something that is (close to) the final image.
- Computation mostly “enhances” captured image (e.g., deblur).

Computational photography

Computational photography



- Generalized optics encode world into intermediate representation.
- Generalized computation decodes representation into multiple images.

Computational photography

- optics
- aperture
- depth of field
- field of view
- exposure
- noise
- color filter arrays
- image processing pipeline

Computational photography

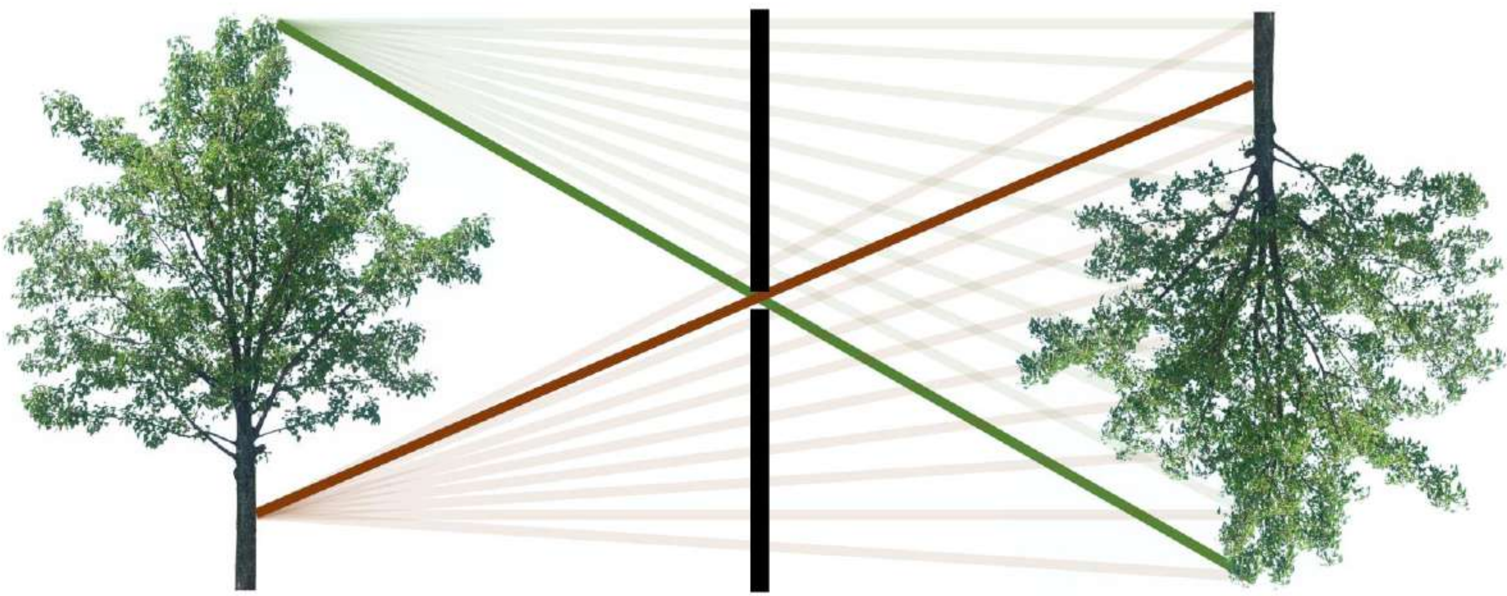
Early example: mosaicing

real world | generalized optics | coded representation of real world | generalized computation | final image(s)

- Color filter array encodes color into a mosaic.
- Demosaicing decodes color into RGB image.

Computational photography

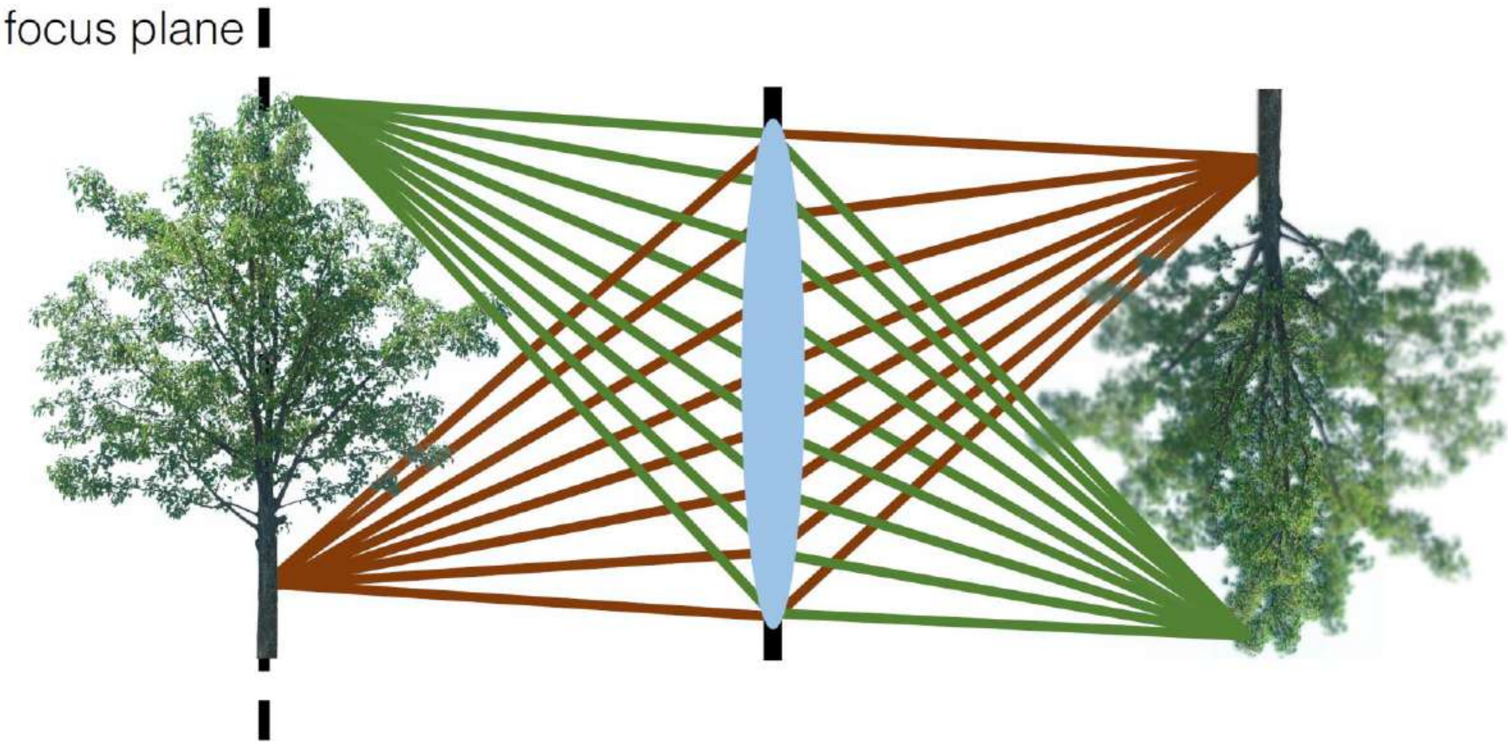
Pinhole camera



- Everything is in focus.
- Very light inefficient.

Computational photography

Lens camera

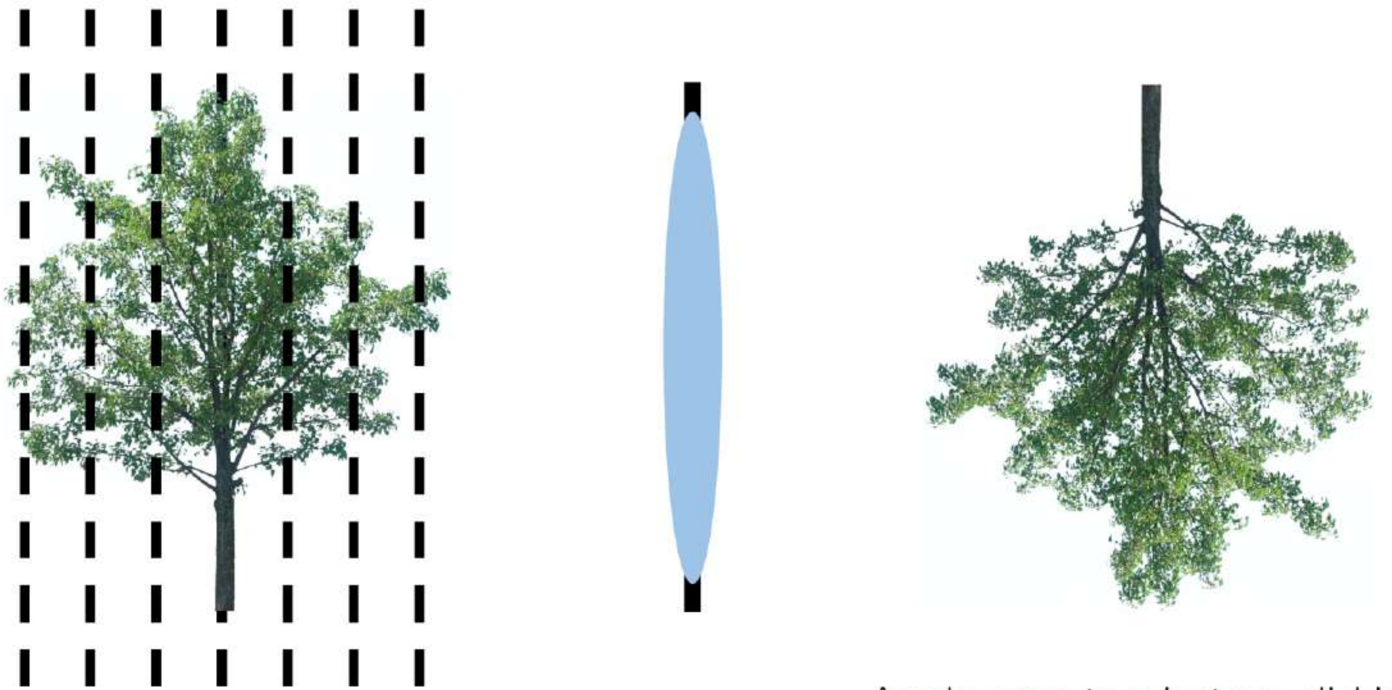


- Only one plane is in focus.
- Very light efficient.

How can we get an all in-focus image?

Computational photography

Focal stack



- Capture images focused at multiple planes.
- Merge them into a single all in-focus image.

Analogous to what we did in HDR
- Focal stack instead of exposure stack.

Computational photography

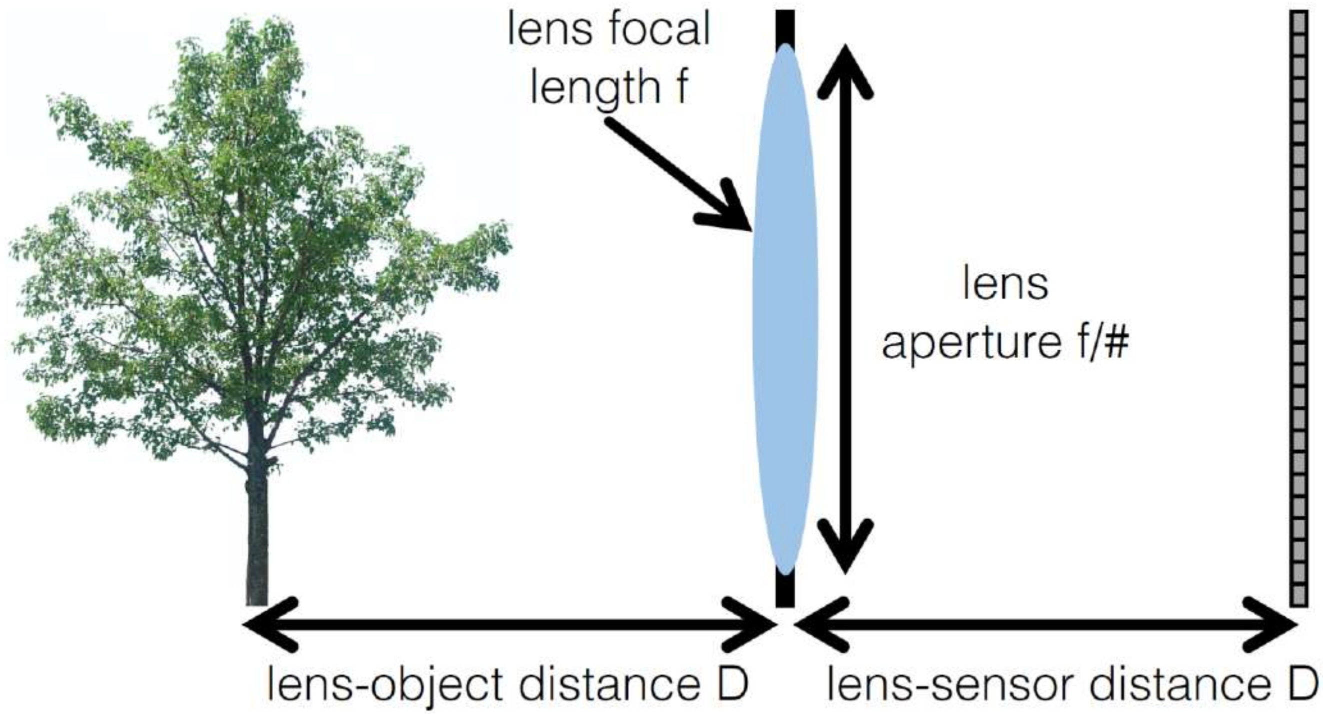
Focal stack imaging

1. Capture a focal stack

2. Merge into an all in-focus image

Computational photography

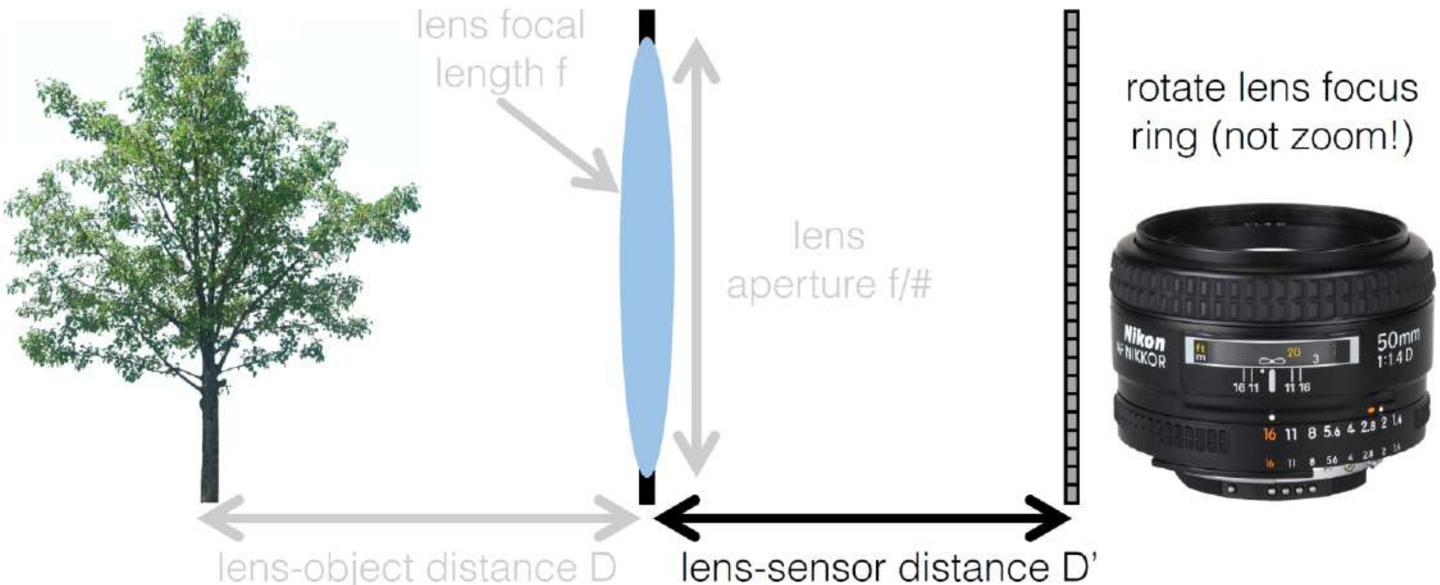
How do you capture a focal stack?



Which of these parameters would you change (and how)?

Computational photography

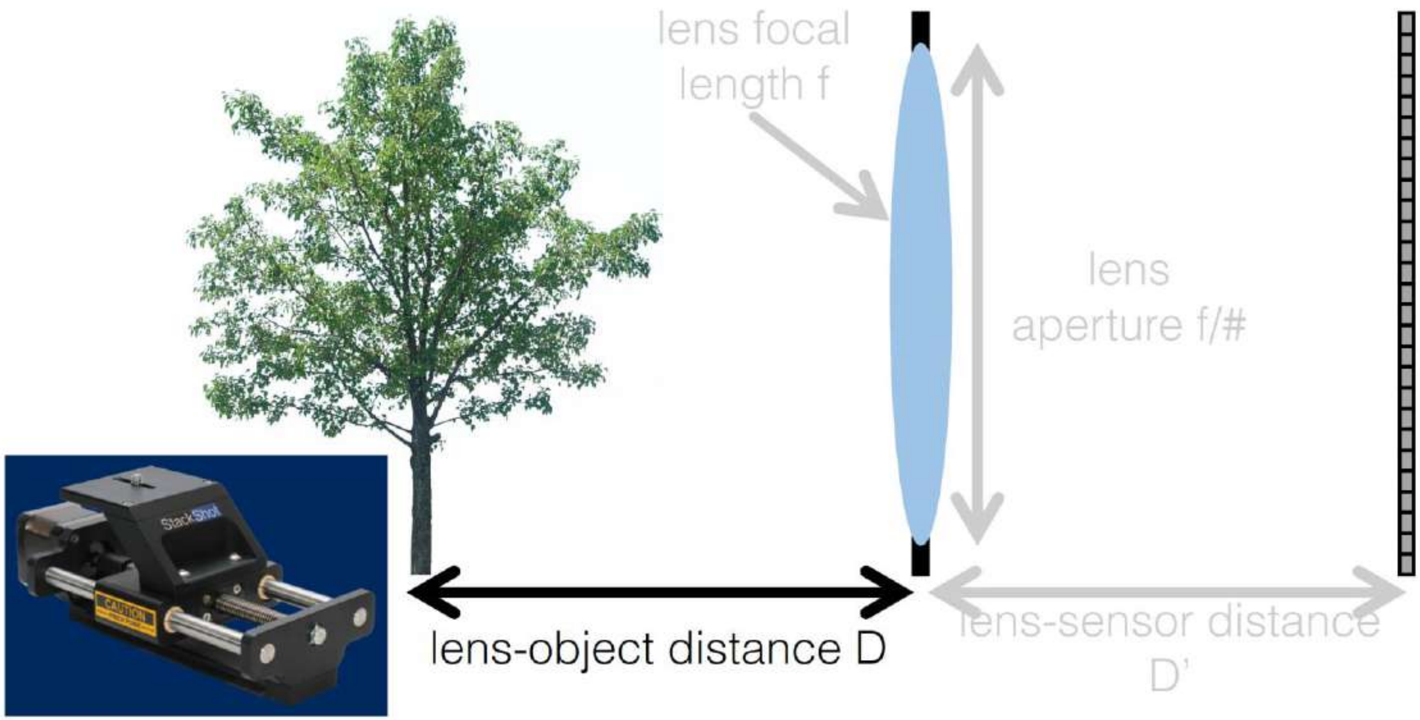
How do you capture a focal stack?



Which of these parameters would you change (and how)?

Computational photography

How do you capture a focal stack?



Which of these parameters would you change (and how)?

Computational photography

How do you merge a focal stack?

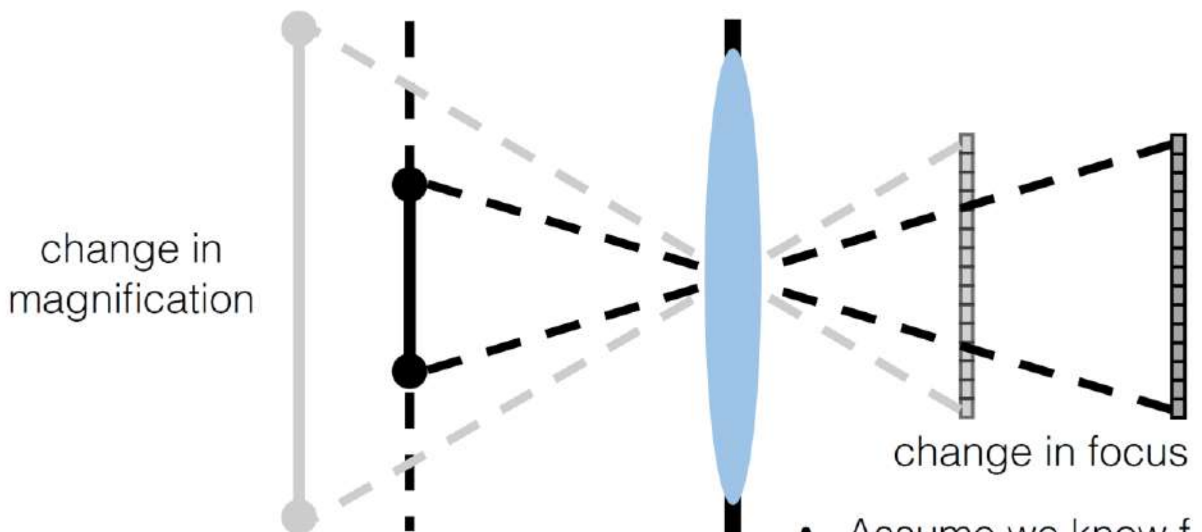
1. Align images
2. Assign per-pixel weights representing “in-focus”-ness
3. Compute image average

Computational photography

Image alignment

Why do we need to align the images?

- When we change focus distance, we also change field of view (magnification).
- Also, scene may not be static (but we will be ignoring this for now).



$$m = \frac{f}{D' - f}$$

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

resize using these equations

- Assume we know f and all D' values.
- How do can we align the images?

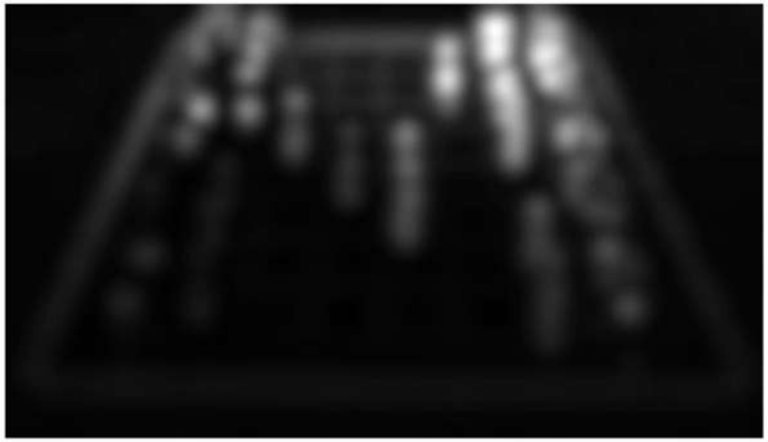
Computational photography

Weight assignment

How do we measure how much "in-focus" each pixel is?

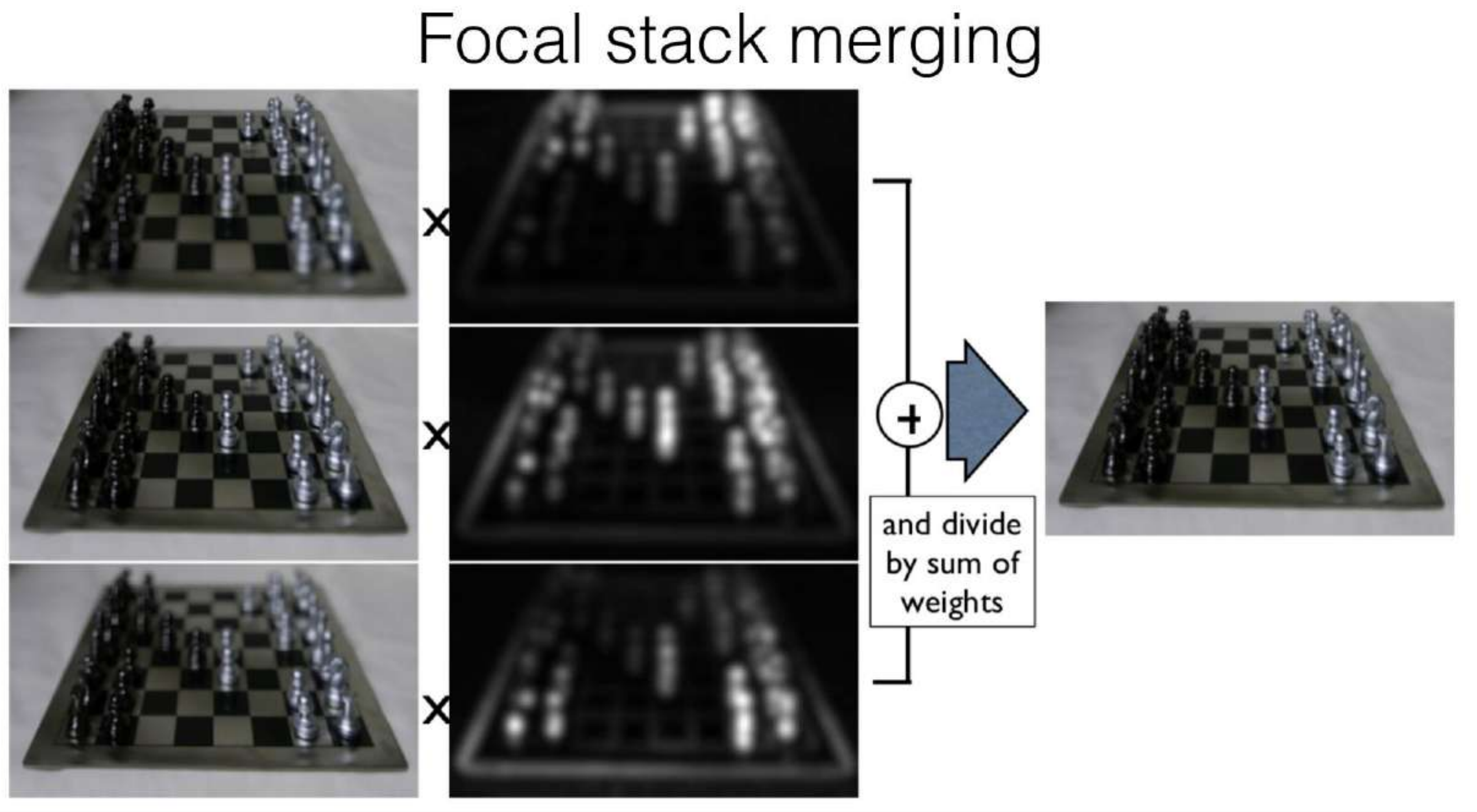
- Measure local sharpness.

run Laplacian operator



do some Gaussian blurring

Computational photography



Computational photography

example image from stack

all in-focus image

middle image from stack

all in-focus image

Computational photography

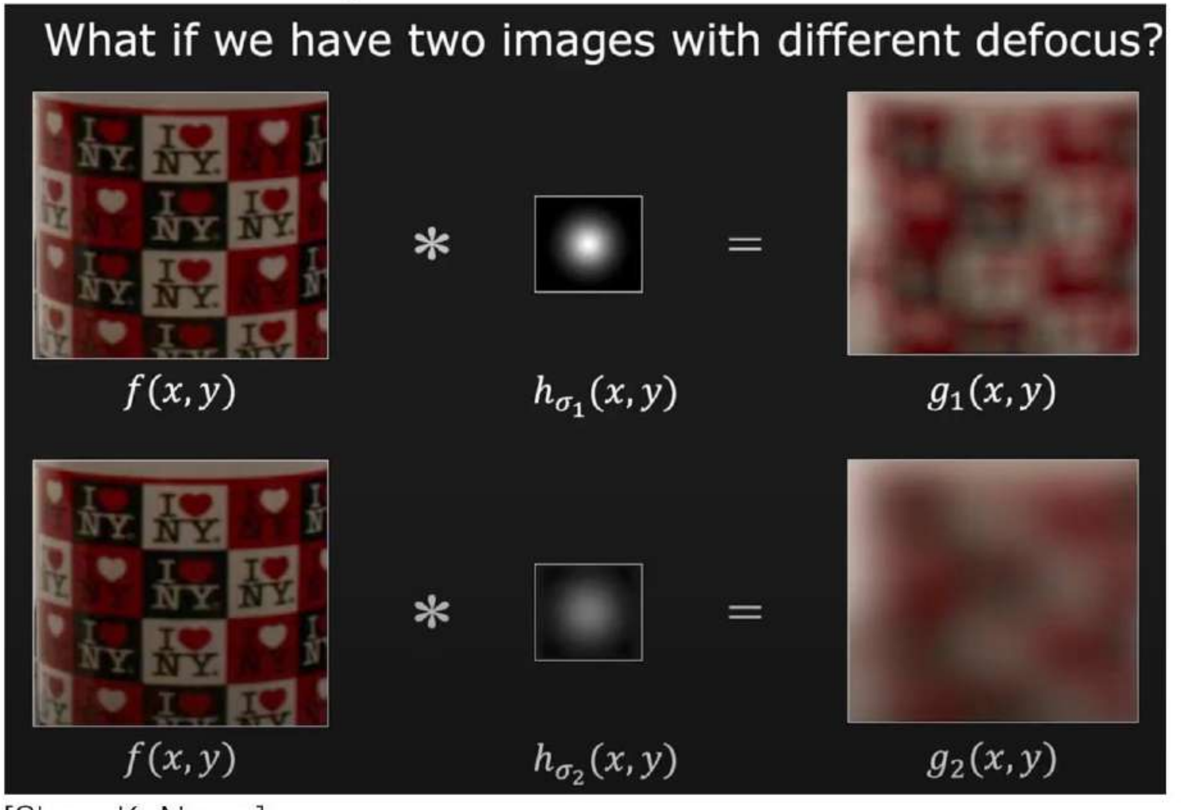
Confocal stereo

- Capture a 2D stack by varying both focus and aperture
- Analyze each pixel's focus-aperture image to find true depth

[Hassinof and Kutulakos, ECCV 2006]

Computational photography

Depth from defocus



[Shree K. Nayar]

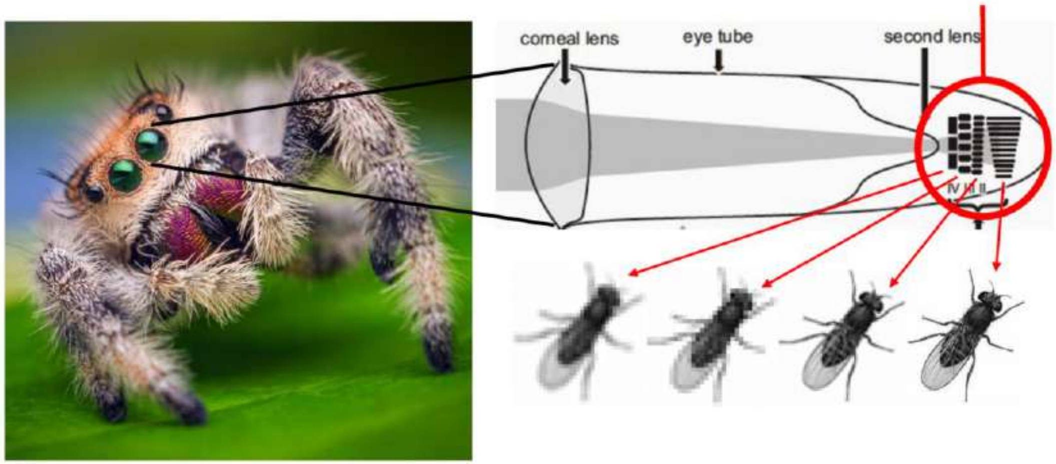
Computational photography

Focal flow

Use a dense focal stack (e.g., from autofocus):

- Assume circle of confusion can be modeled as (typically Gaussian) blur kernel with varying σ.
- Estimate optical flow between successive frames in focal stack.
- Relate optical flow change to depth.

Requires very little computation (no convolutions) but only works on textured patches (why?).



Biologically-inspired from jumping spiders!

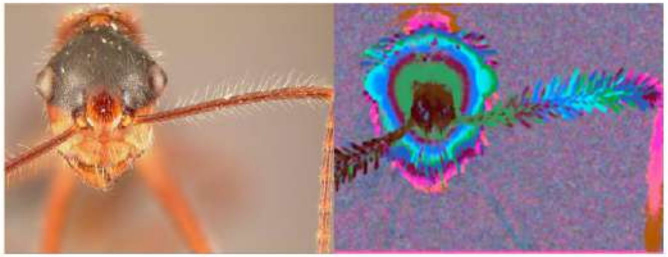
[Alexander et al., ECCV 2016; Guo et al., ICCV 2017, PNAS 2019]

Computational photography

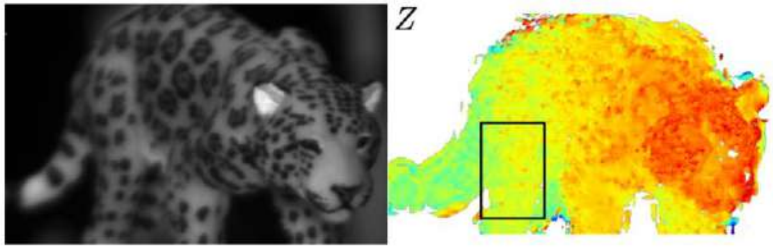
Depth inference techniques

Confocal stereo:
- Requires focus-aperture stack.
- Gives per-pixel depth estimates.

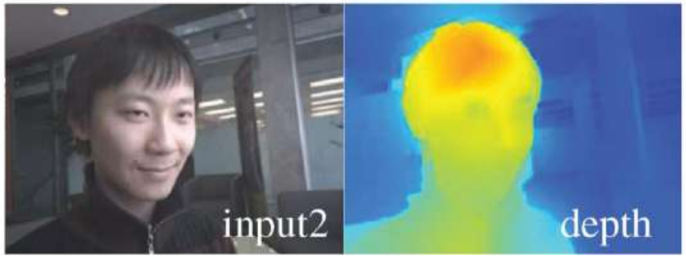
Depth from focus:
- Requires focus stack.
- Gives per-patch depth estimates.



Focal flow:
- Requires dense focus stack.
- Efficient per-patch depth estimates.



Depth from defocus:
- Requires only two images.
- Gives per-patch depth estimates.



Light field photography

Light
Field
Theory

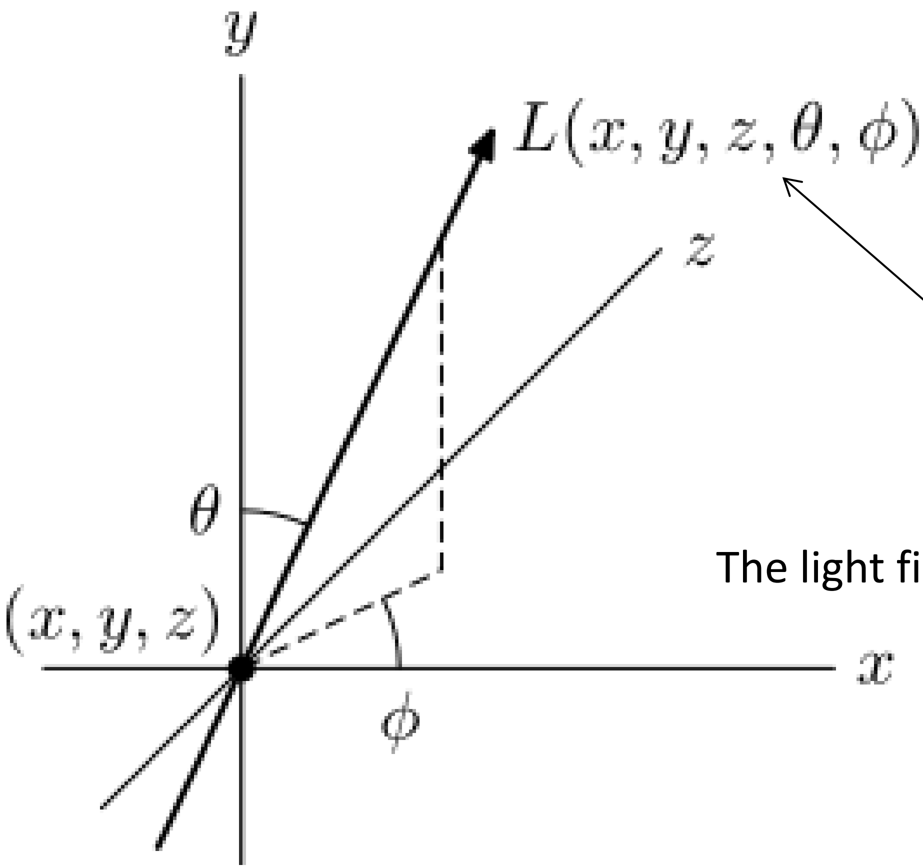
Photographs do not record most of the information about the light entering the camera.

Light field cameras shows how light coming from one part of the lens differs from the light coming from another. (Vectors of light, where they originate, and how they are manipulated using lenses)

Light fields are defined in a couple different ways, one is as the geometrical distribution of light rays flowing in space.

The total geometric distribution of light as a plenoptic function over the 5D space of rays – 3D for each spatial position and 2D for each direction of flow.

Light field photography

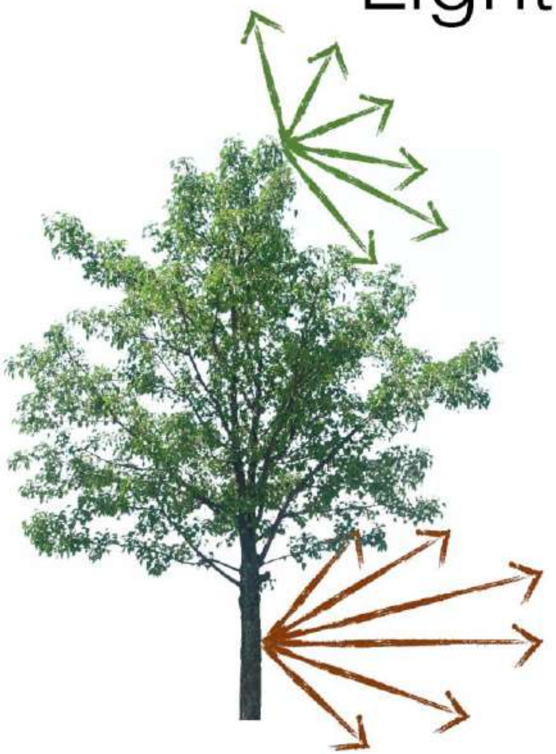


5D Plenoptic Function

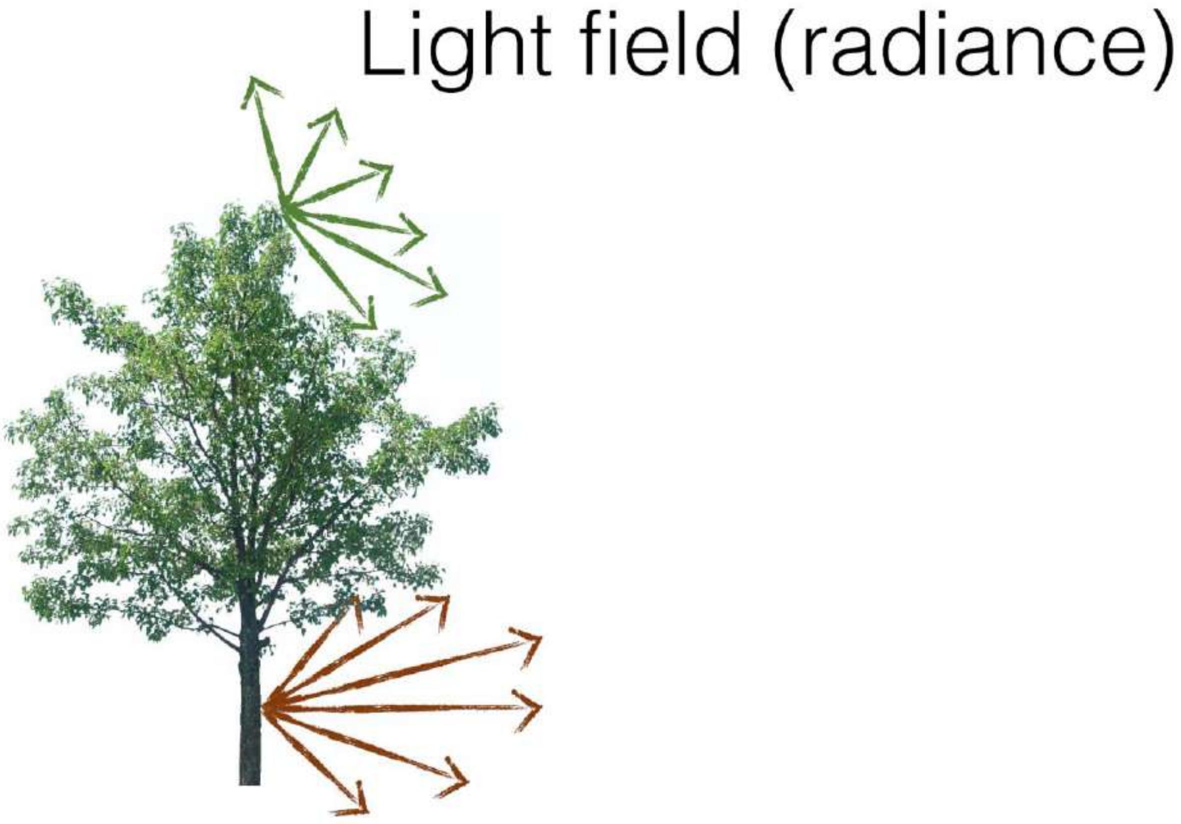
The light field at any point in space as a 5D function.

Light field photography

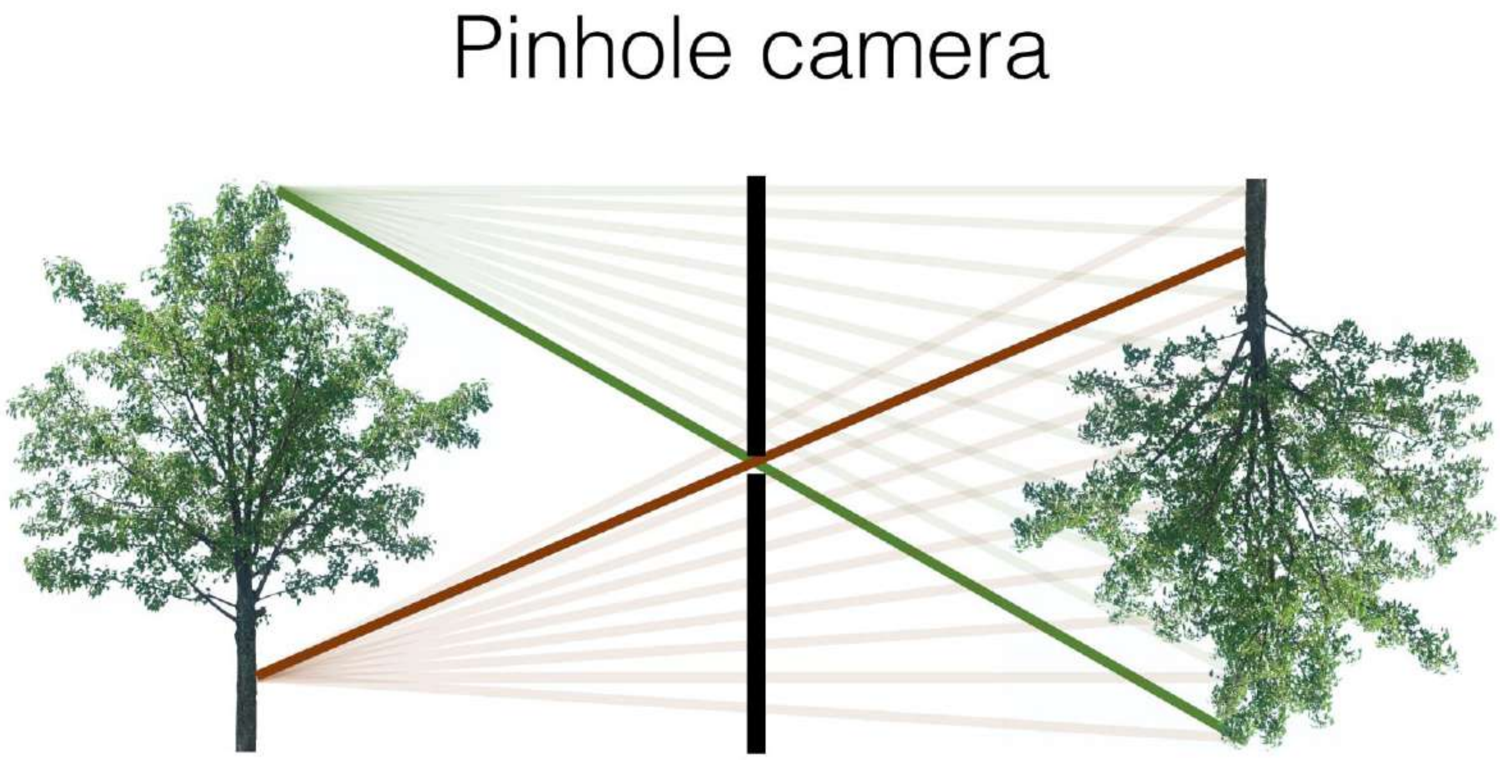
Light field (radiance)



Light field photography

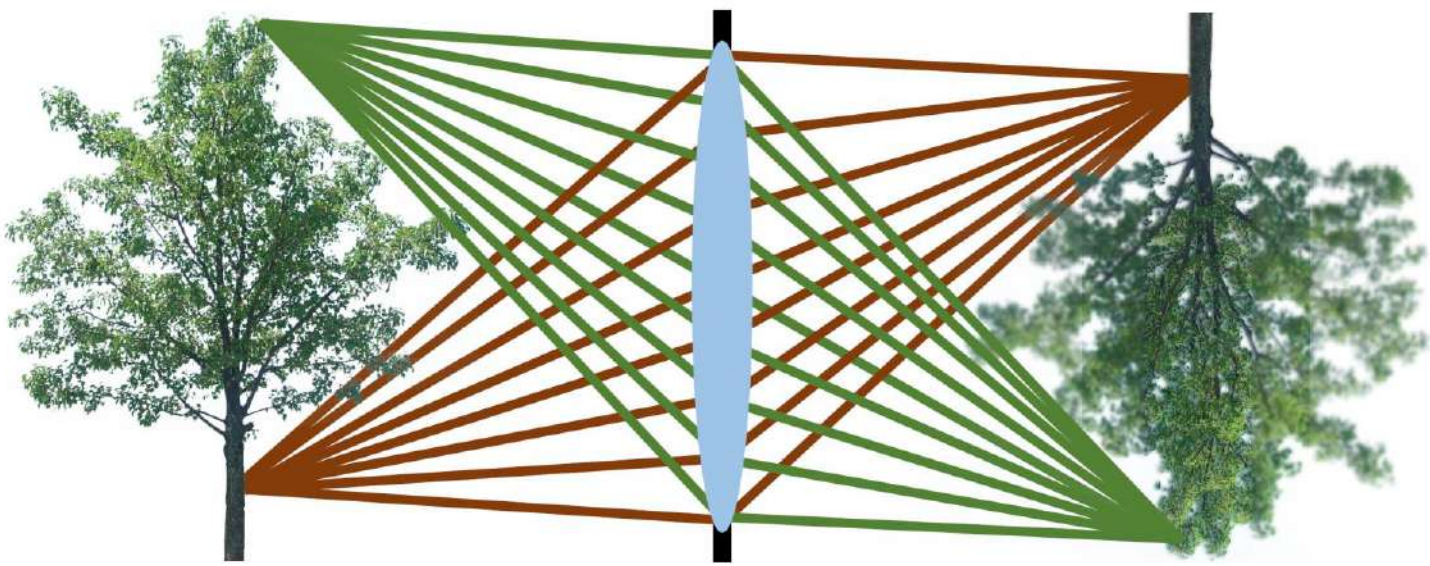


Light field photography



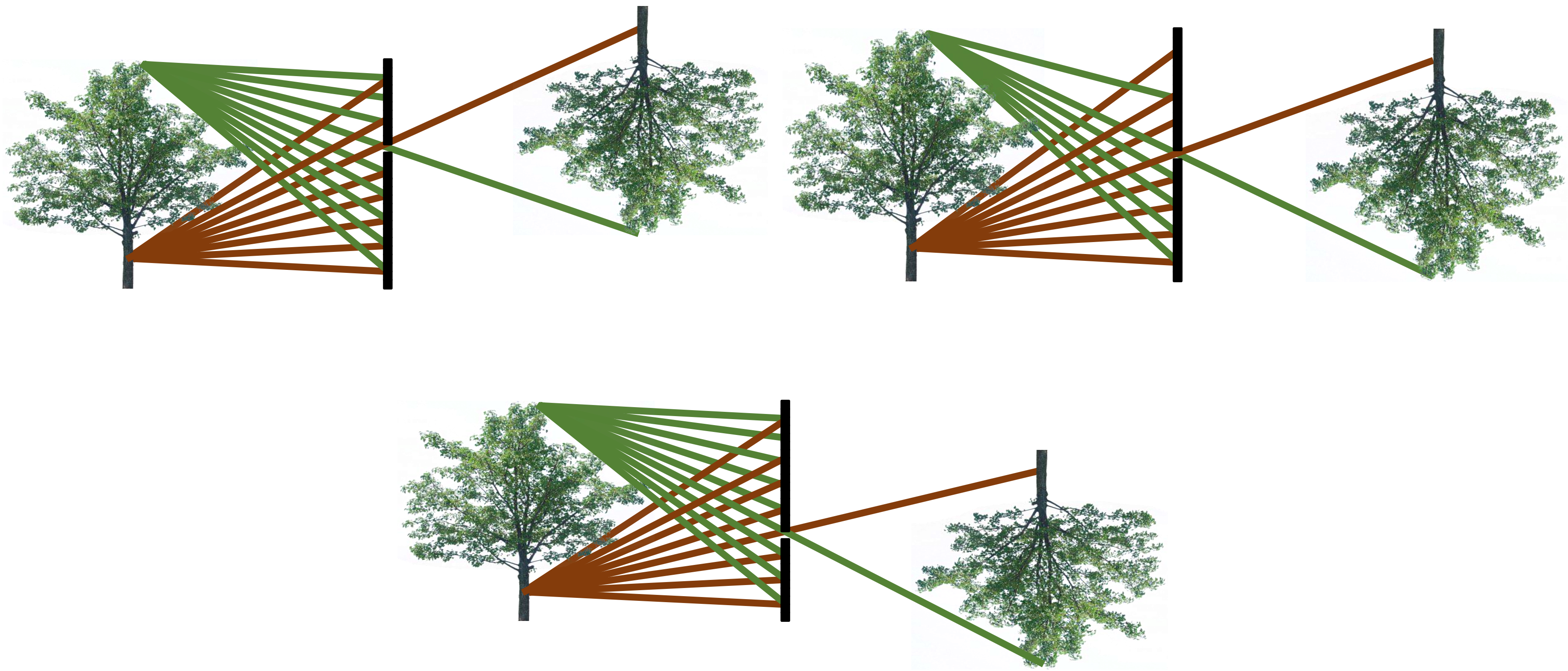
Light field photography

Measuring rays



Light field photography

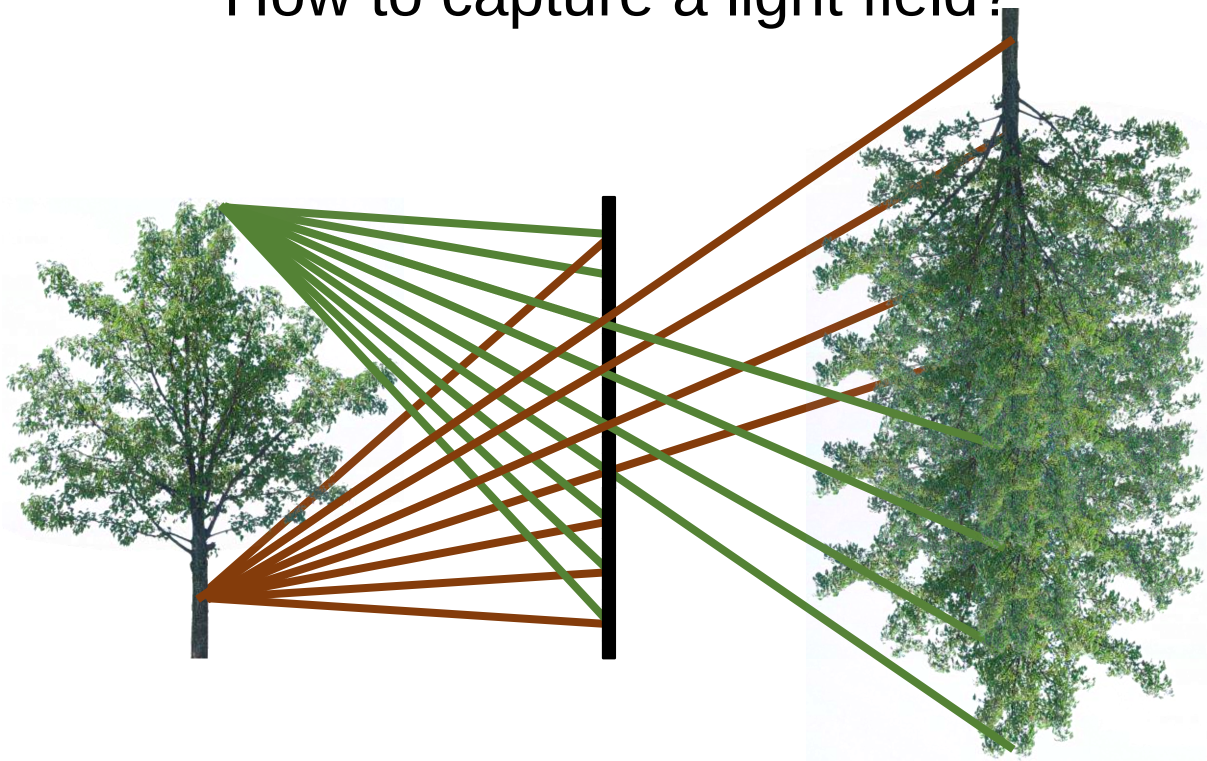
Measuring rays



The same set of rays can be captured by using a pinhole camera from multiple viewpoints

Light field photography

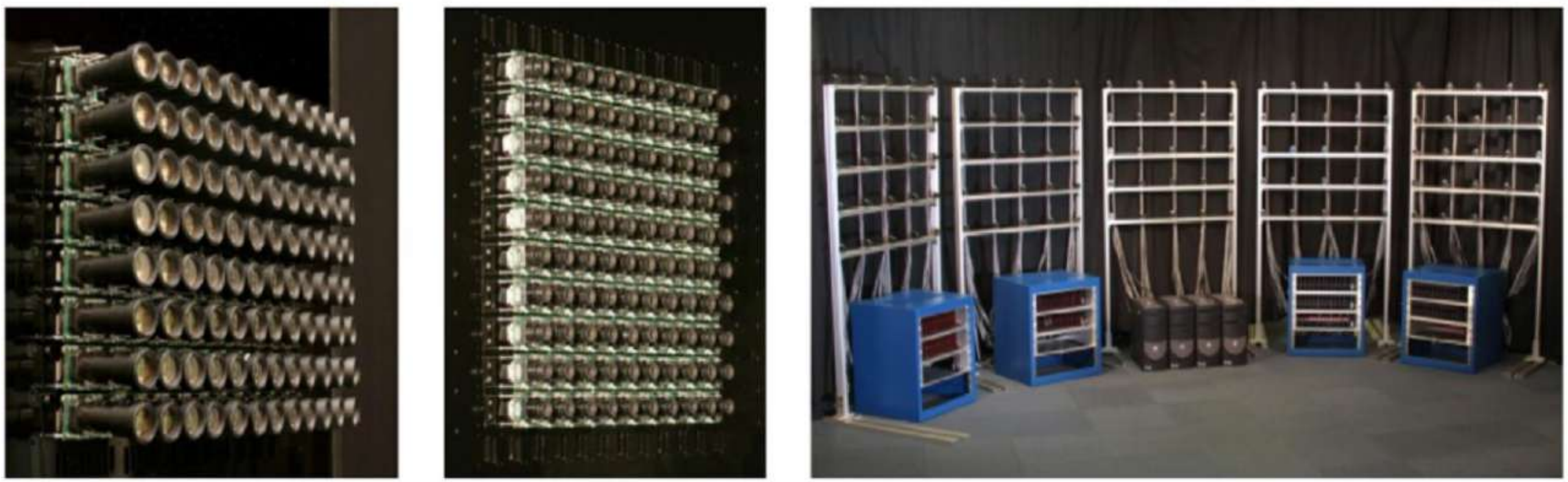
How to capture a light field?



Light field photography

Option 1: use multiple cameras

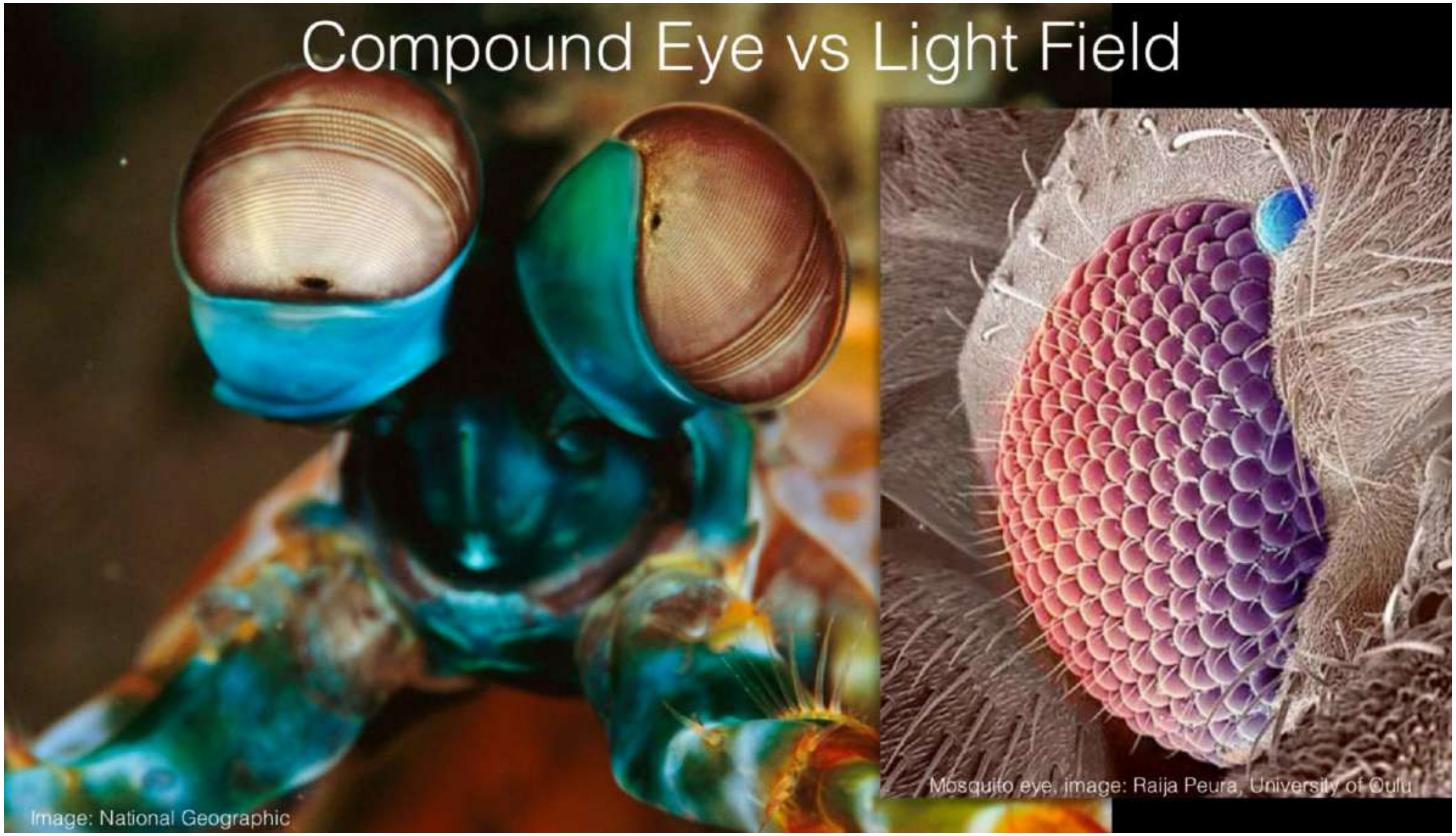
Stanford camera array



("synthetic aperture")

[Willburn et al., SIGGRAPH 2005]

Light field photography



Light field photography

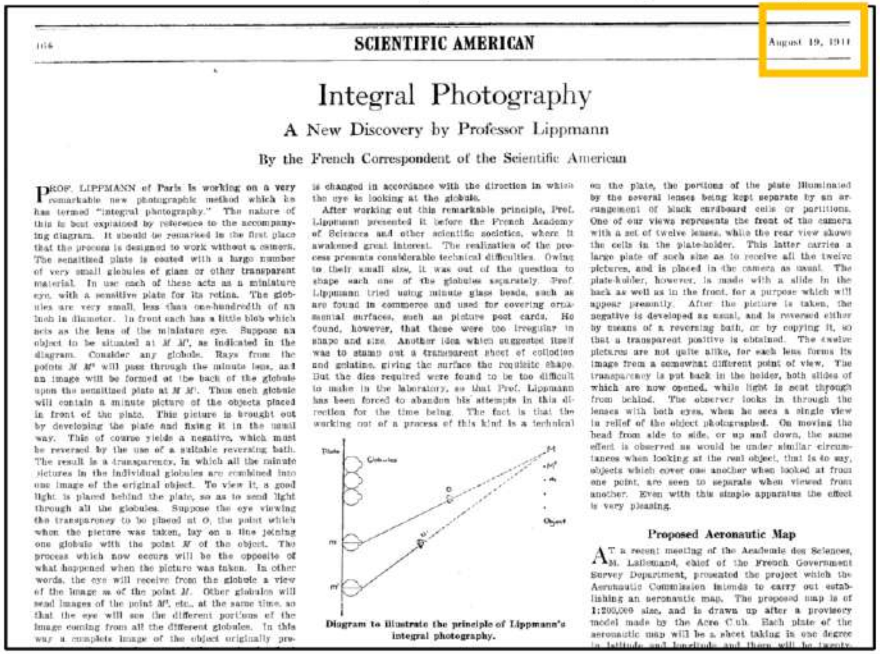
Option 2: take multiple images with one camera

Single camera mounted on LEGO motor.

Light field photography

Option 3: use a plenoptic camera

plenoptic = plenus (Latin for “full”) + optic (Greek for “seeing”, in this case)



116 SCIENTIFIC AMERICAN August 19, 1911

Integral Photography

A New Discovery by Professor Lippmann

By the French Correspondent of the Scientific American

PROF. LIPPMANN of Paris is working on a very remarkable new photographic method which he has termed "integral photography." The nature of this is best explained by reference to the accompanying diagram. It should be remarked in the first place that the process is designed to work without a camera. The sensitized plate is coated with a large number of very small globules of glass or other transparent material. In use each of these acts as a miniature eye, with a sensitive plate for its retina. The globules are very small, less than one-hundredth of an inch in diameter. In front each has a little blob which acts as the lens of the miniature eye. Suppose an object to be situated at M M', as indicated in the diagram. Consider any globule. Rays from the points M M' will pass through the minute lens, and an image will be formed at the back of the globule upon the sensitized plate at M M'. Thus each globule will contain a minute picture of the objects placed in front of the plate. This picture is brought out by developing the plate and fixing it in the usual way. This of course yields a negative, which must be reversed by the use of a suitable reversing bath. The result is a transparency, in which all the minute pictures in the individual globules are combined into one image of the original object. To view it, a good light is placed behind the plate, so as to send light through all the globules. Suppose the eye viewing the transparency to be placed at O, the point which when the picture was taken, lay on a line joining one globule with the point M of the object. The process which now occurs will be the opposite of what happened when the picture was taken. In other words, the eye will receive from the globule a view of the image m of the point M. Other globules will send images of the point M', etc., at the same time, so that the eye will see the different portions of the image coming from all the different globules. In this way a complete image of the object originally pre- [...] is changed in accordance with the direction in which the eye is looking at the globule.

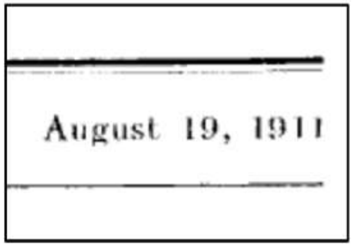
After working out this remarkable principle, Prof. Lippmann presented it before the French Academy of Sciences and other scientific societies, where it awakened great interest. The realization of the process presents considerable technical difficulties. Owing to their small size, it was out of the question to shape each one of the globules separately. Prof. Lippmann tried using minute glass beads, such as are found in commerce and used for covering ornamental surfaces, such as picture post cards. He found, however, that these were too irregular in shape and size. Another idea which suggested itself was to stamp out a transparent sheet of collodion and gelatine, giving the surface the requisite shape. But the dies required were found to be too difficult to make in the laboratory, so that Prof. Lippmann has been forced to abandon his attempts in this direction for the time being. The fact is that the working out of a process of this kind is a technical [...] on the plate, the portions of the plate illuminated by the several lenses being kept separate by an arrangement of black cardboard cells or partitions. One of our views represents the front of the camera with a set of twelve lenses, while the rear view shows the cells in the plate-holder. This latter carries a large plate of such size as to receive all the twelve pictures, and is placed in the camera as usual. The plate-holder, however, is made with a slide in the back as well as in the front, for a purpose which will appear presently. After the picture is taken, the negative is developed as usual, and is reversed either by means of a reversing bath, or by copying it, so that a transparent positive is obtained. The twelve pictures are not quite alike, for each lens forms its image from a somewhat different point of view. The transparency is put back in the holder, both slides of which are now opened, while light is sent through from behind. The observer looks in through the lenses with both eyes, when he sees a single view in relief of the object photographed. On moving the head from side to side, or up and down, the same effect is observed as would be under similar circumstances when looking at the real object, that is to say, objects which cover one another when looked at from one point, are seen to separate when viewed from another. Even with this simple apparatus the effect is very pleasing.

Diagram to illustrate the principle of Lippmann's integral photography.

Proposed Aeronautic Map

AT a recent meeting of the Academie des Sciences, M. Lallemand, chief of the French Government Survey Department, presented the project which the Aeronautic Commission intends to carry out establishing an aeronautic map. The proposed map is of 1:200,000 size, and is drawn up after a provisory model made by the Aero Club. Each plate of the aeronautic map will be a sheet taking in one degree in latitude and longitude and there will be twenty-

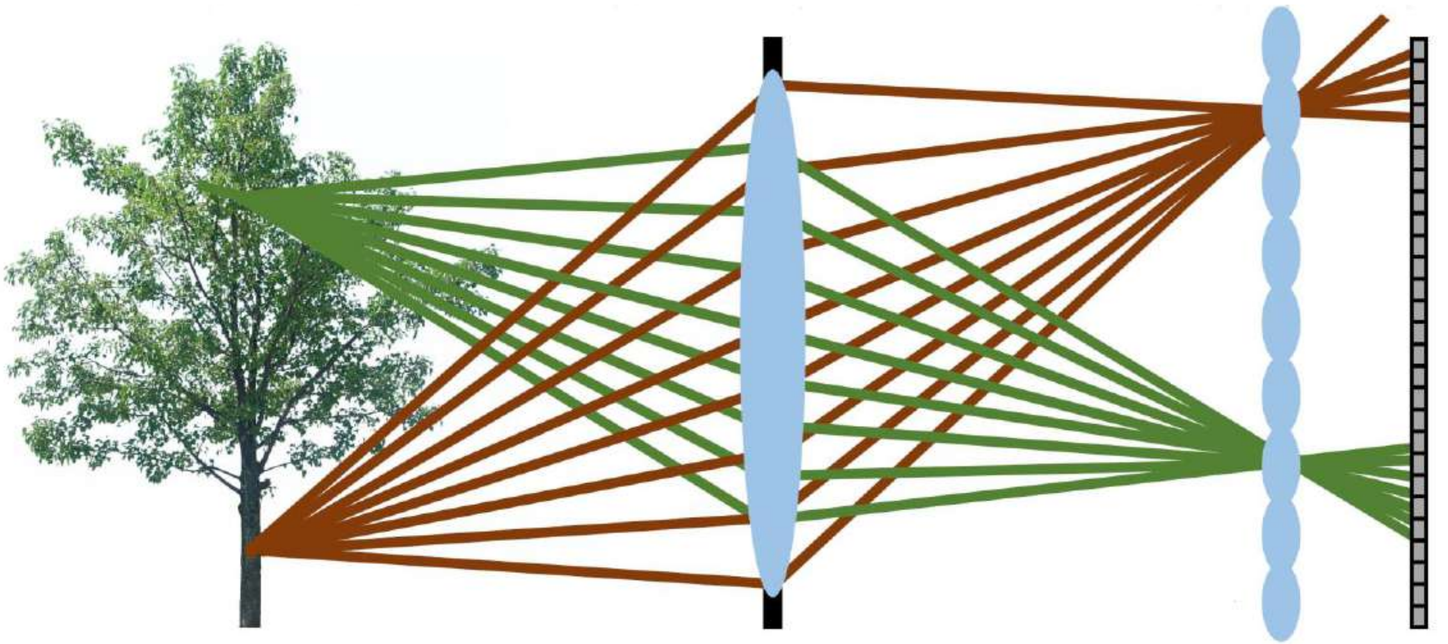
First conceptualized by Gabriel Lippmann, who called it integral photography.



Notice the date: more than a century ago.

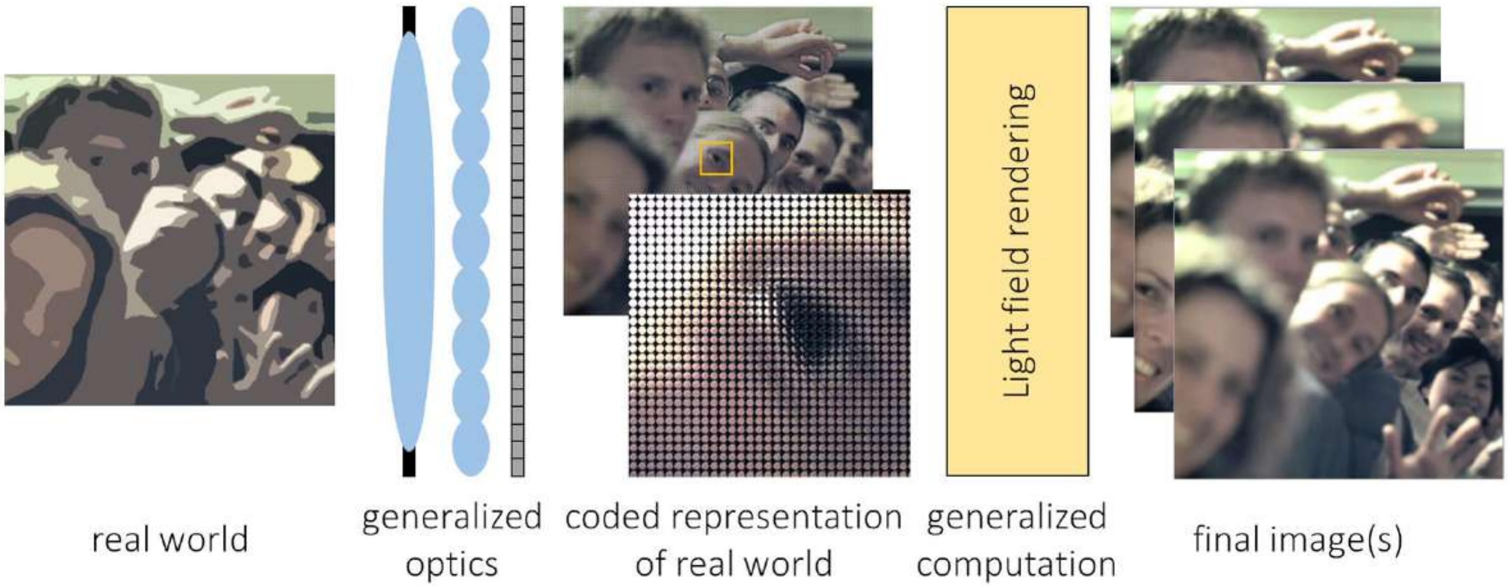
Light field photography

Option 3: use a plenoptic camera



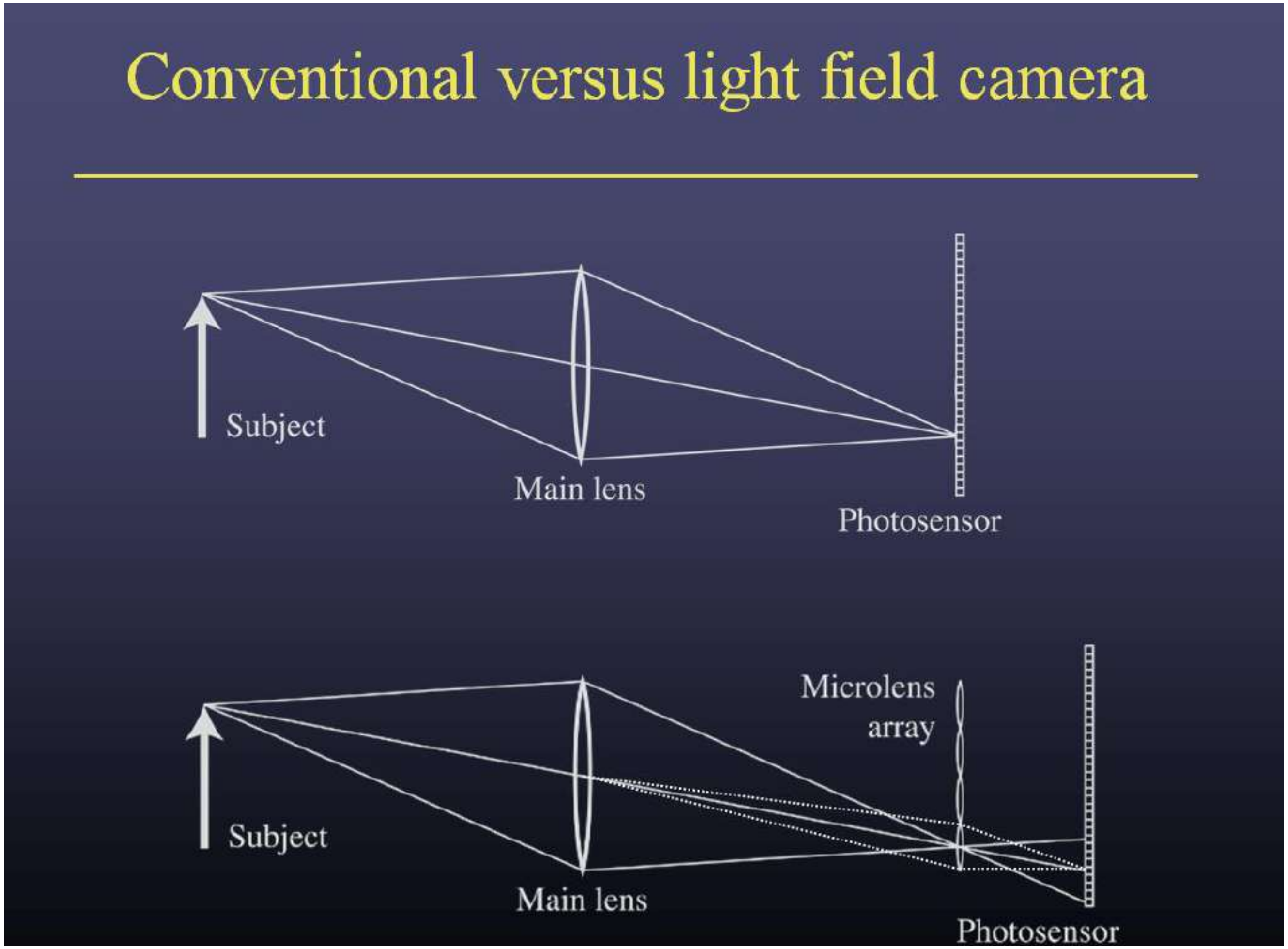
Light field photography

Option 3: use a plenoptic camera

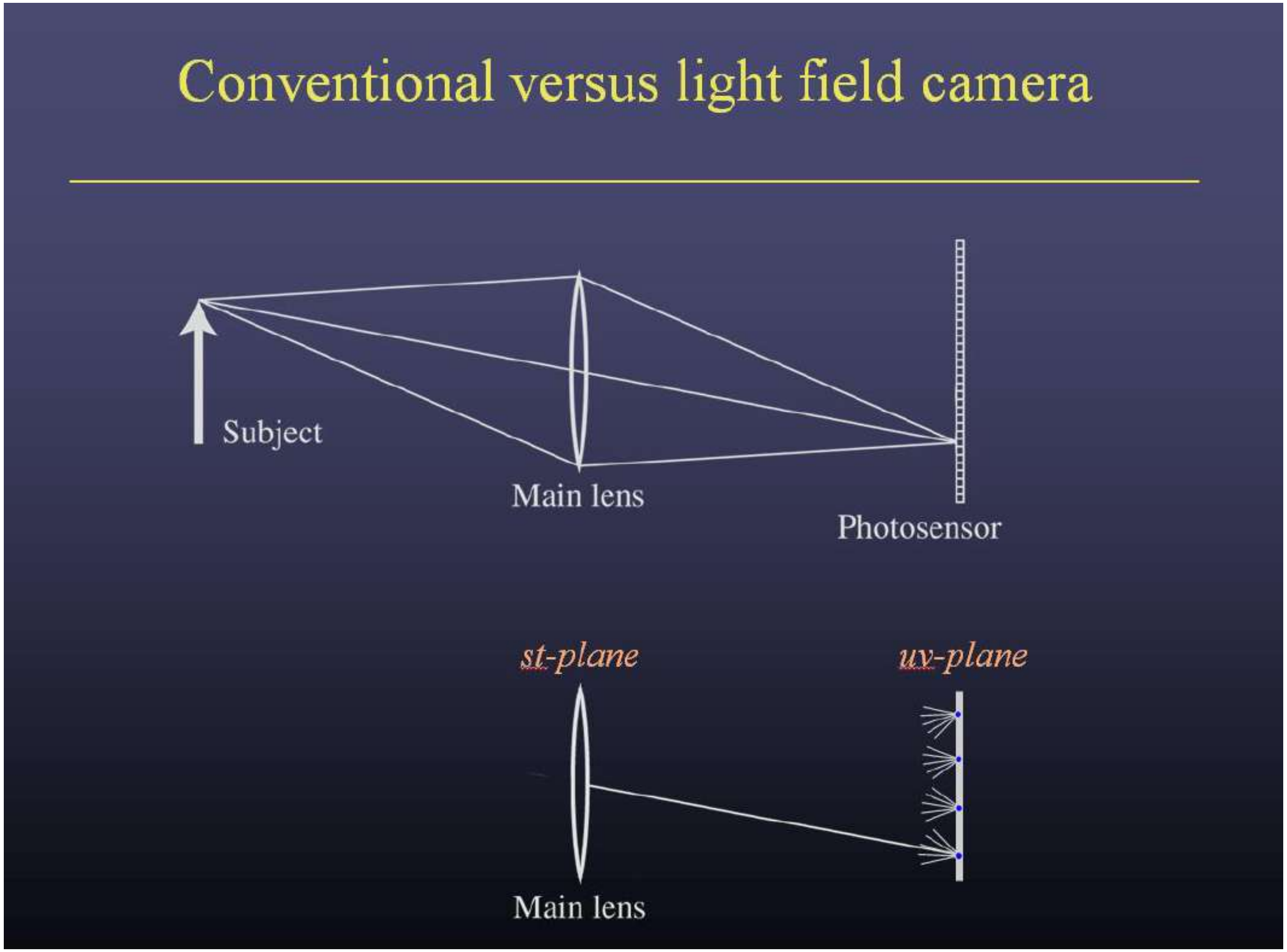


- Plenoptic camera encodes world into light field.
- Light field rendering decodes light field into refocused or multi-viewpoint images.

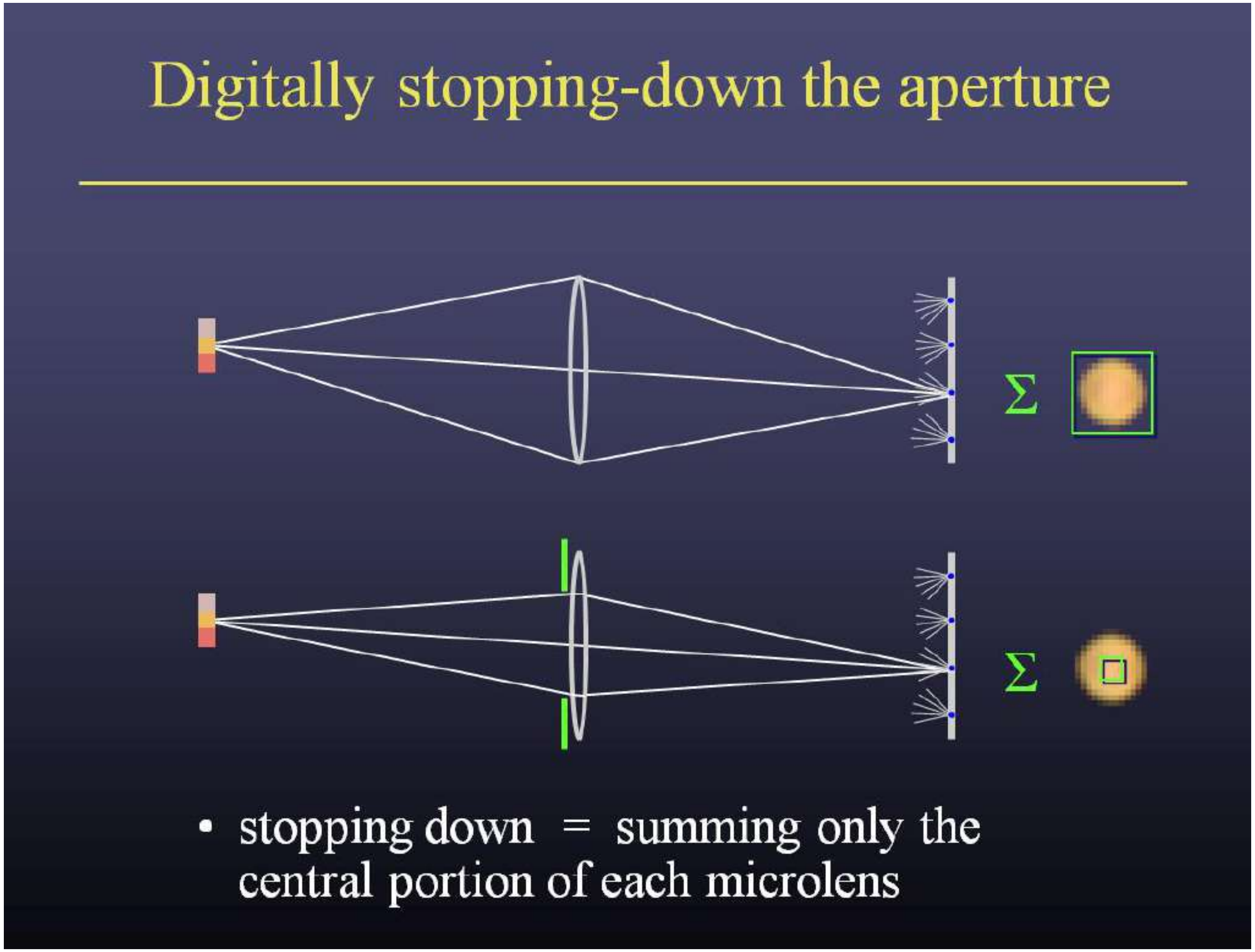
Light field photography



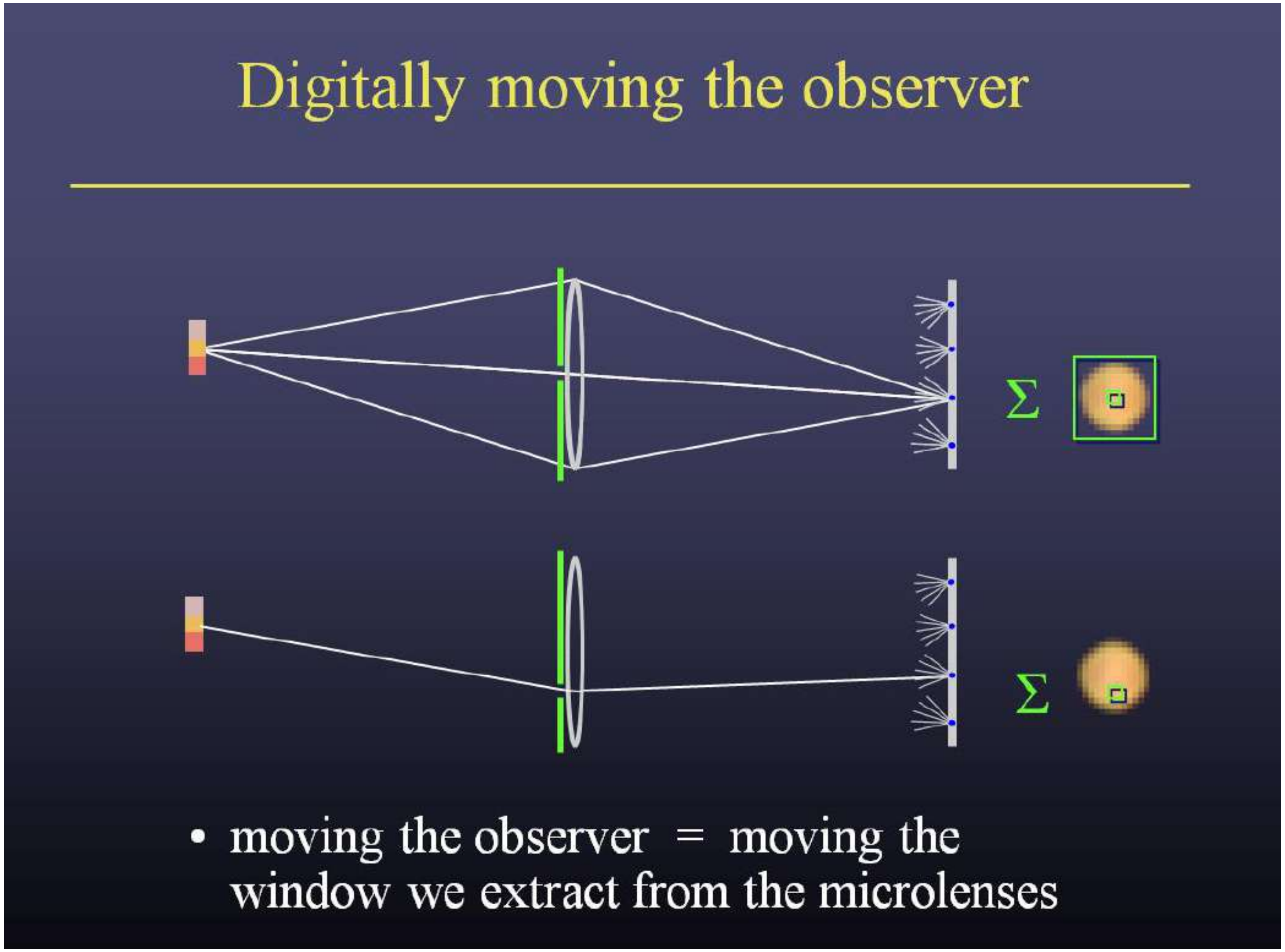
Light field photography



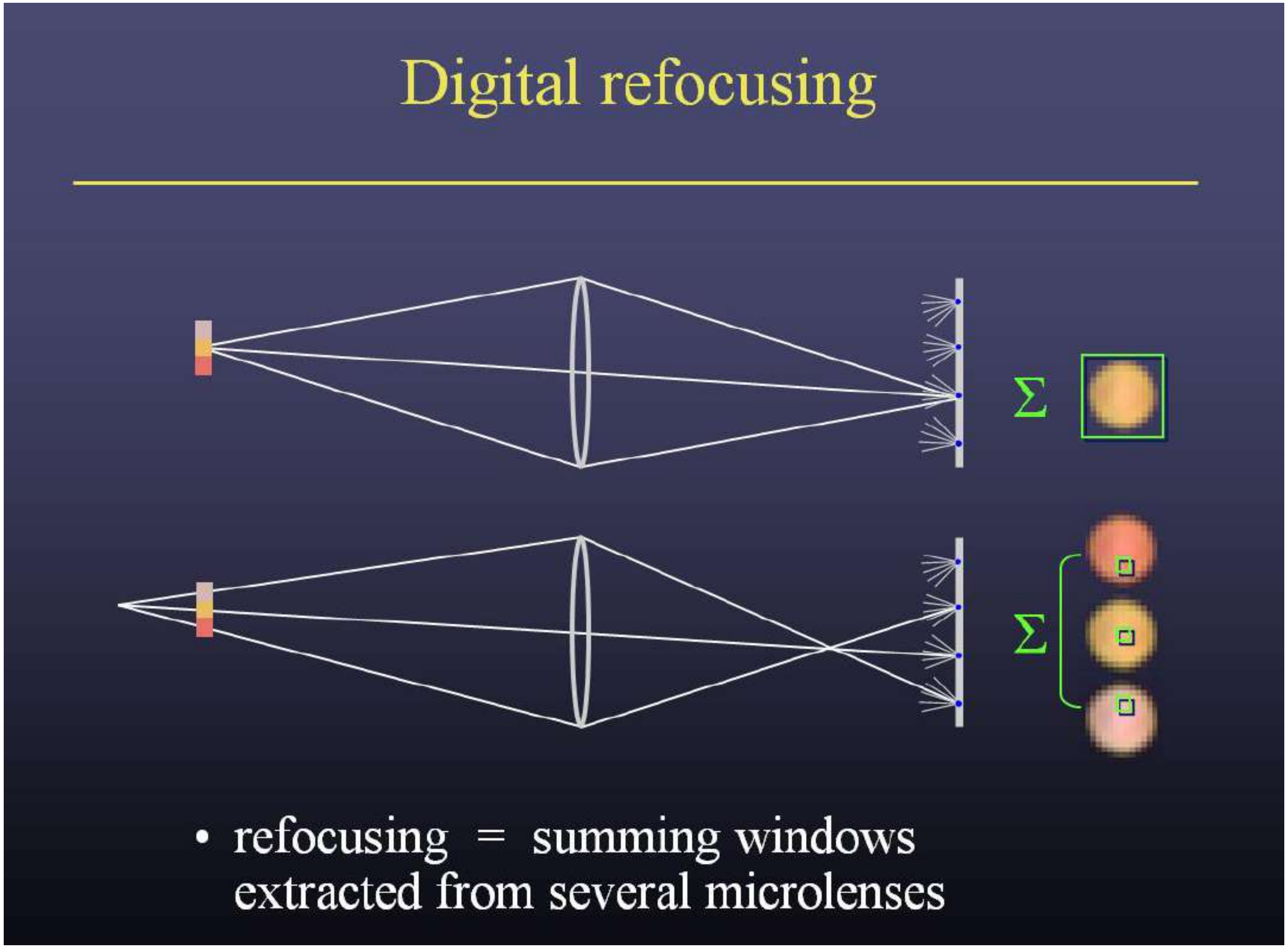
Light field photography



Light field photography



Light field photography



Light field photography

Digital refocusing

refocusing

all focused

Light field photography

Light field photography using a handheld plenoptic camera

Lytro camera

Lytro Illum pro camera

Light field photography

What is Lytro camera?

- Plenoptic camera that has a main photographic lens, a micro-lens array and a digital photosensor
- Captures the light field of an image to replicate it with the ability to refocus the image afterwards

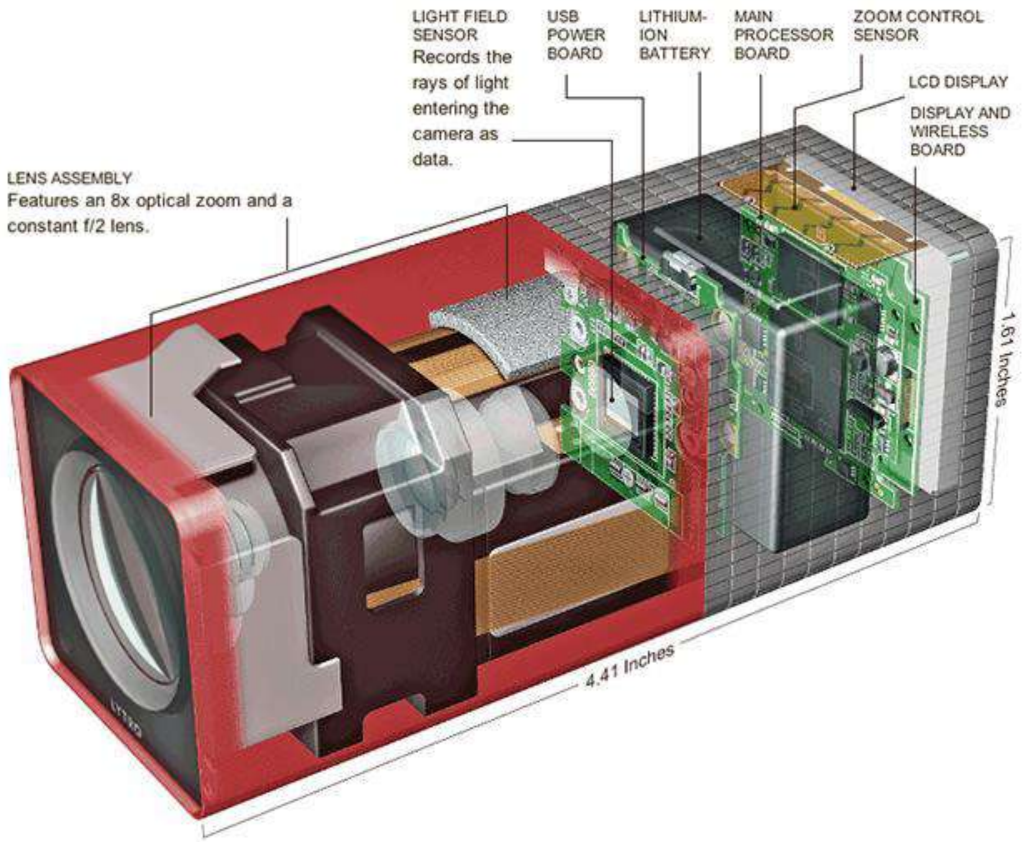
Light field photography

What is inside the Lytro Camera?

- Lens Array
- Light Field Sensor
- Light Field Engine
- Marvell Avastar 88W8787 SoC - a chip that offers WiFi and Bluetooth capabilities

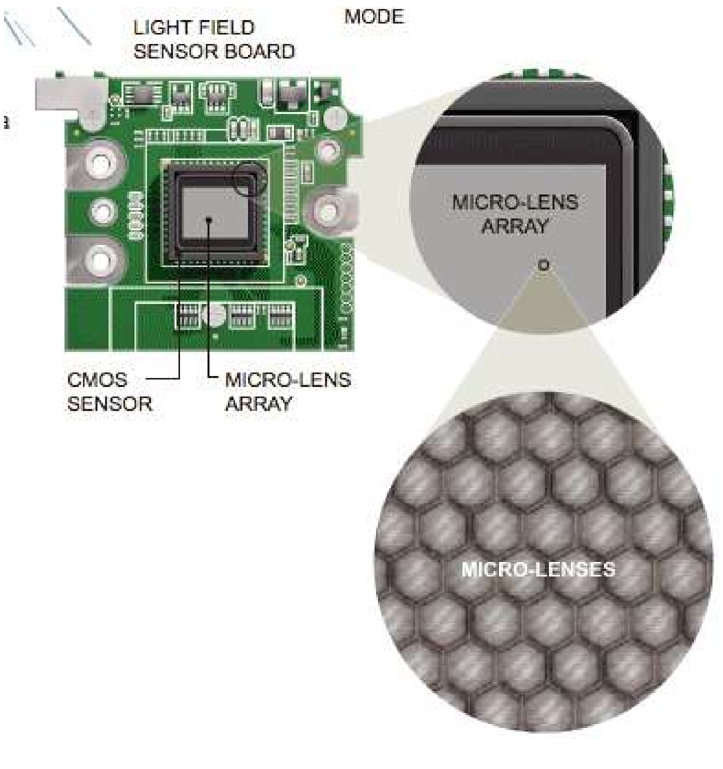
Light field photography

What is inside the Lytro Camera?



Light field photography

Light Field Sensor

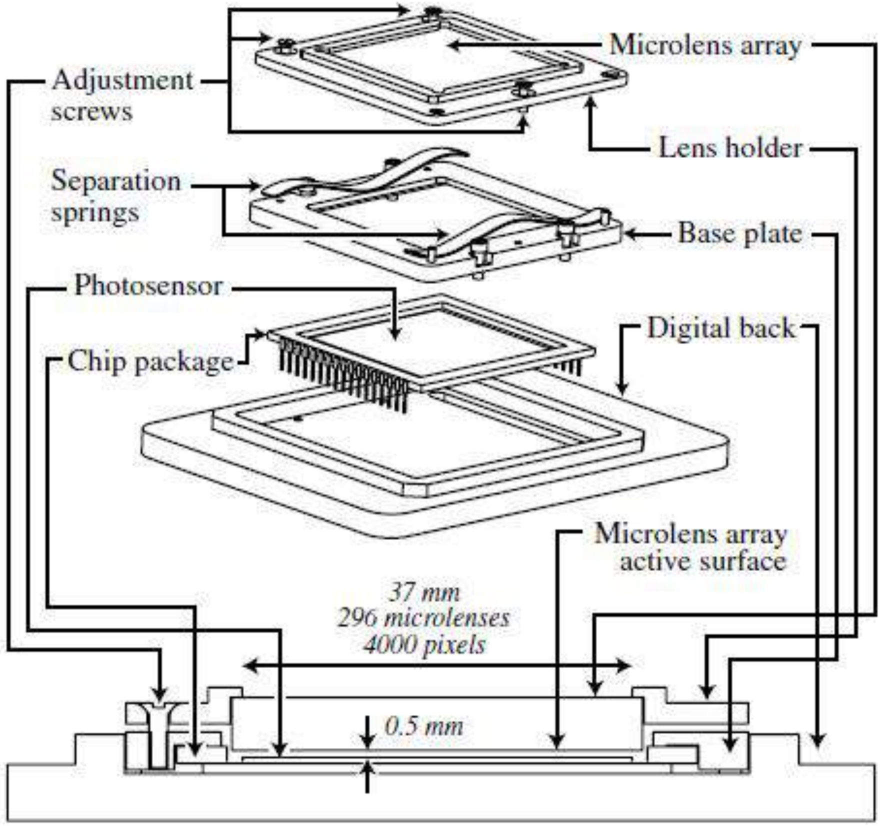


- CMOS Sensor: common in cameras, convert light into electrons and place it into a cell. It then reads the values in each cell and changes it to a digital value.
- Micro-lens array: attached to a standard sensor, it divides the pixels from the CMOS into areas, showing the image at different angles.

Light field photography

Light field photography

Lytro camera specification



Light field photography

Industrial plenoptic cameras

Plenoptic cameras have become quite popular in lab and industrial settings.

- Much higher resolution, both spatial and angular, than commercial cameras.
- Support interchangeable lenses.
- Can do video.
- <u>Very</u> expensive.

Light field photography

Advantages

- highly portable
- refocusing ability

Disadvantages

- reduced total resolution
- fixed aperture

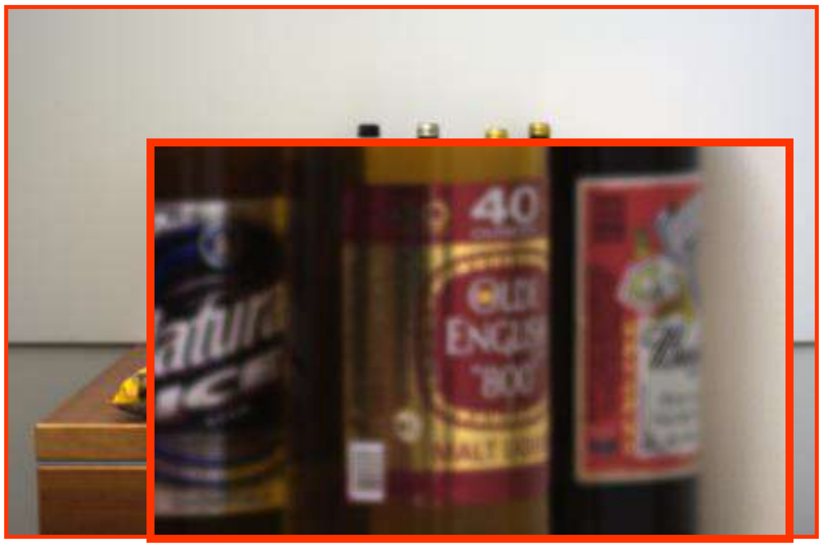
Conventional camera with coded aperture

Output #1: Depth map

Single input image

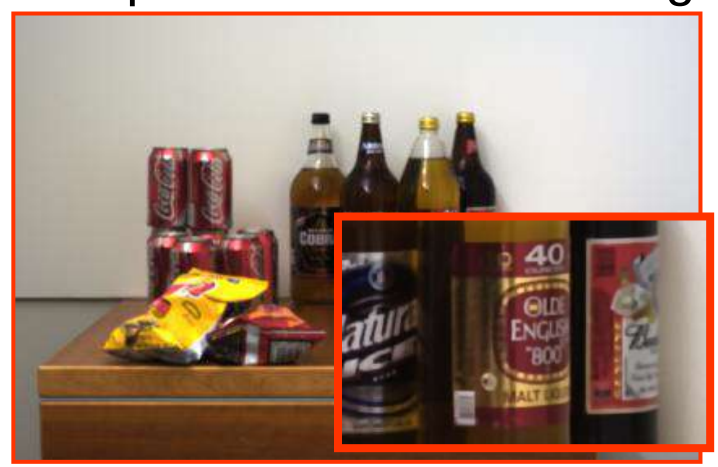
Conventional camera with coded aperture

Single input image



Output #1: Depth map

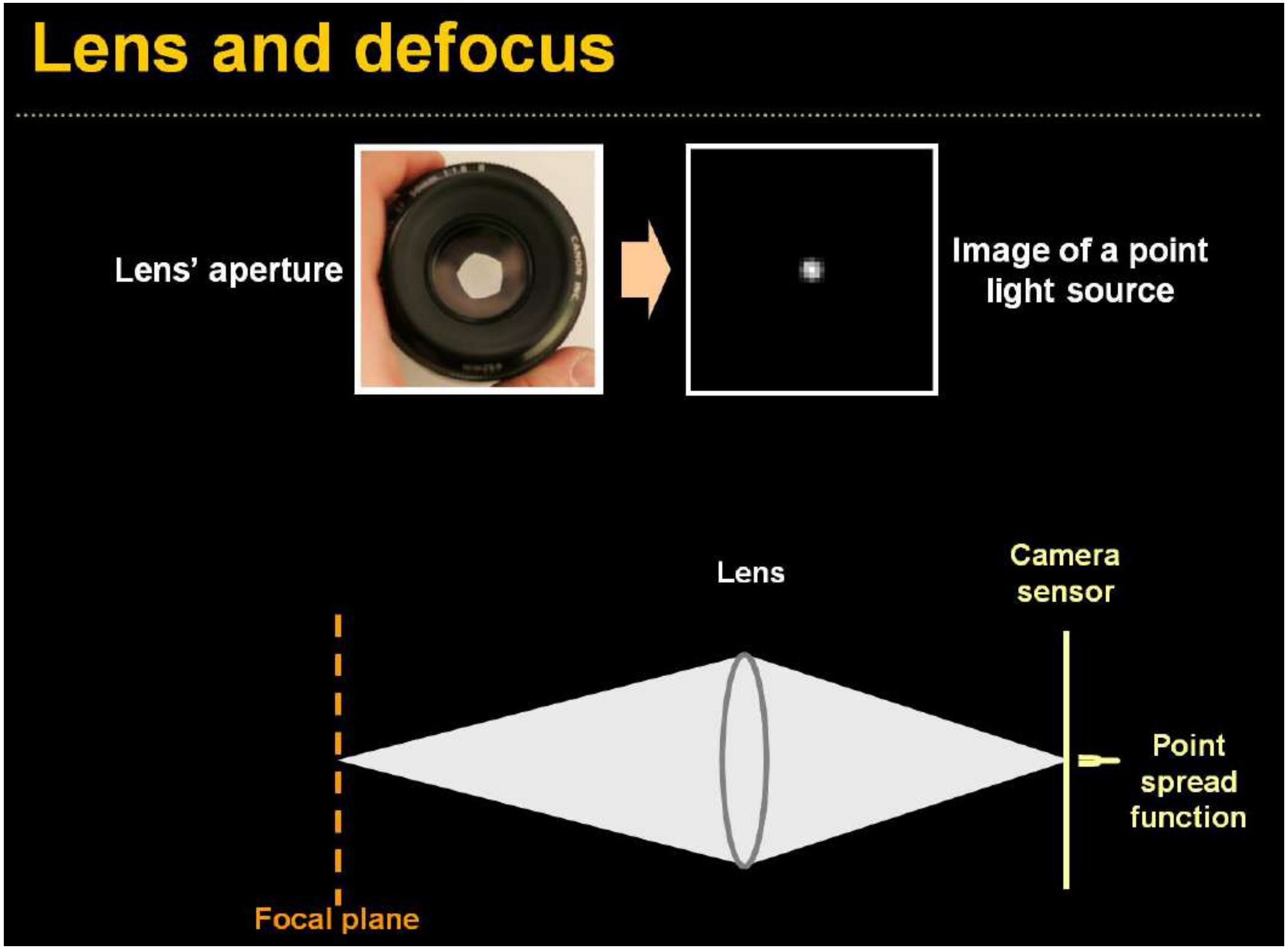
Output #2: All-focused images



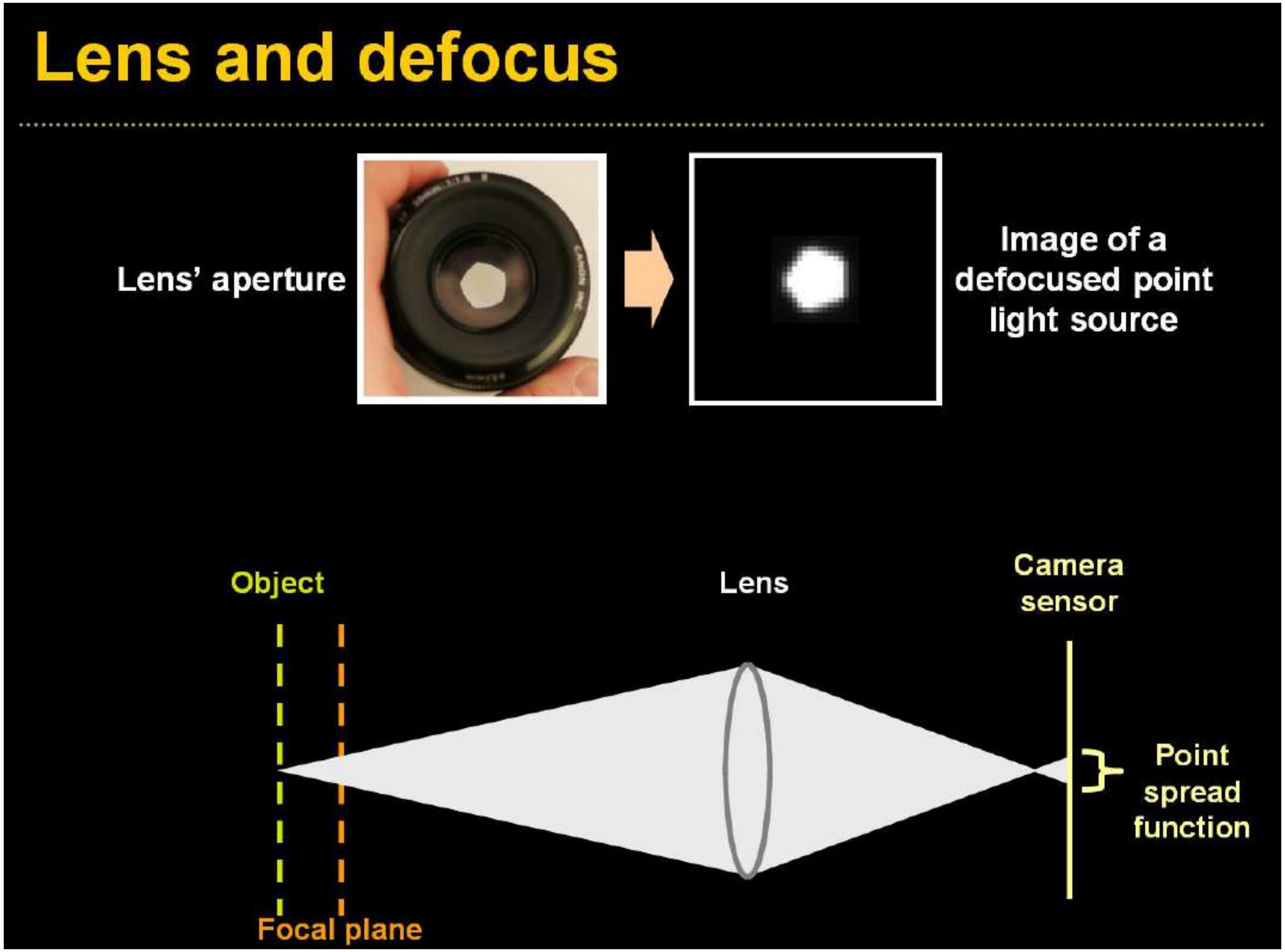
Conventional camera with coded aperture

Conventional camera vs. camera with coded aperture

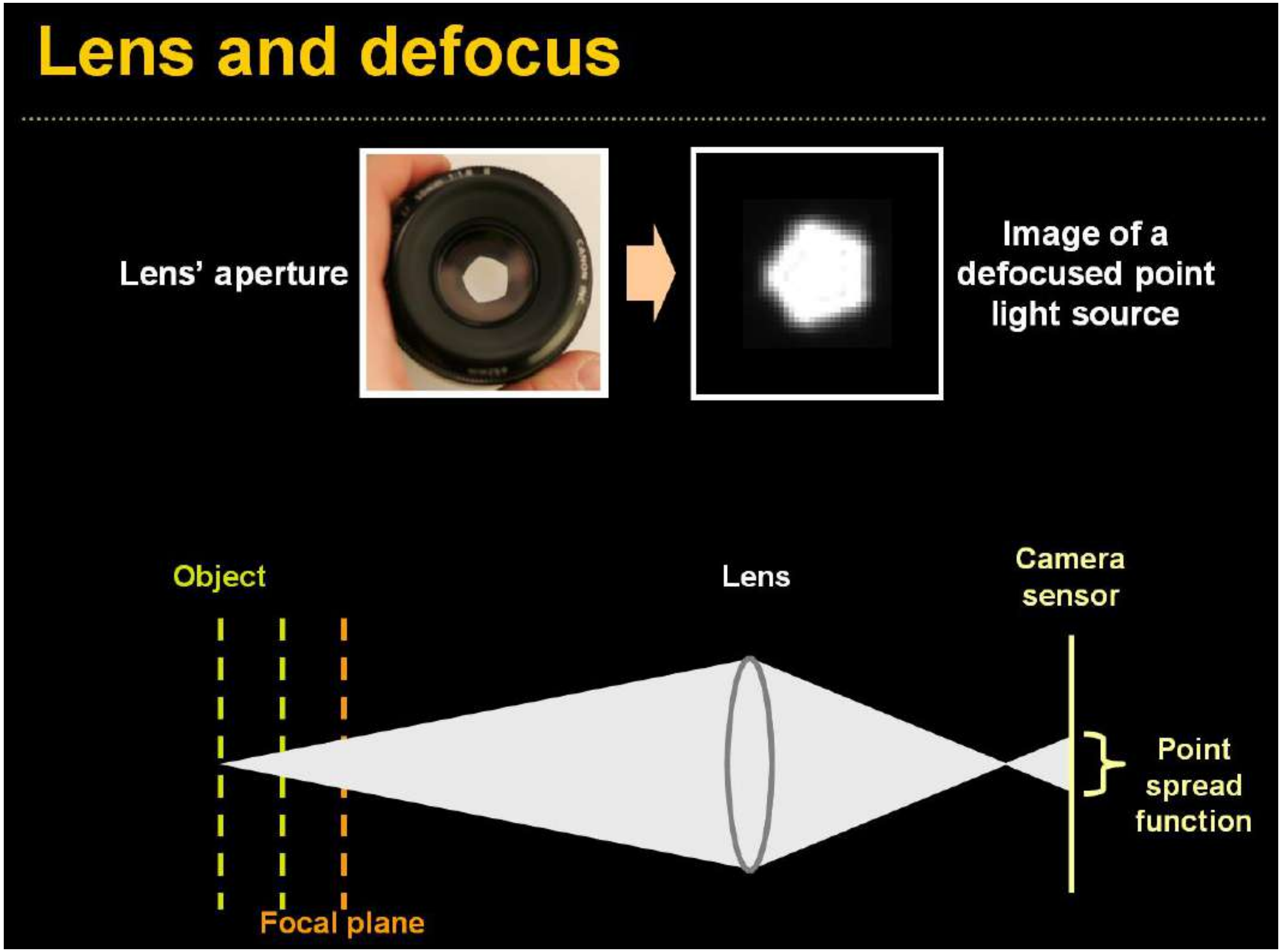
Conventional camera with coded aperture



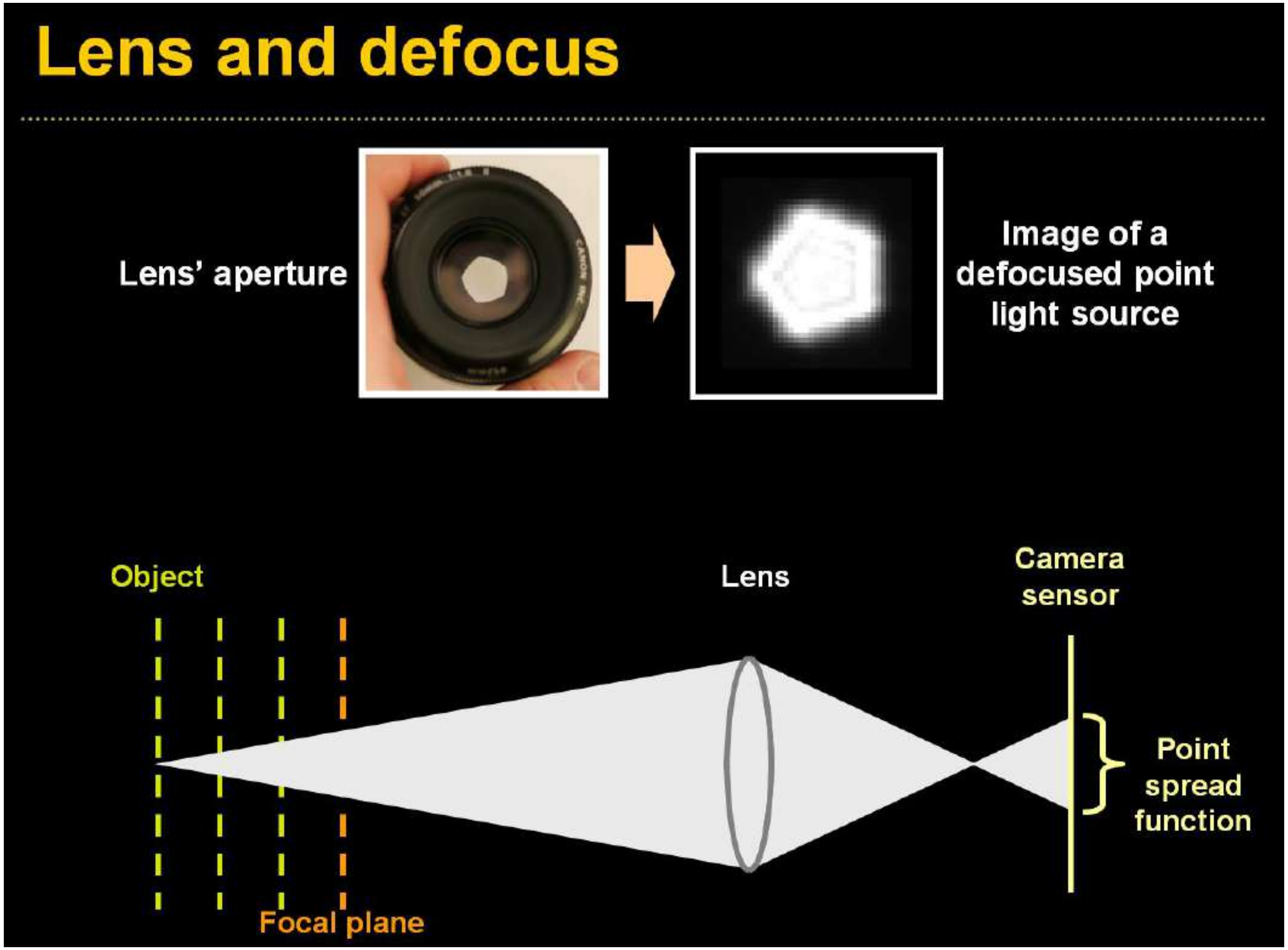
Conventional camera with coded aperture



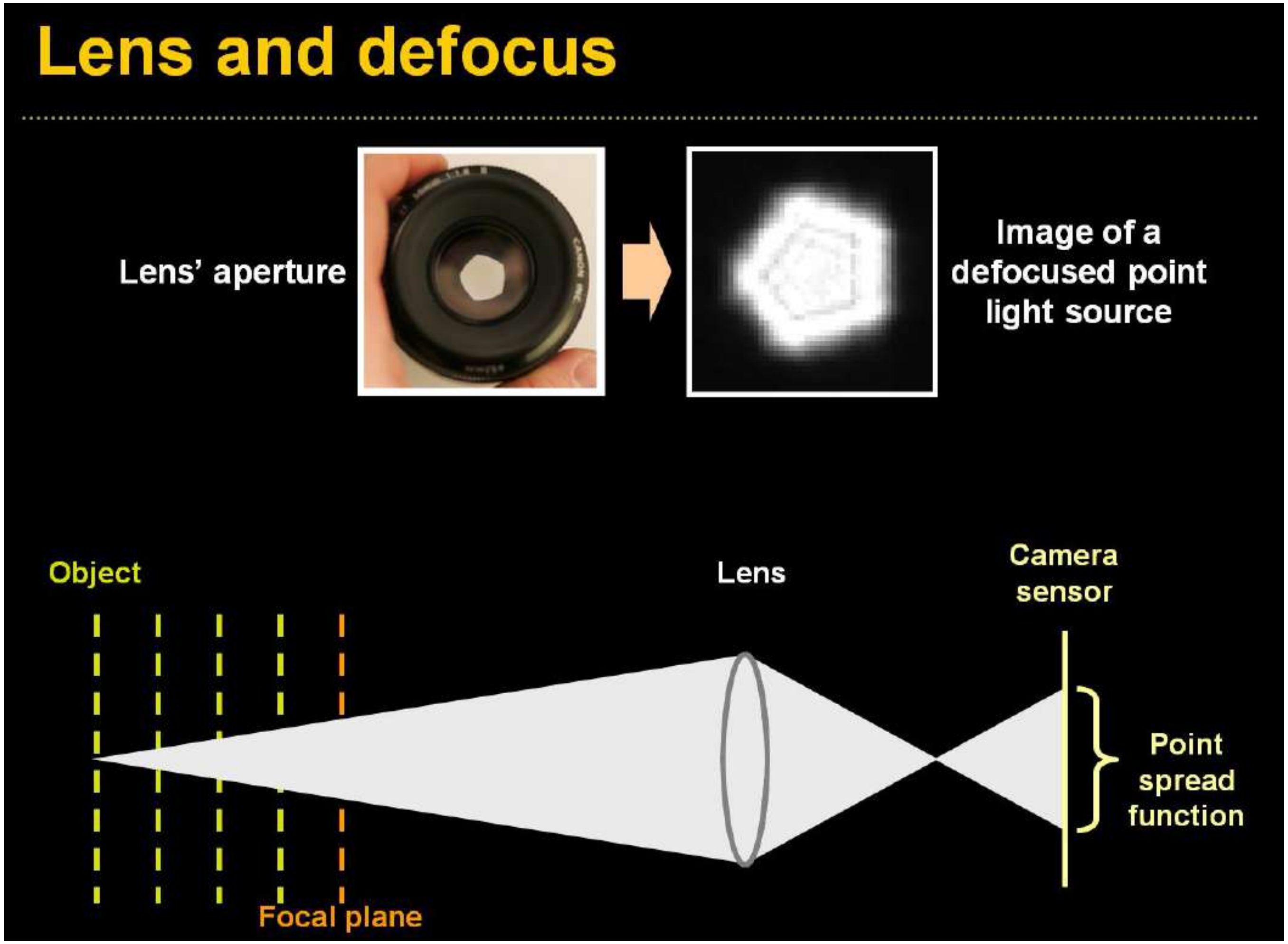
Conventional camera with coded aperture



Conventional camera with coded aperture

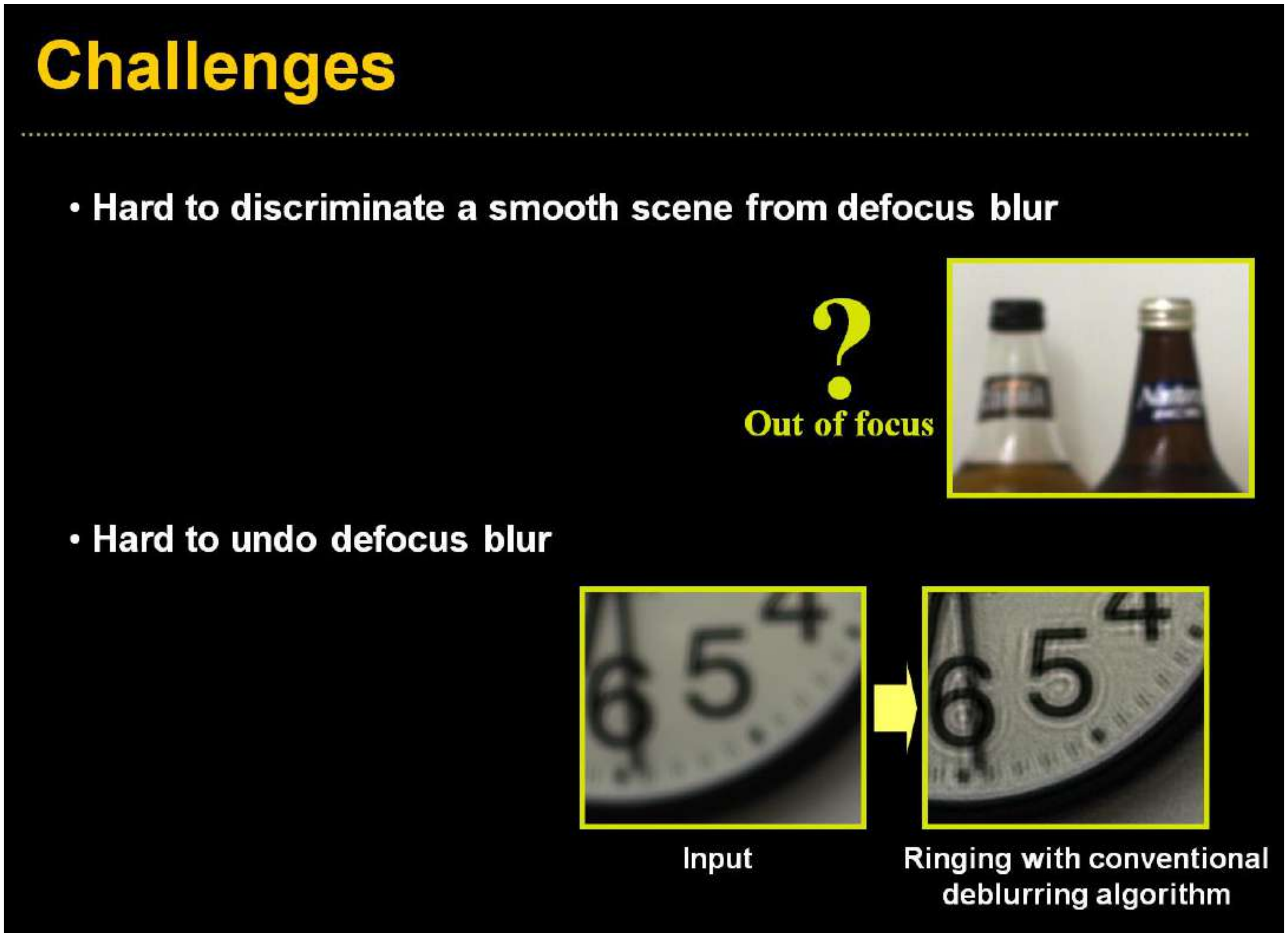


Conventional camera with coded aperture

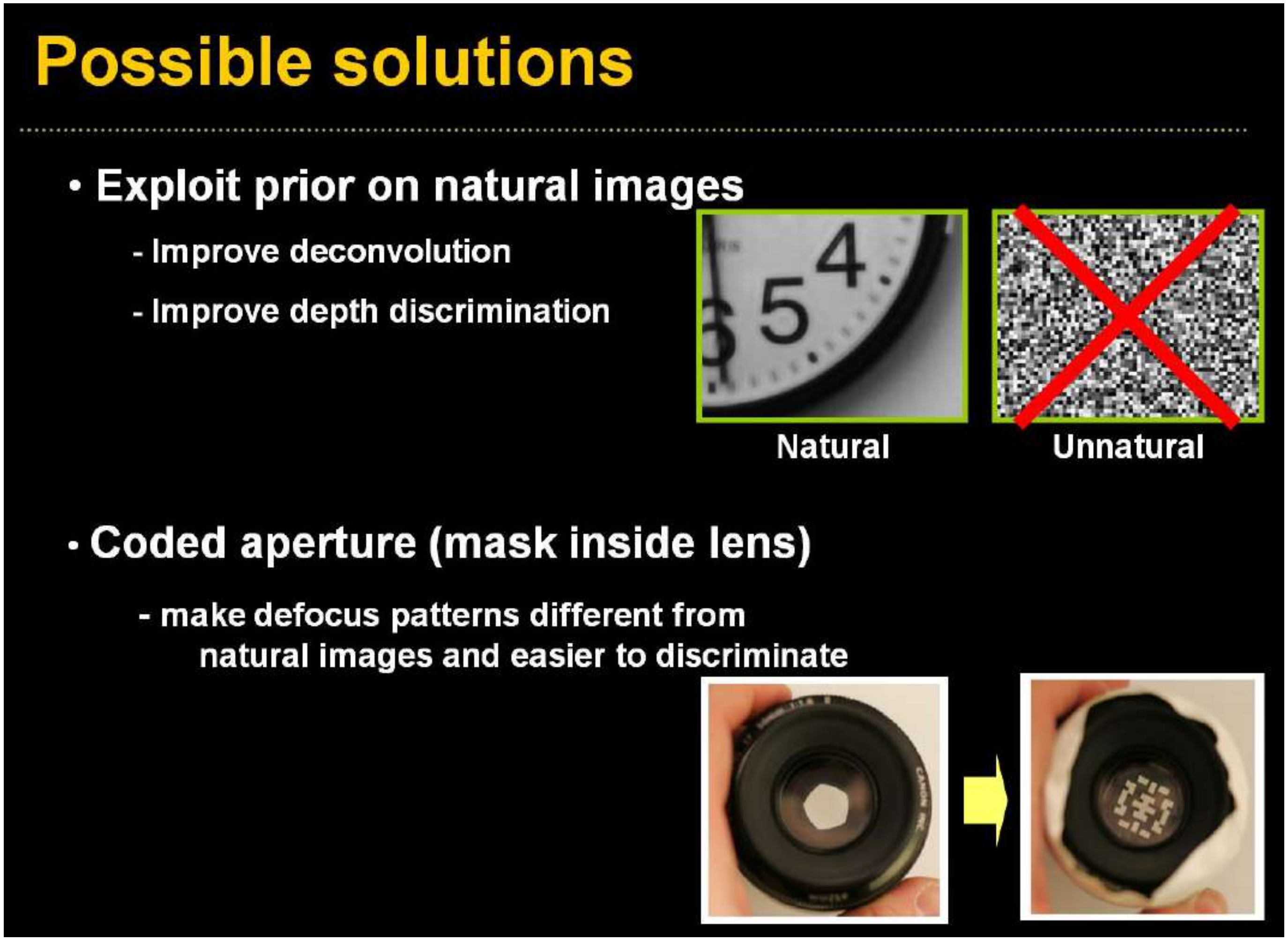


Conventional camera with coded aperture

Conventional camera with coded aperture

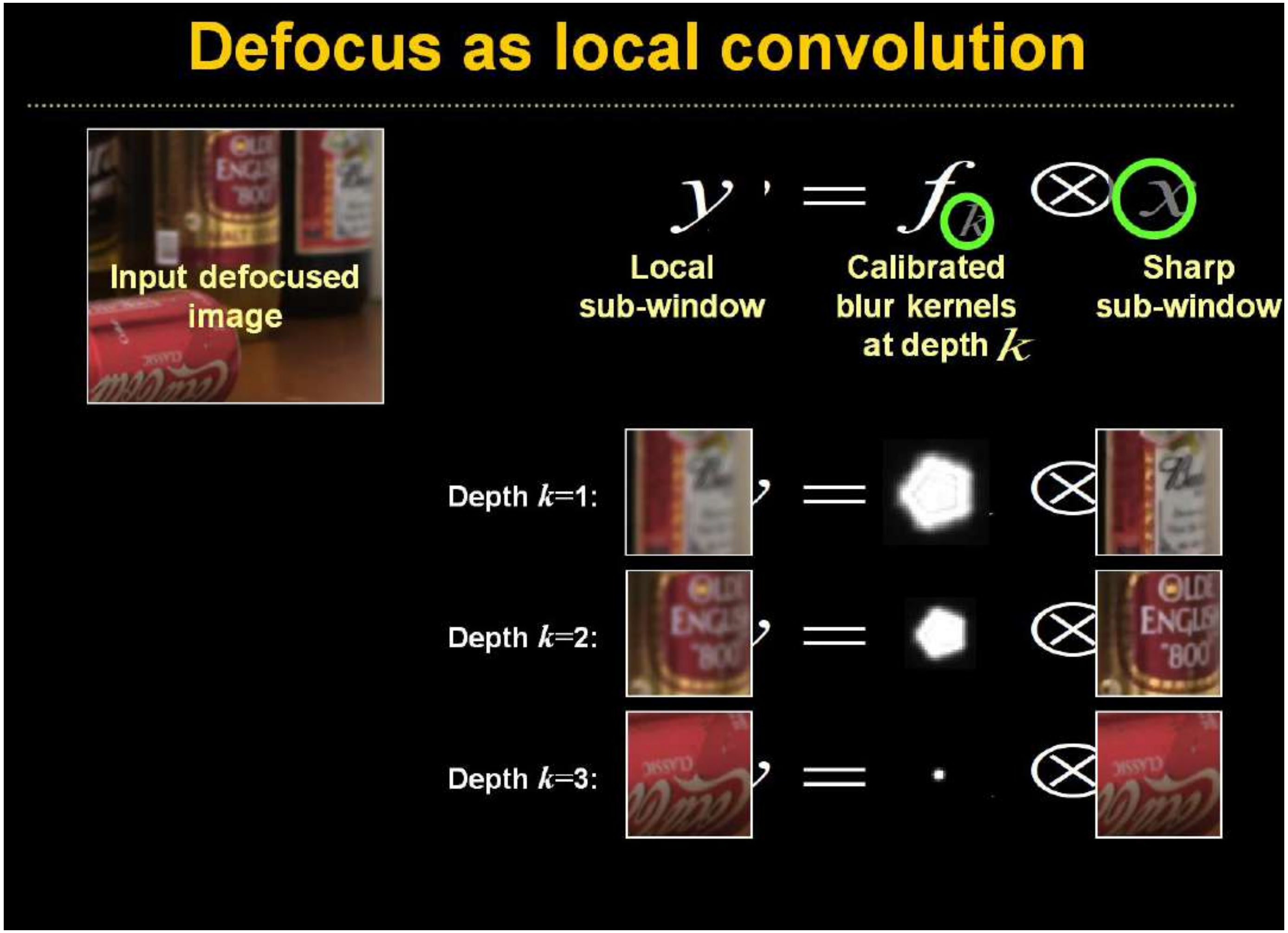


Conventional camera with coded aperture

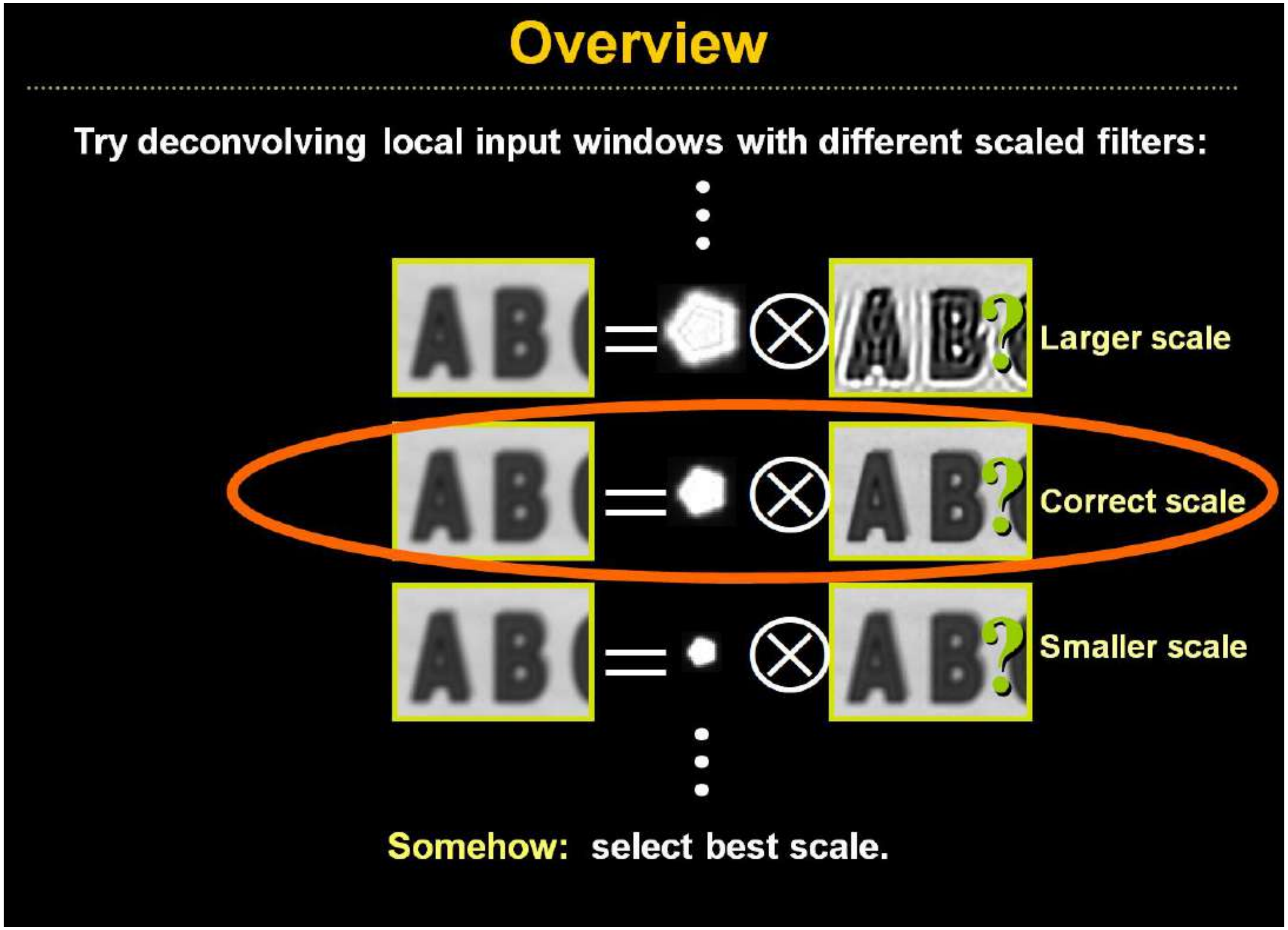


Conventional camera with coded aperture

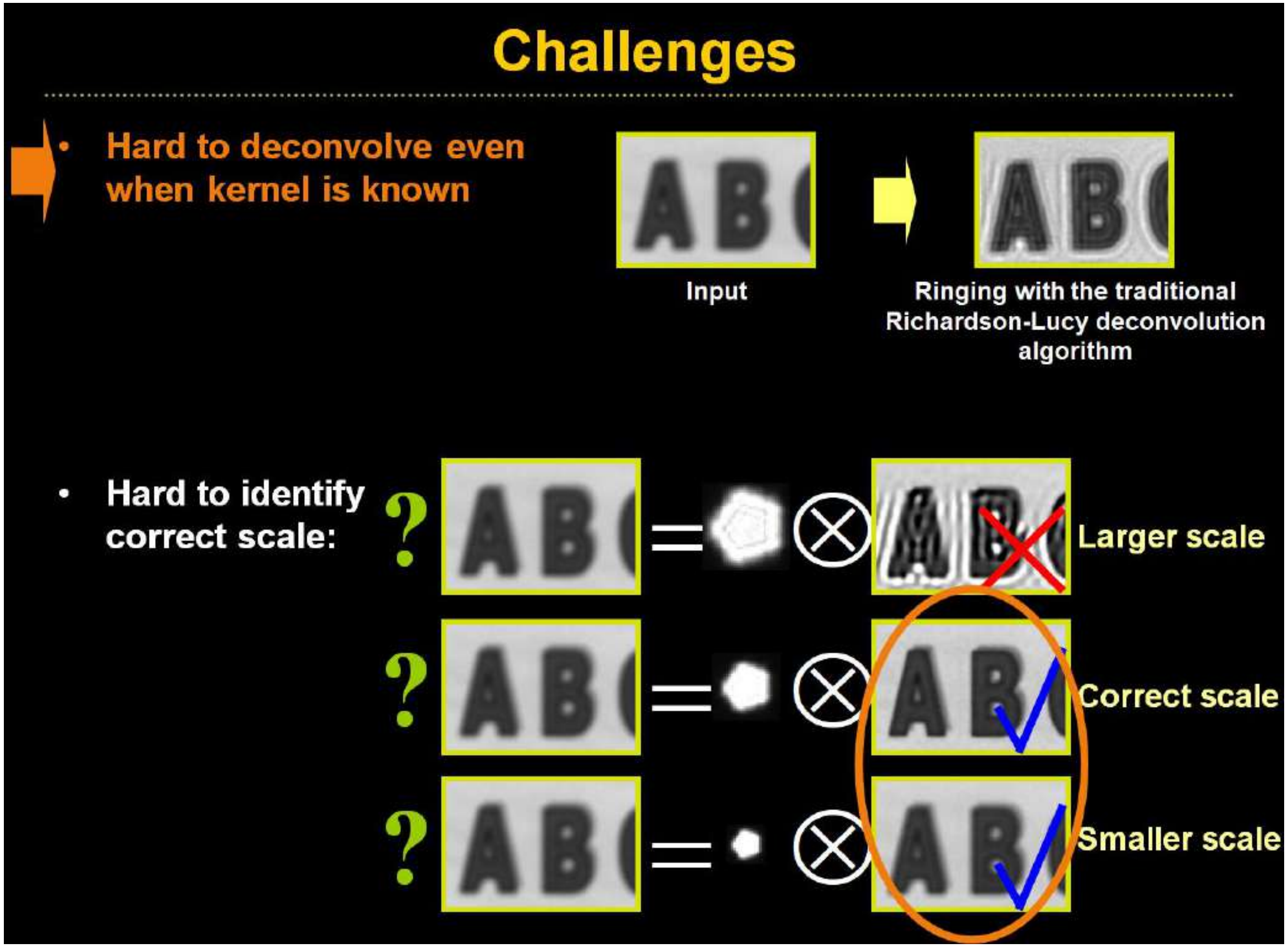
Conventional camera with coded aperture



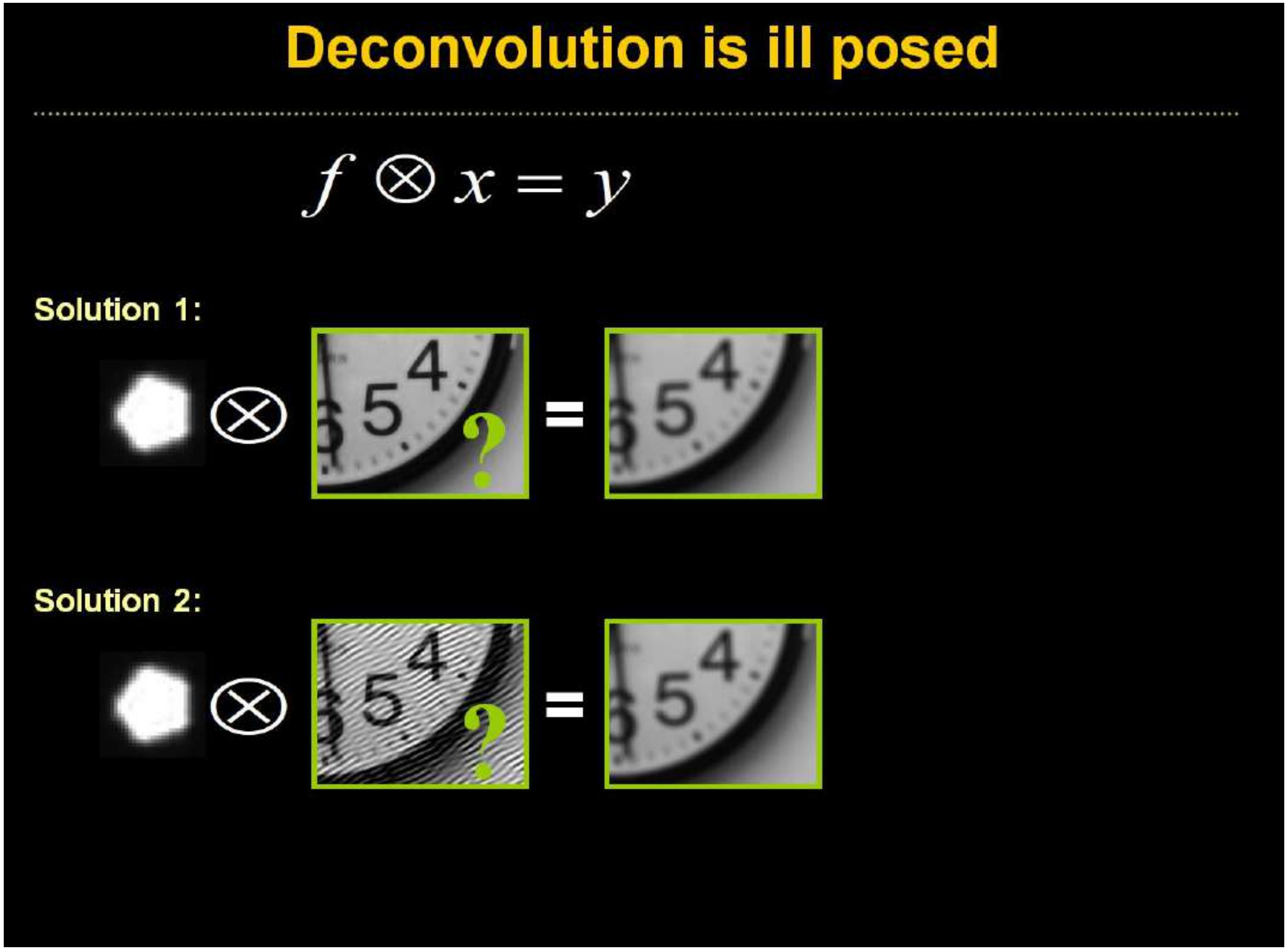
Conventional camera with coded aperture



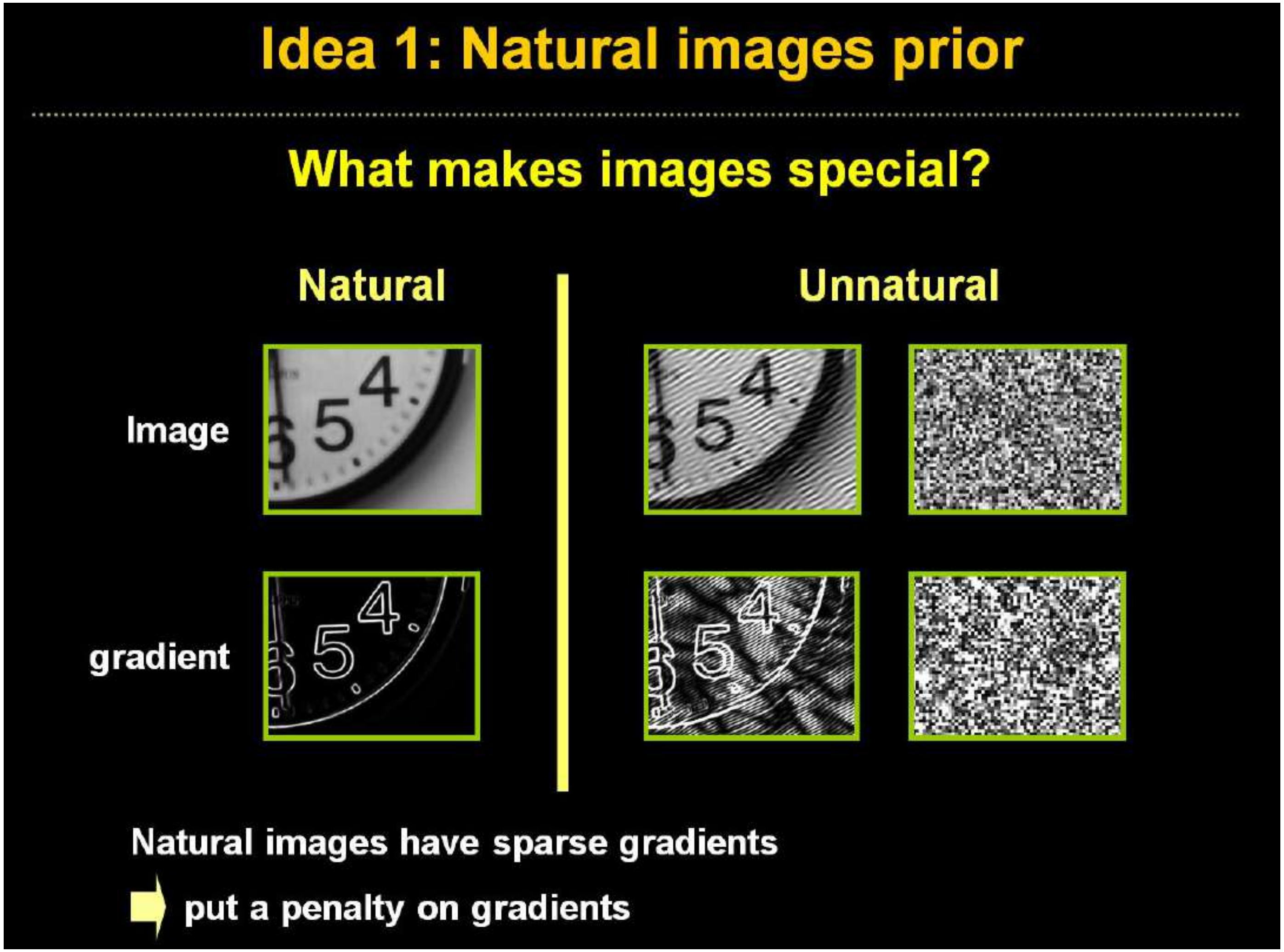
Conventional camera with coded aperture



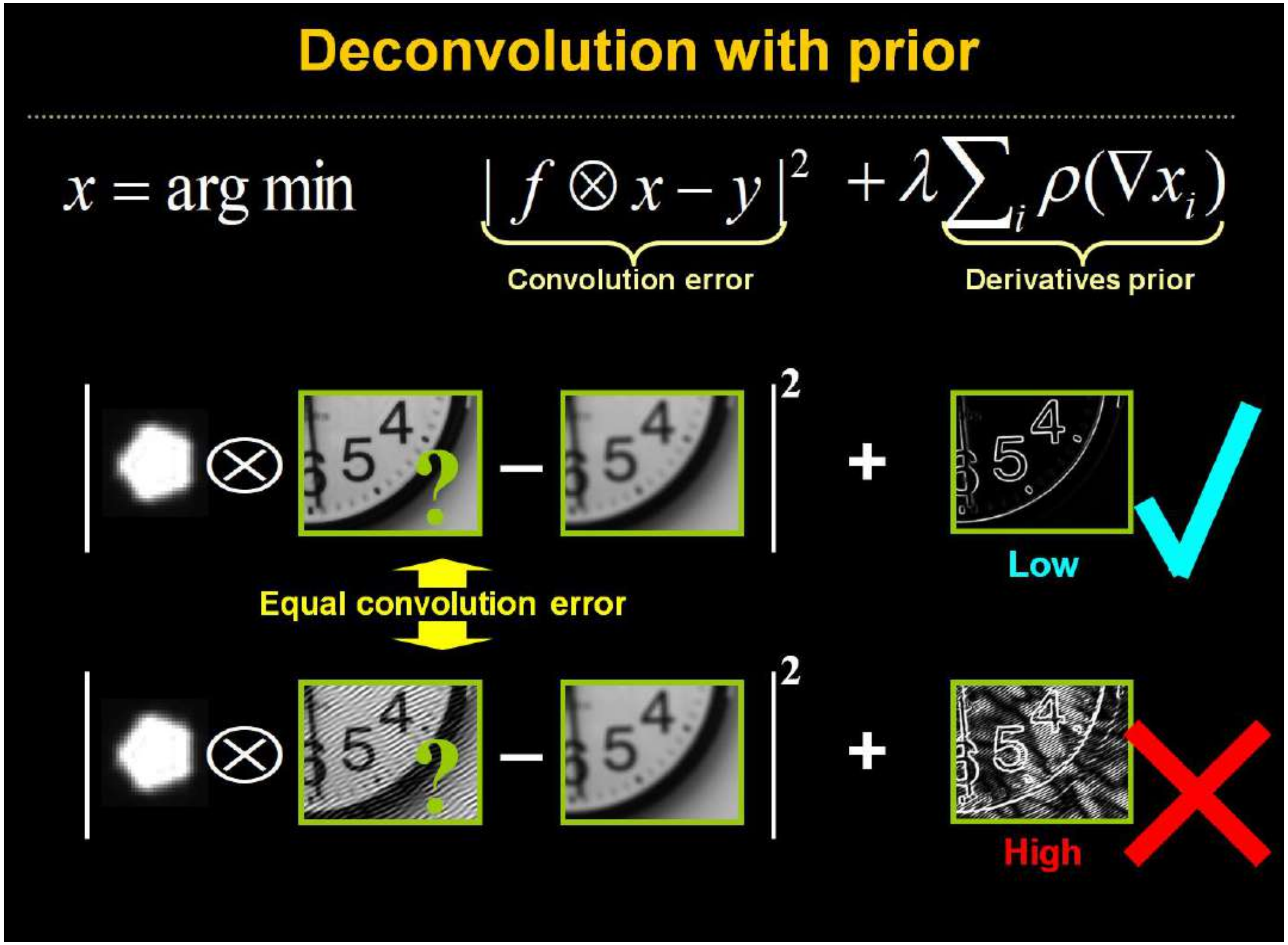
Conventional camera with coded aperture



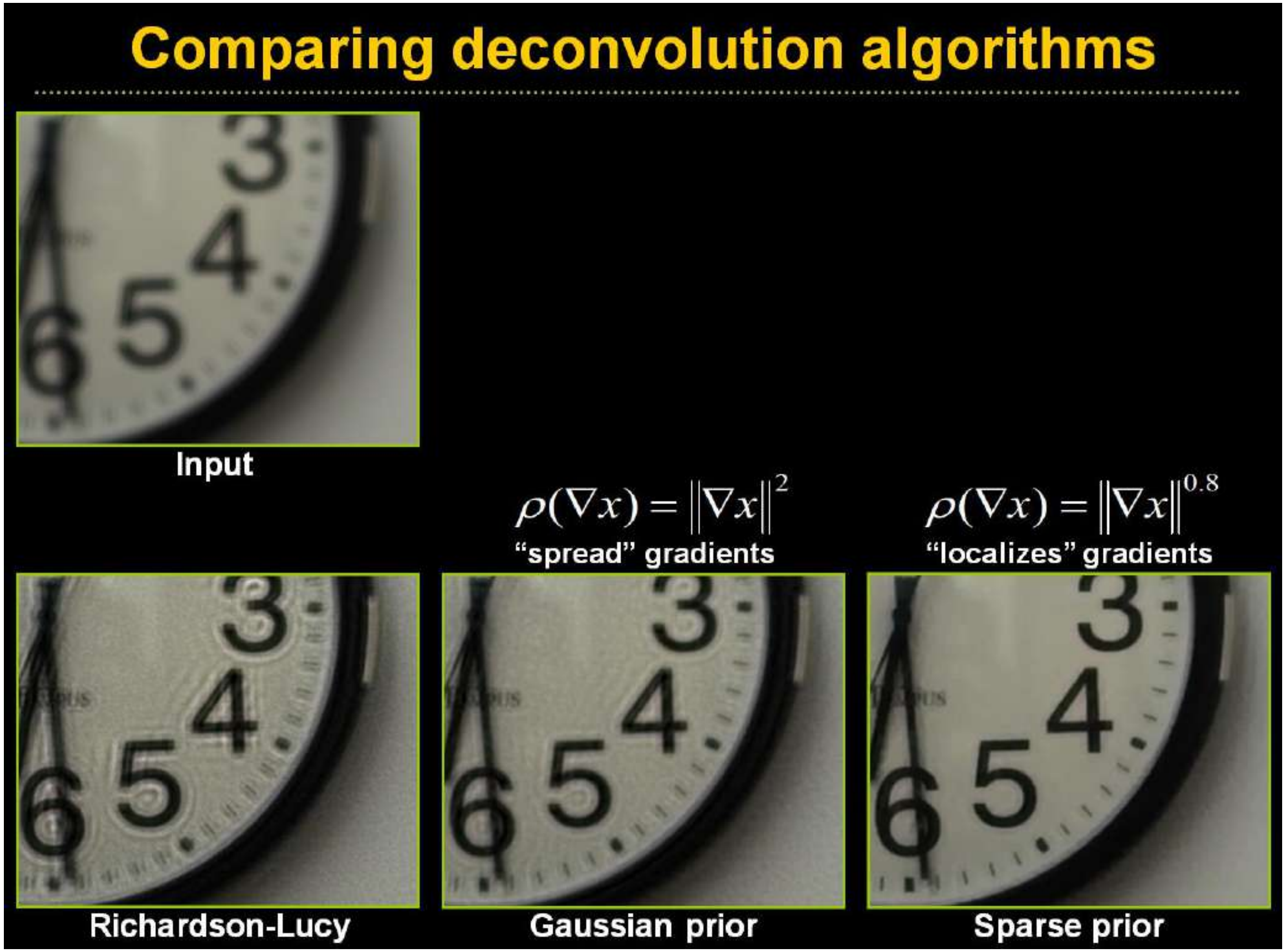
Conventional camera with coded aperture



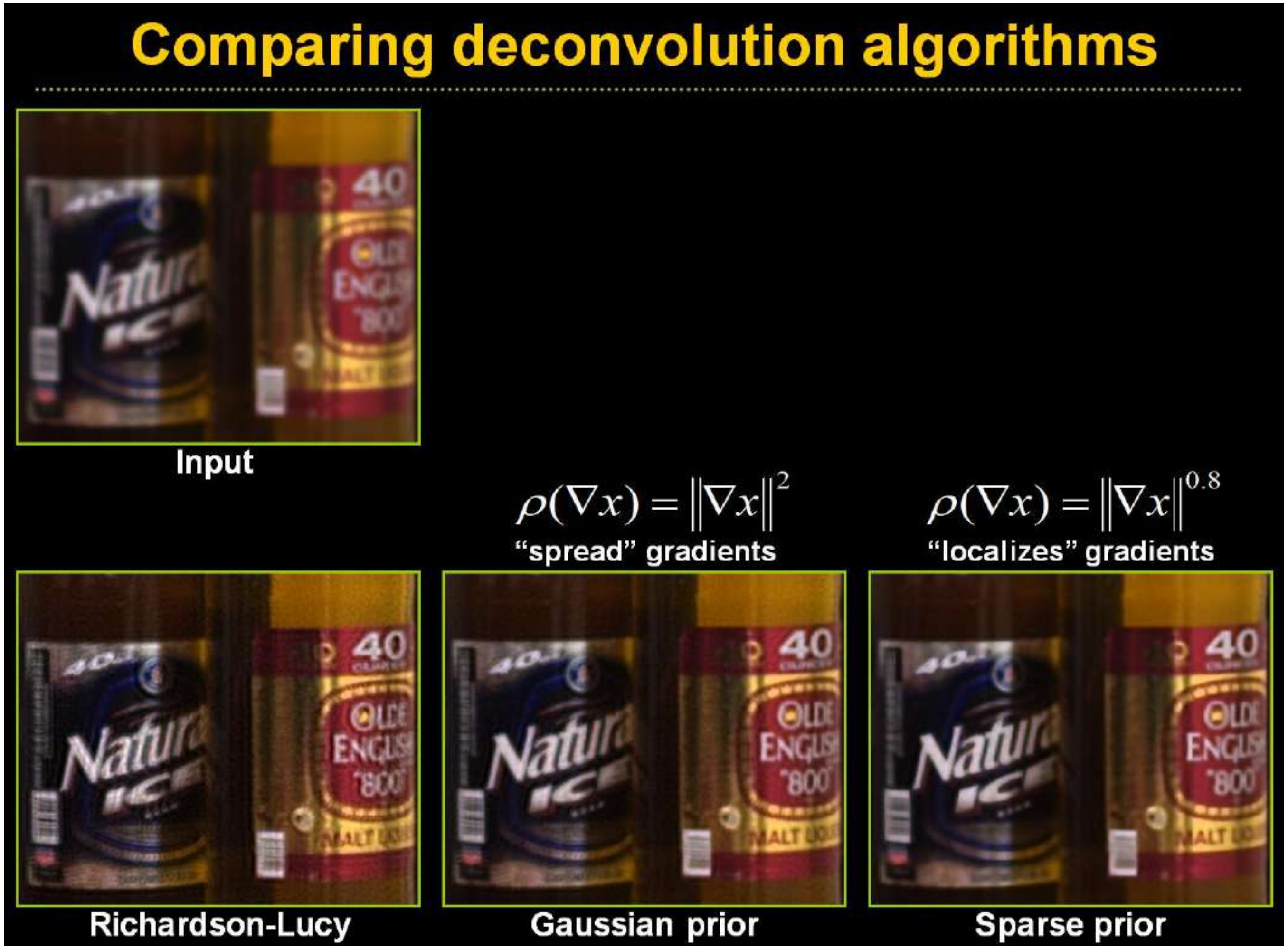
Conventional camera with coded aperture



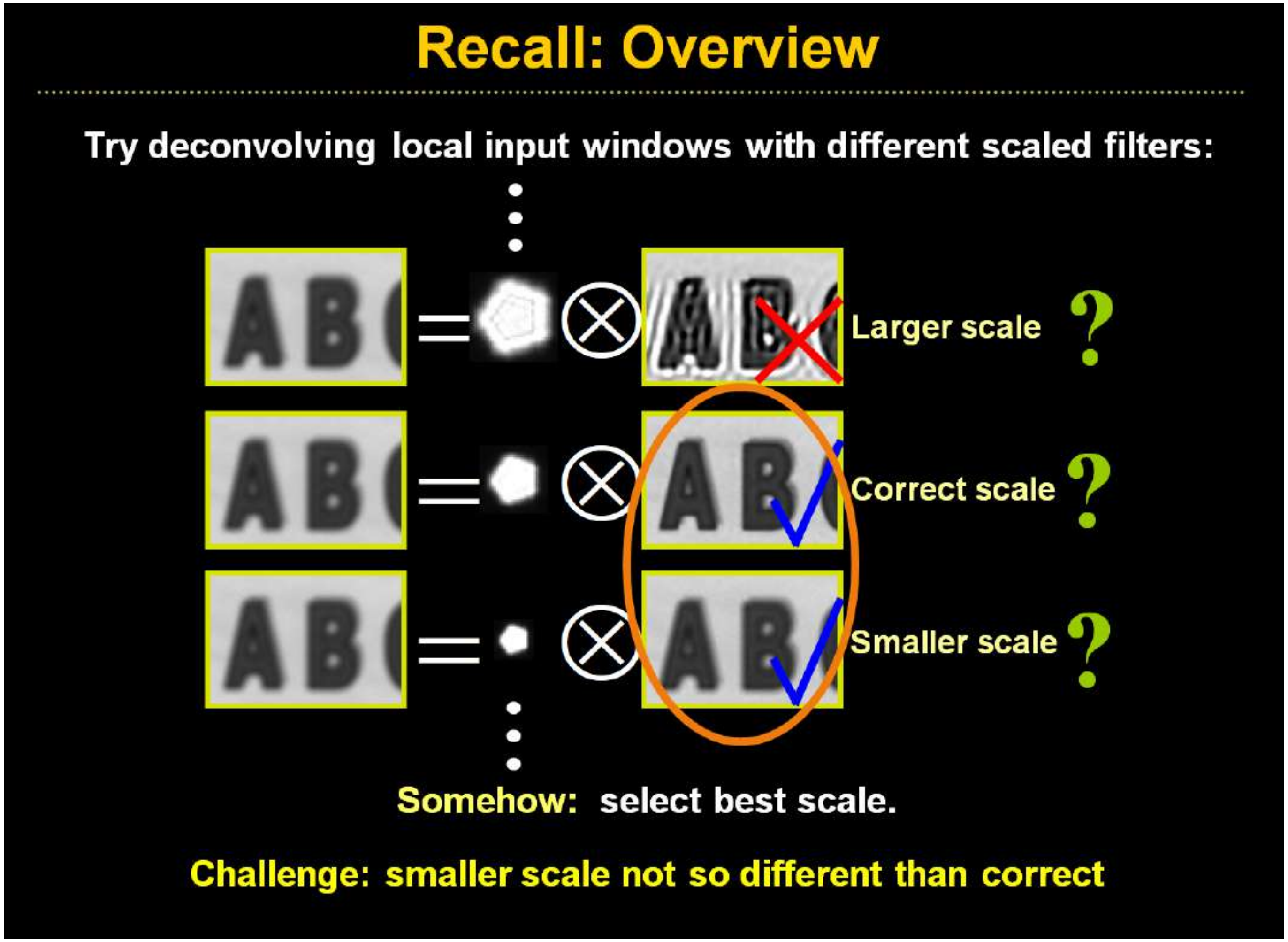
Conventional camera with coded aperture



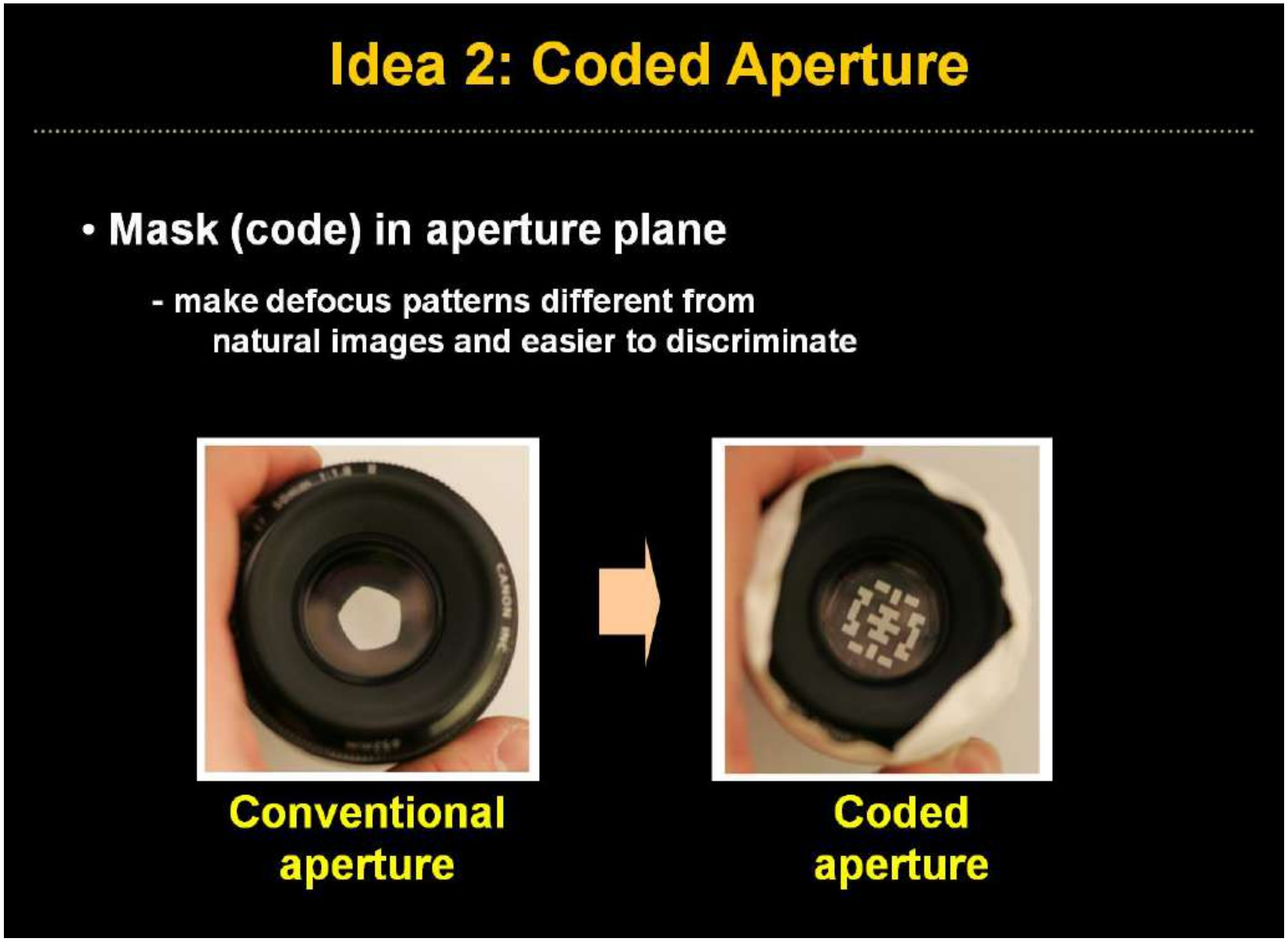
Conventional camera with coded aperture



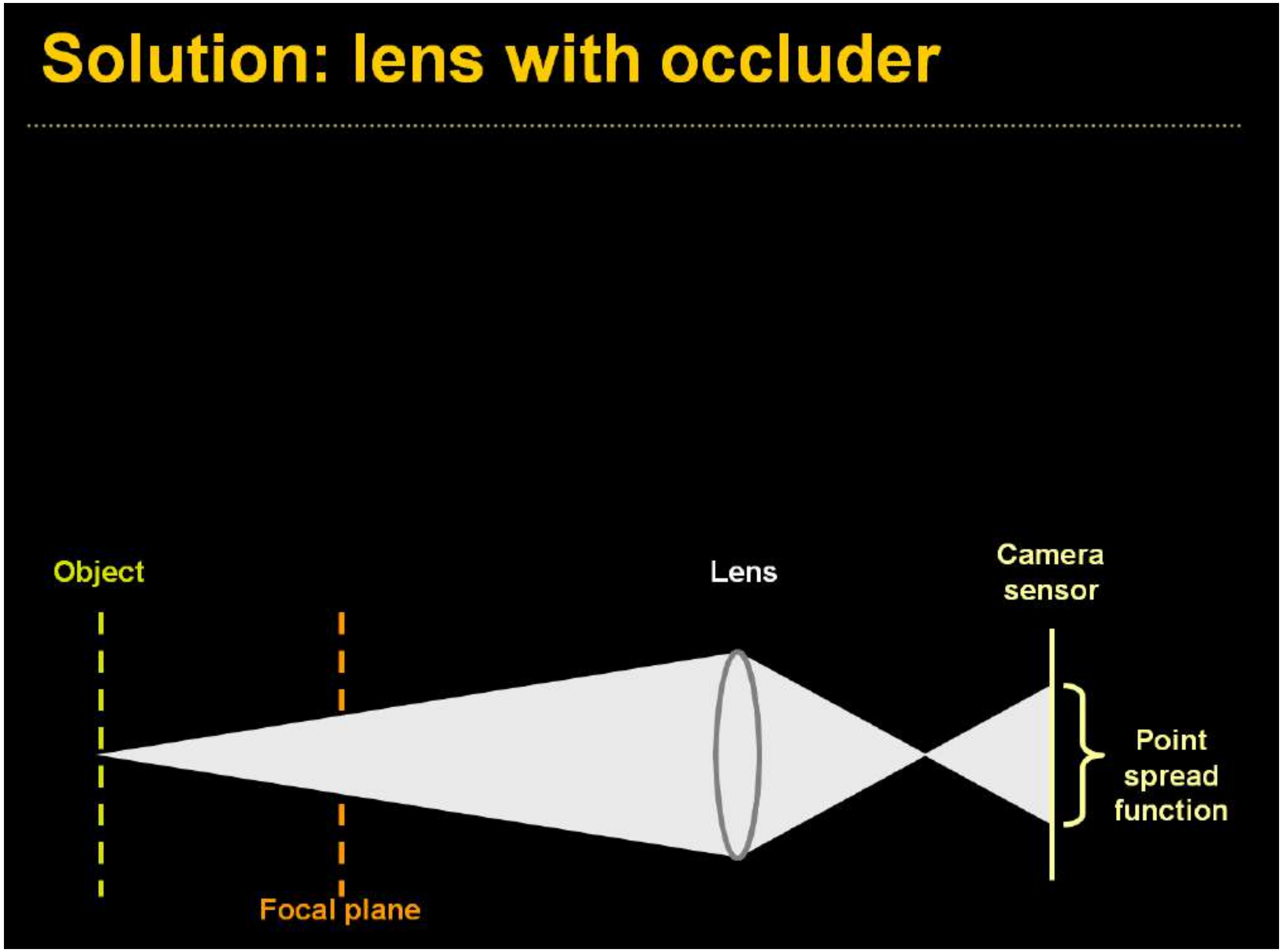
Conventional camera with coded aperture



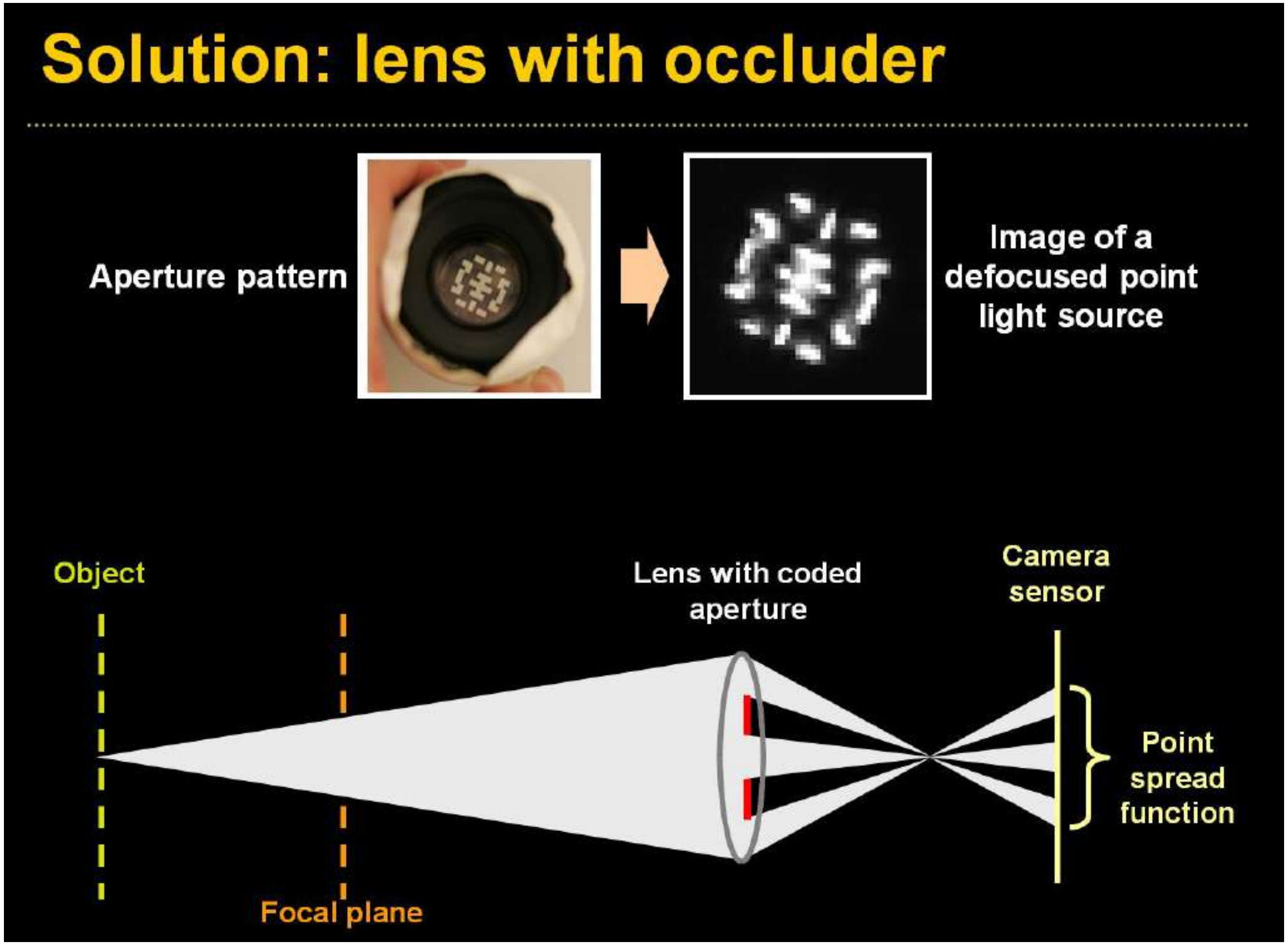
Conventional camera with coded aperture



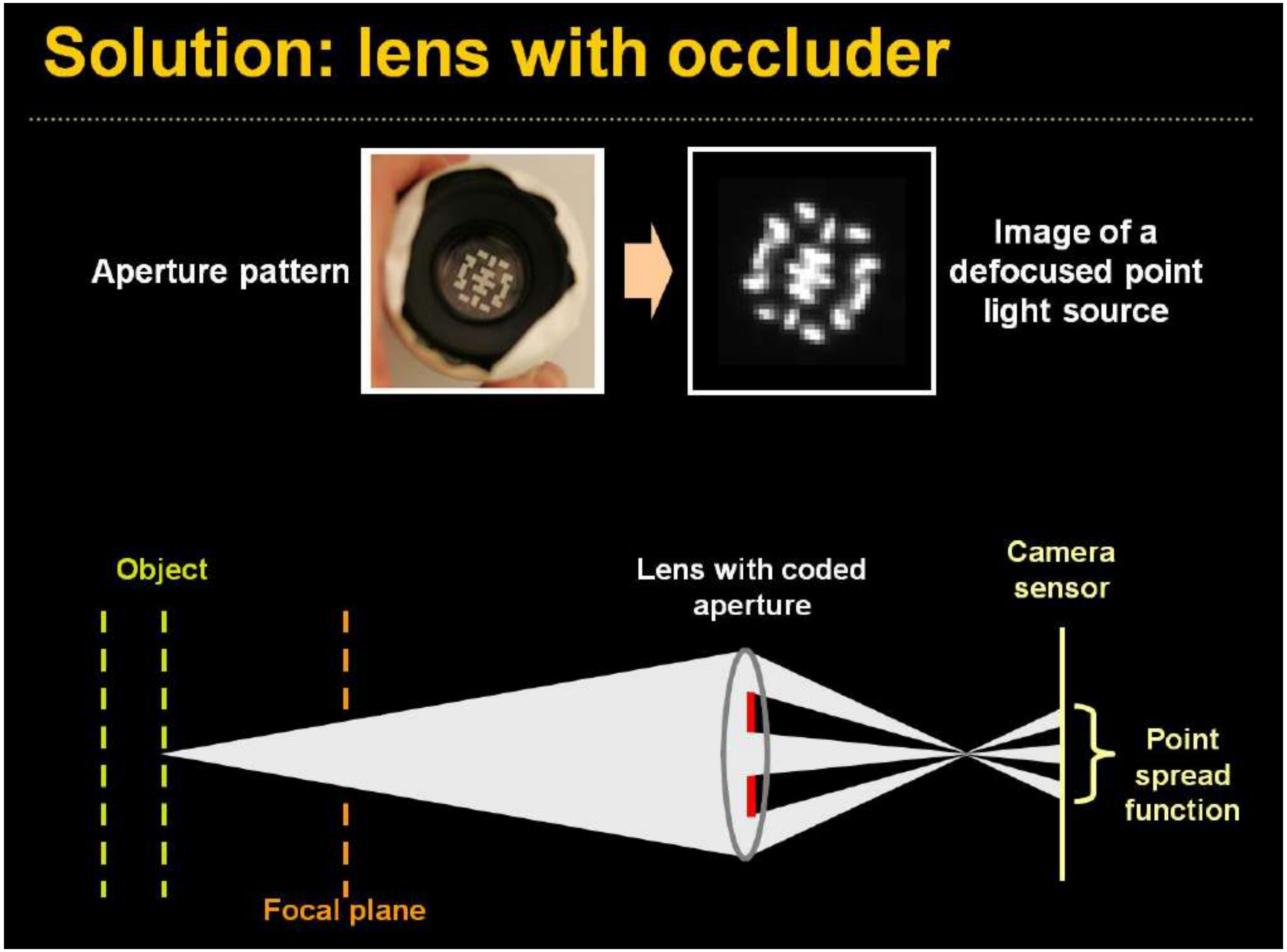
Conventional camera with coded aperture



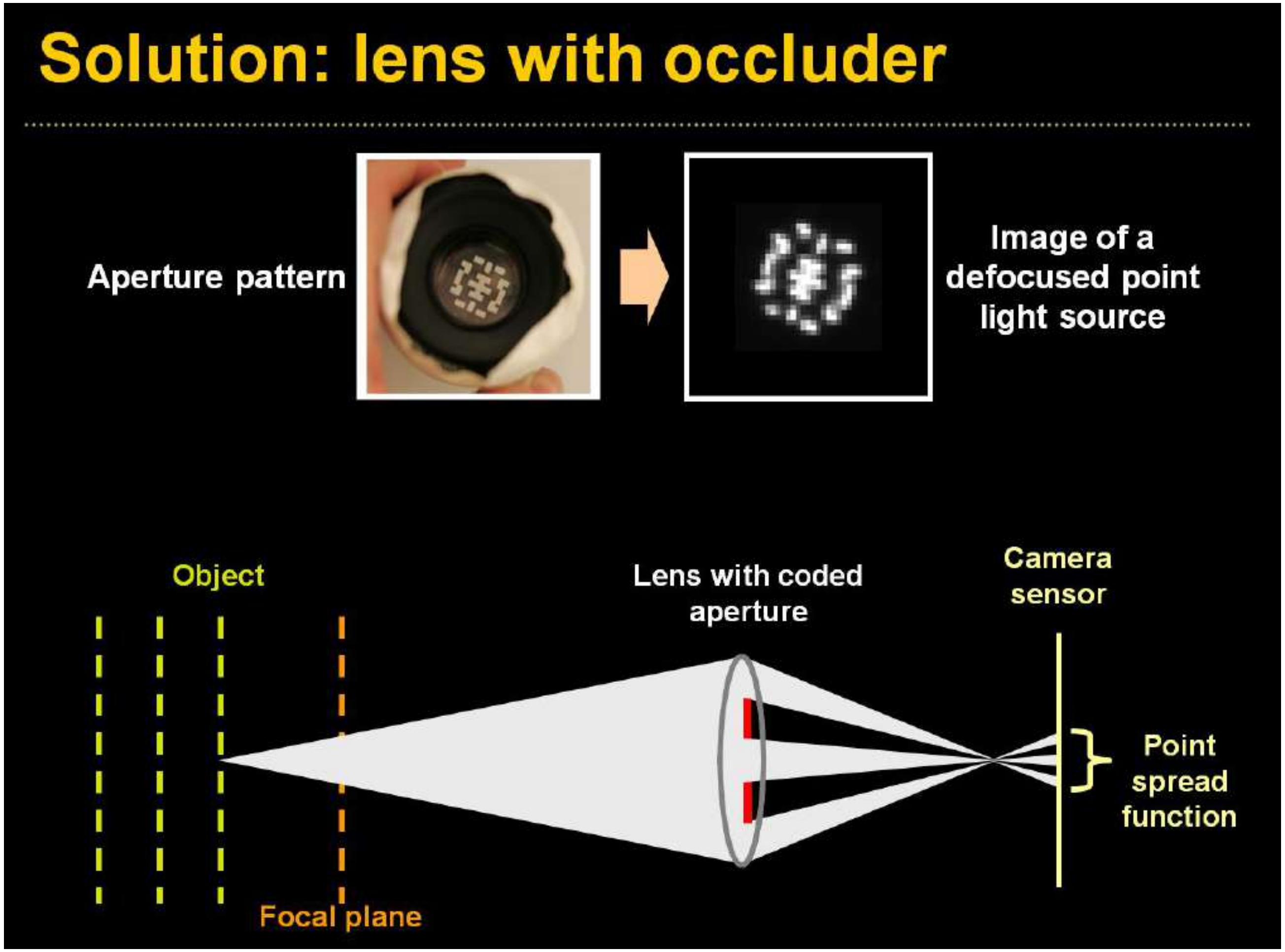
Conventional camera with coded aperture



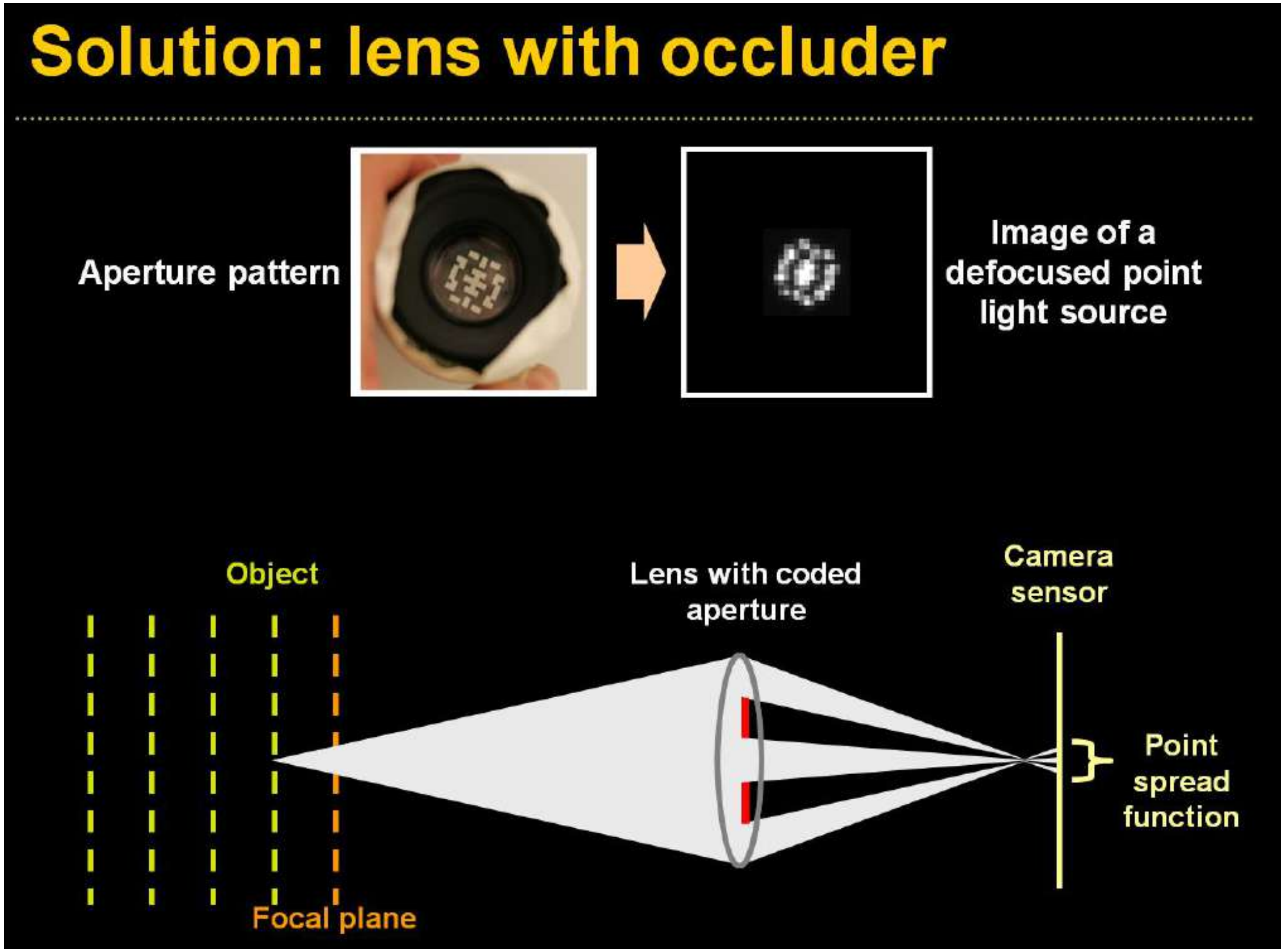
Conventional camera with coded aperture



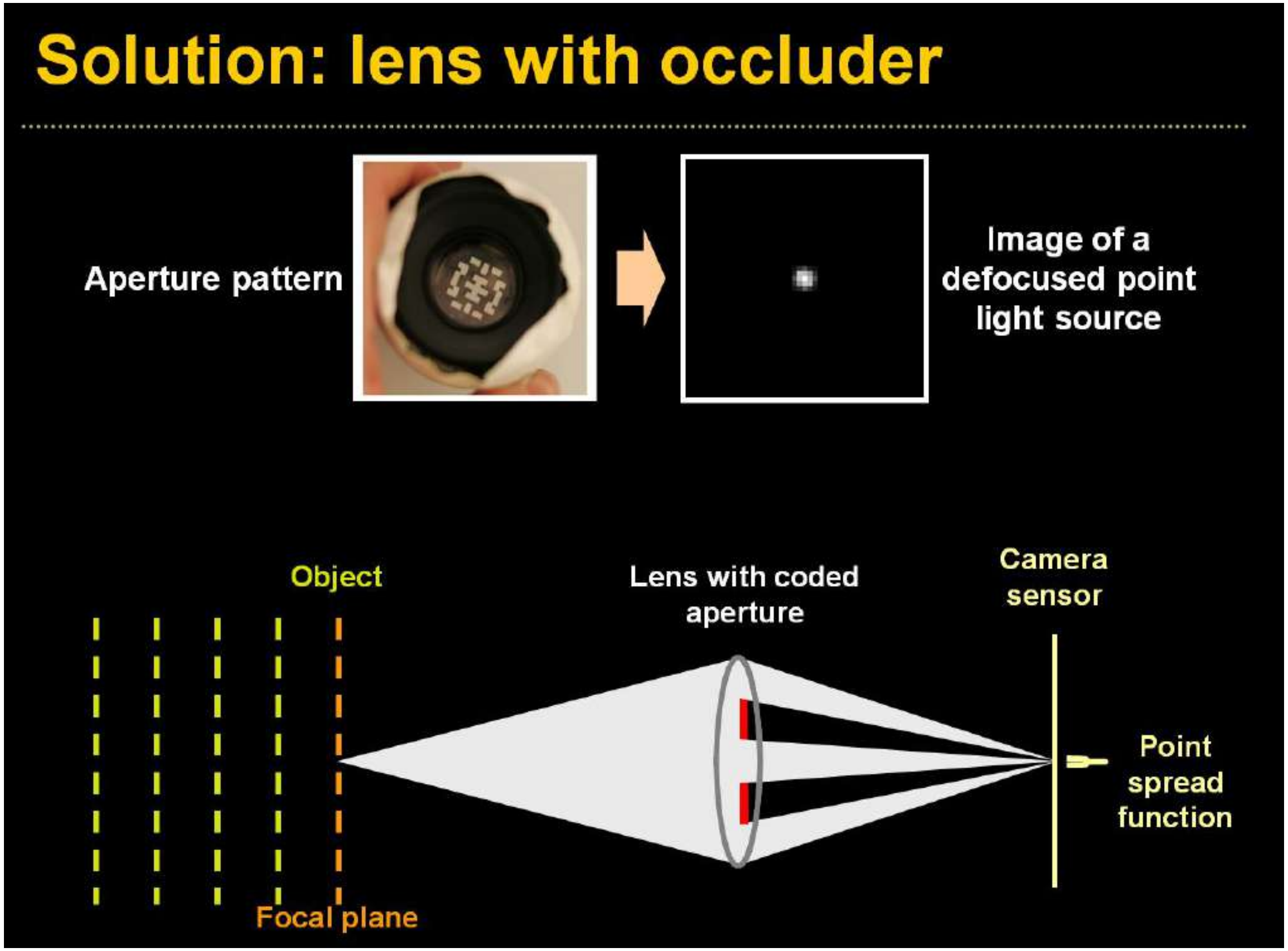
Conventional camera with coded aperture



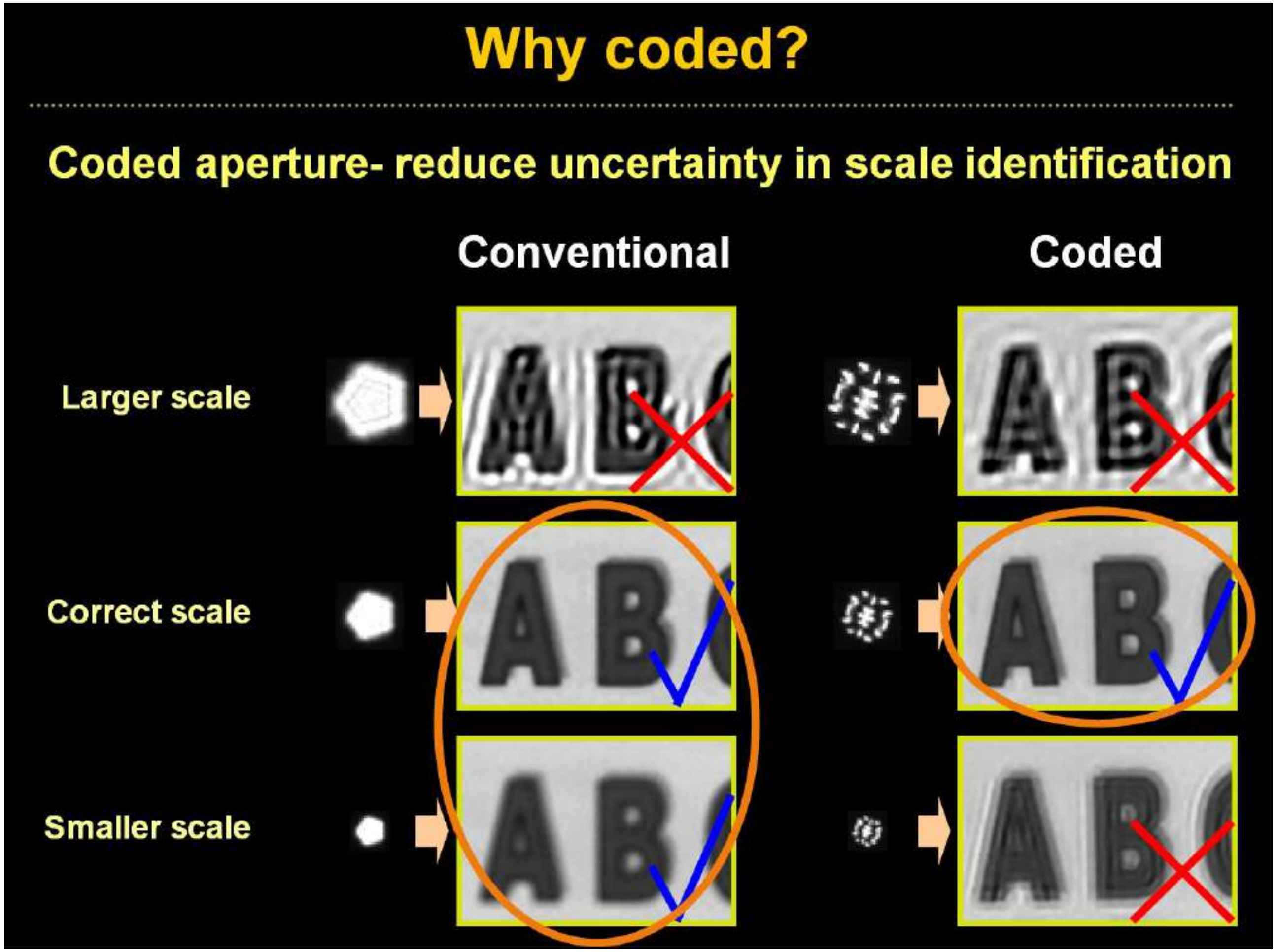
Conventional camera with coded aperture



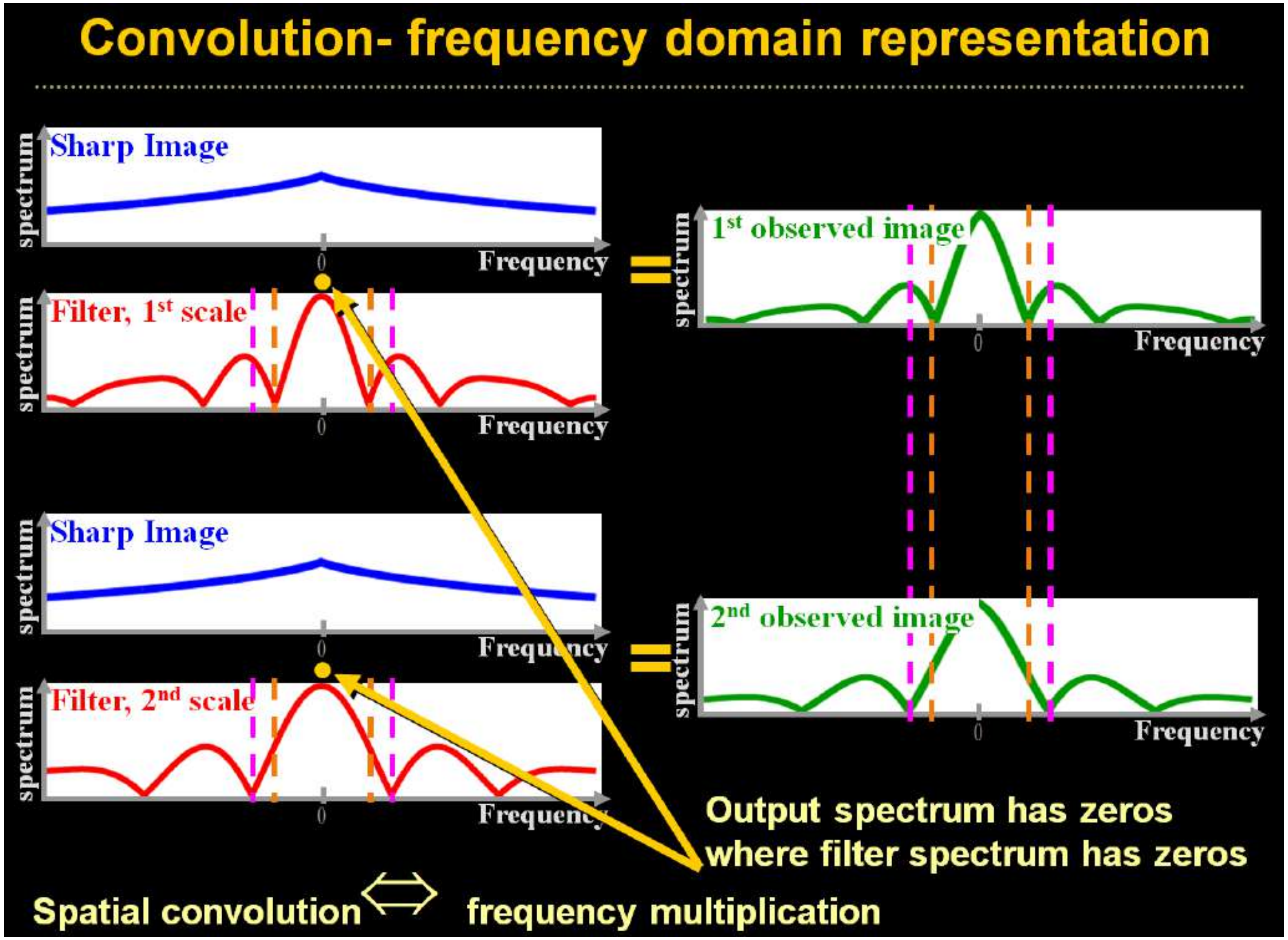
Conventional camera with coded aperture



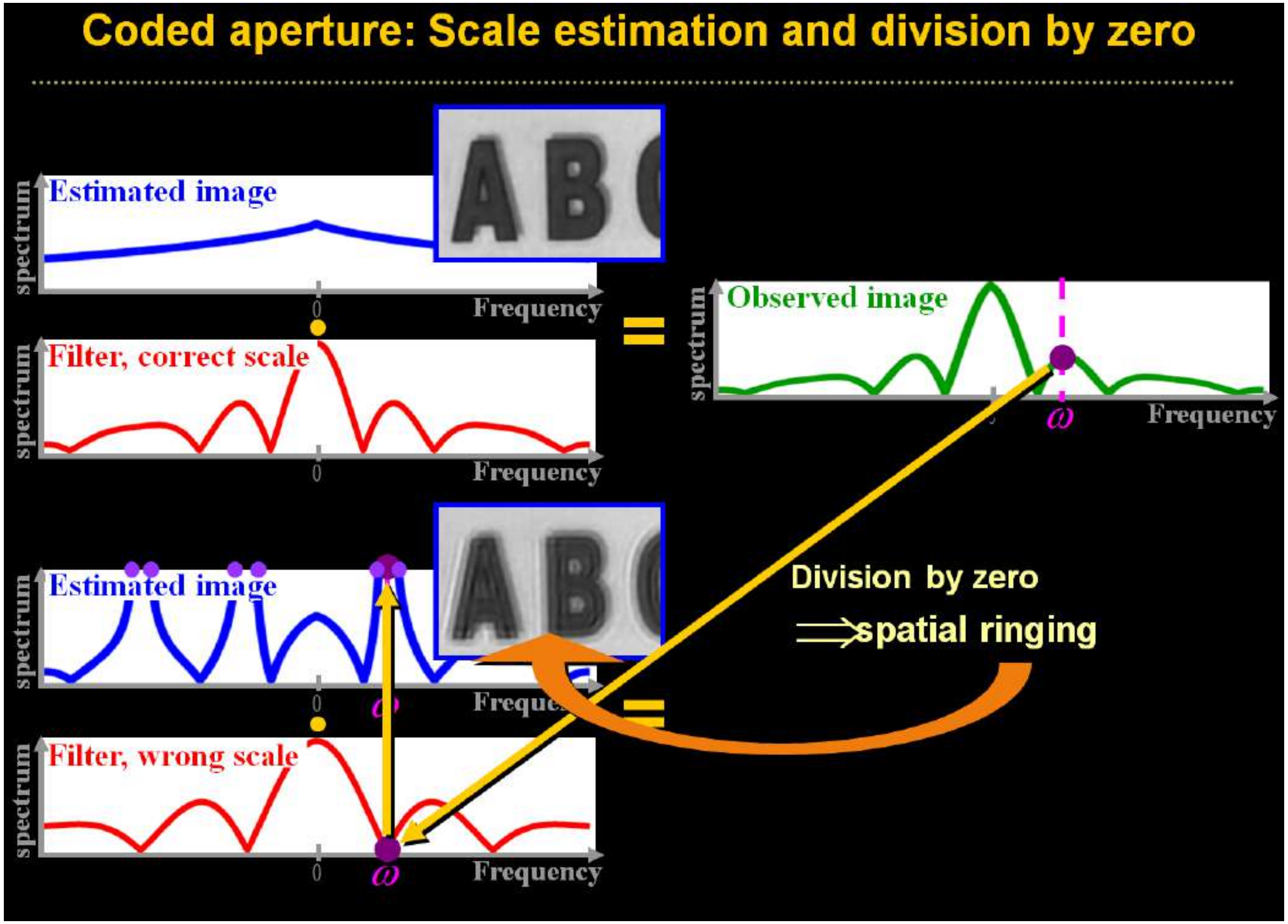
Conventional camera with coded aperture



Conventional camera with coded aperture



Conventional camera with coded aperture

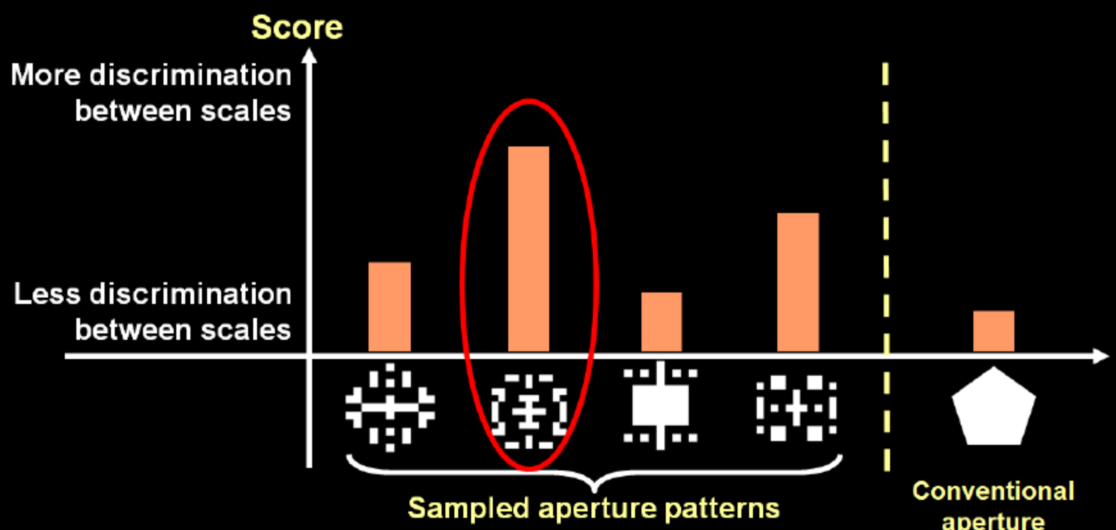


Conventional camera with coded aperture

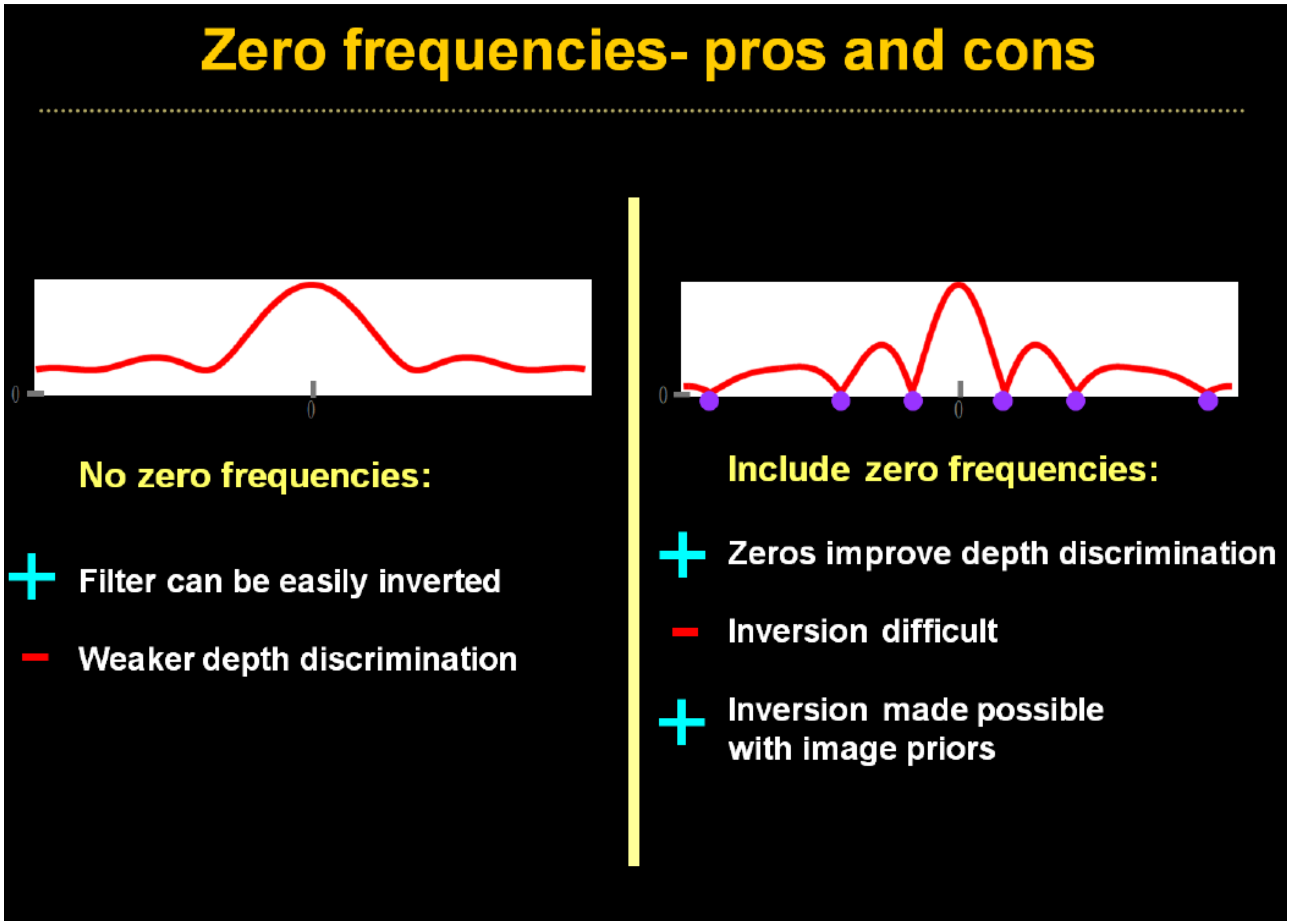
Filter Design

Analytically search for a pattern maximizing discrimination between images at different defocus scales (*KL-divergence*)

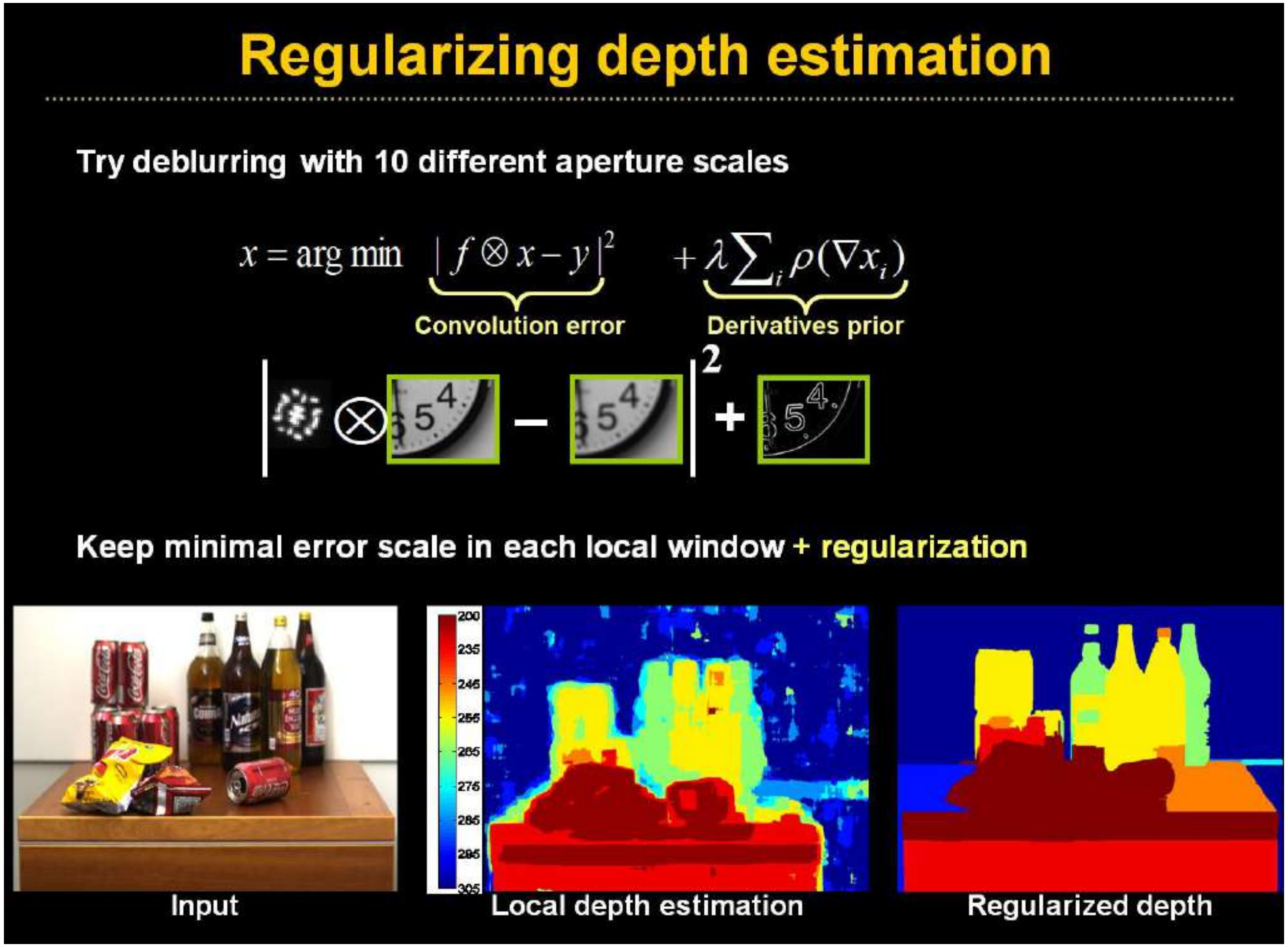
Account for image prior and physical constraints



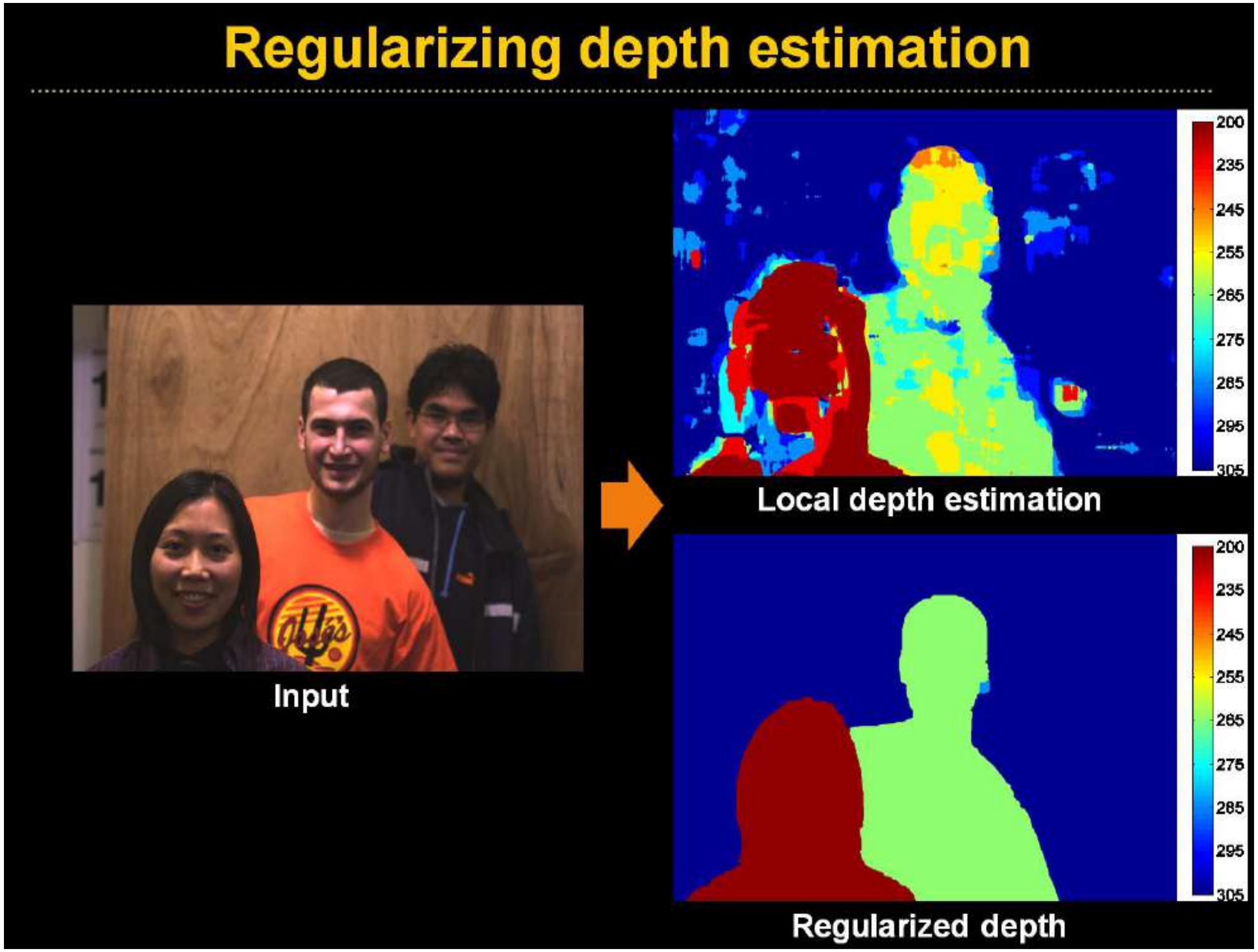
Conventional camera with coded aperture



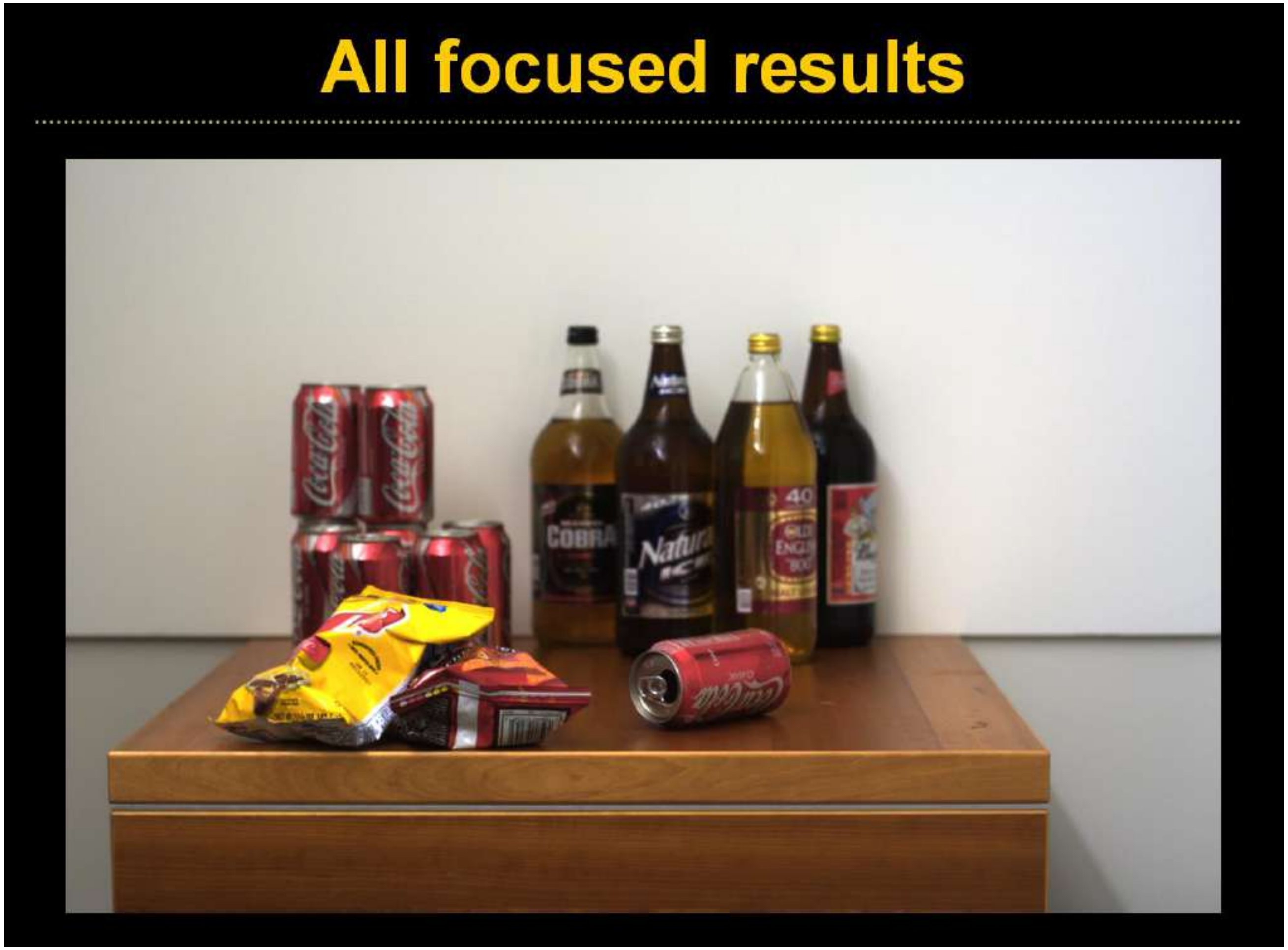
Conventional camera with coded aperture



Conventional camera with coded aperture



Conventional camera with coded aperture



Conventional camera with coded aperture

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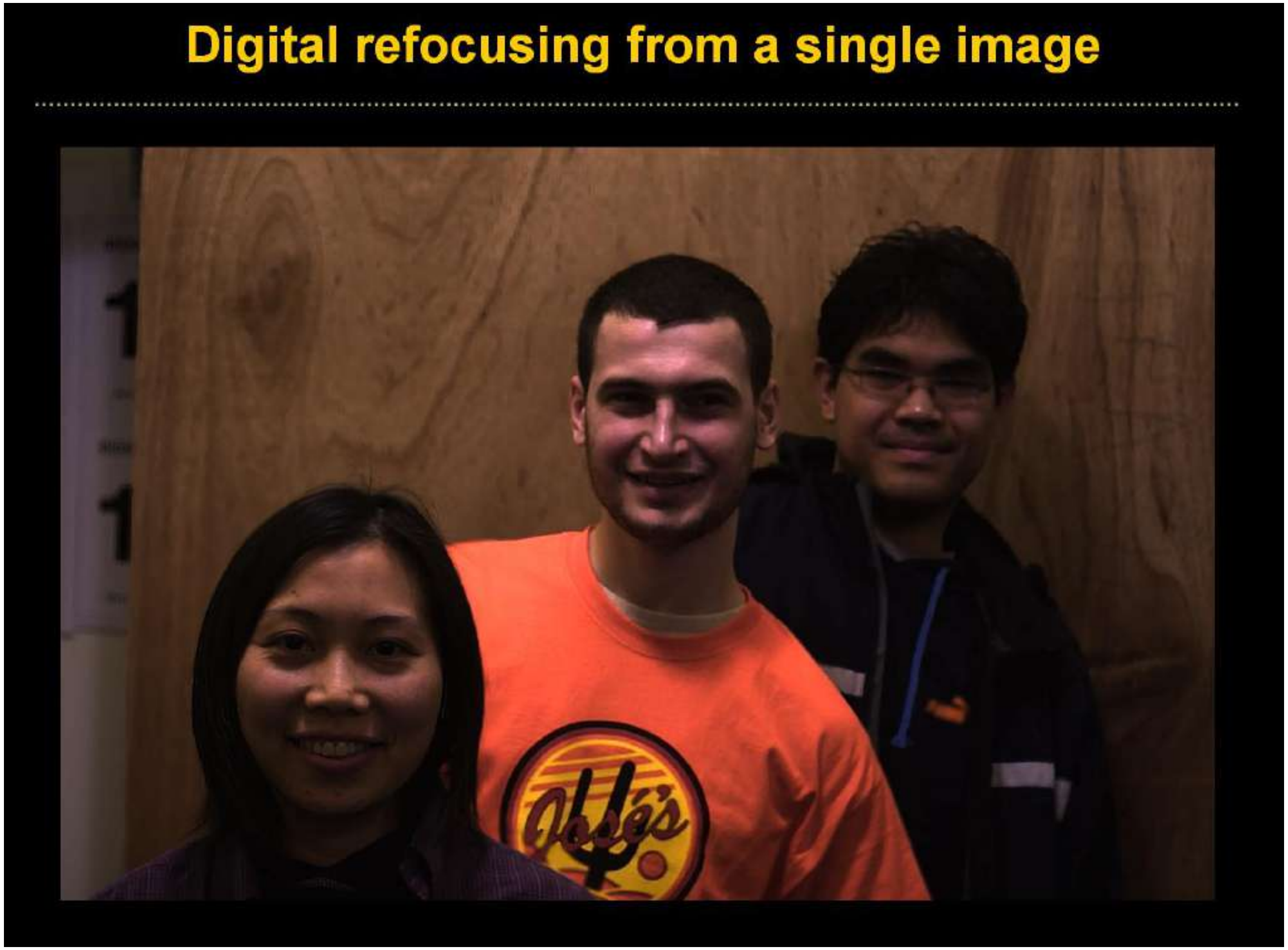
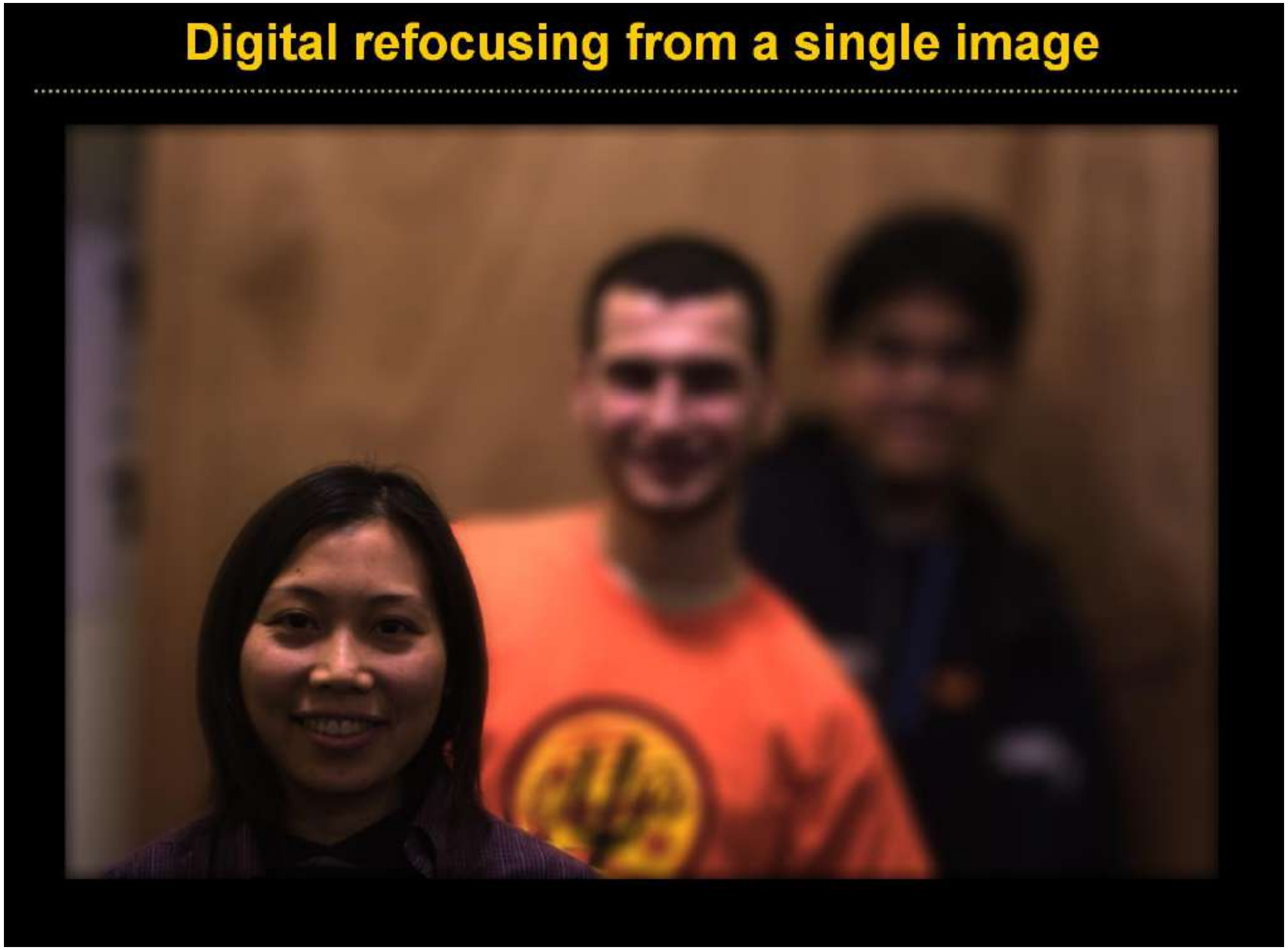
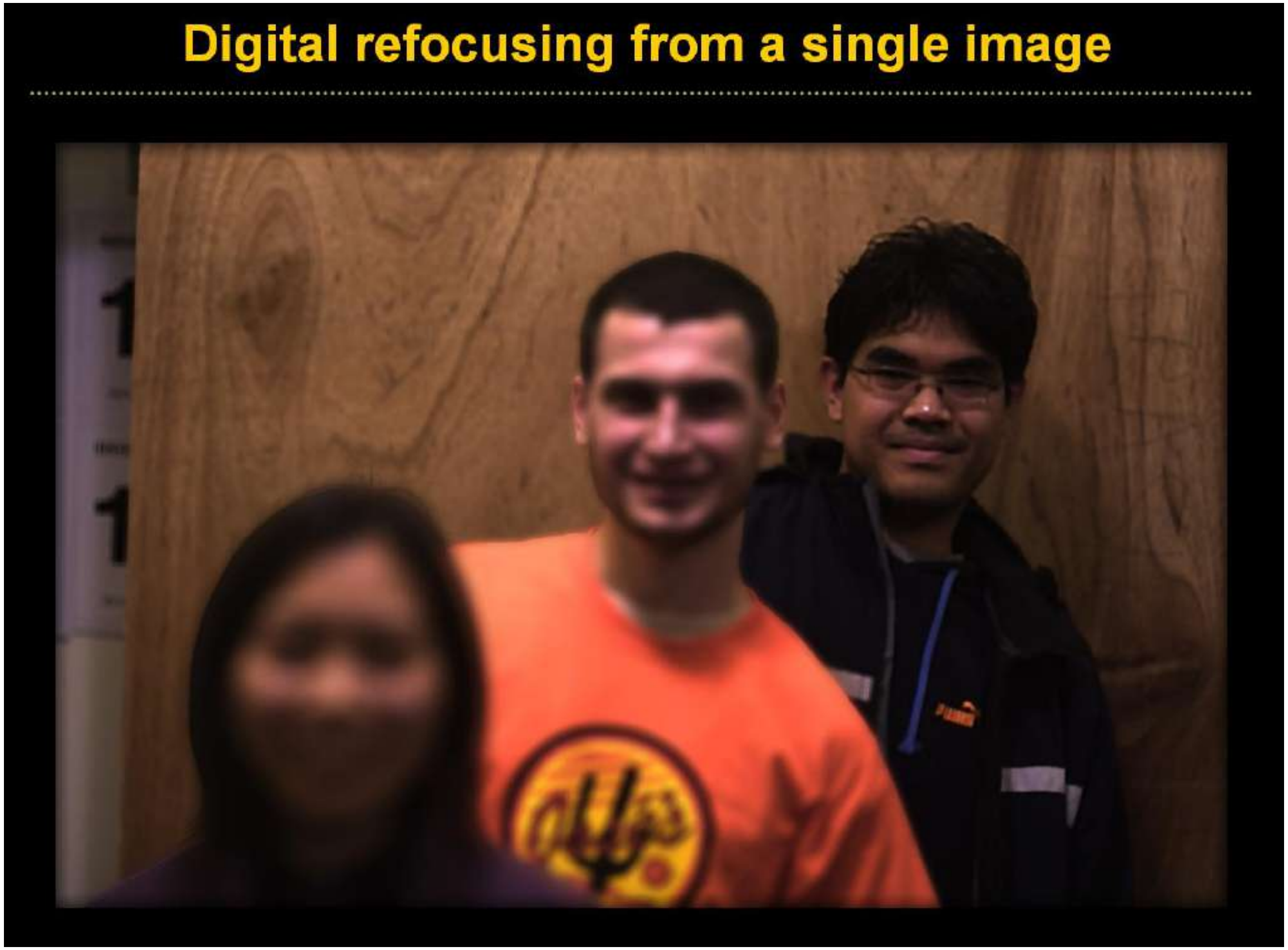


Conventional camera with coded aperture

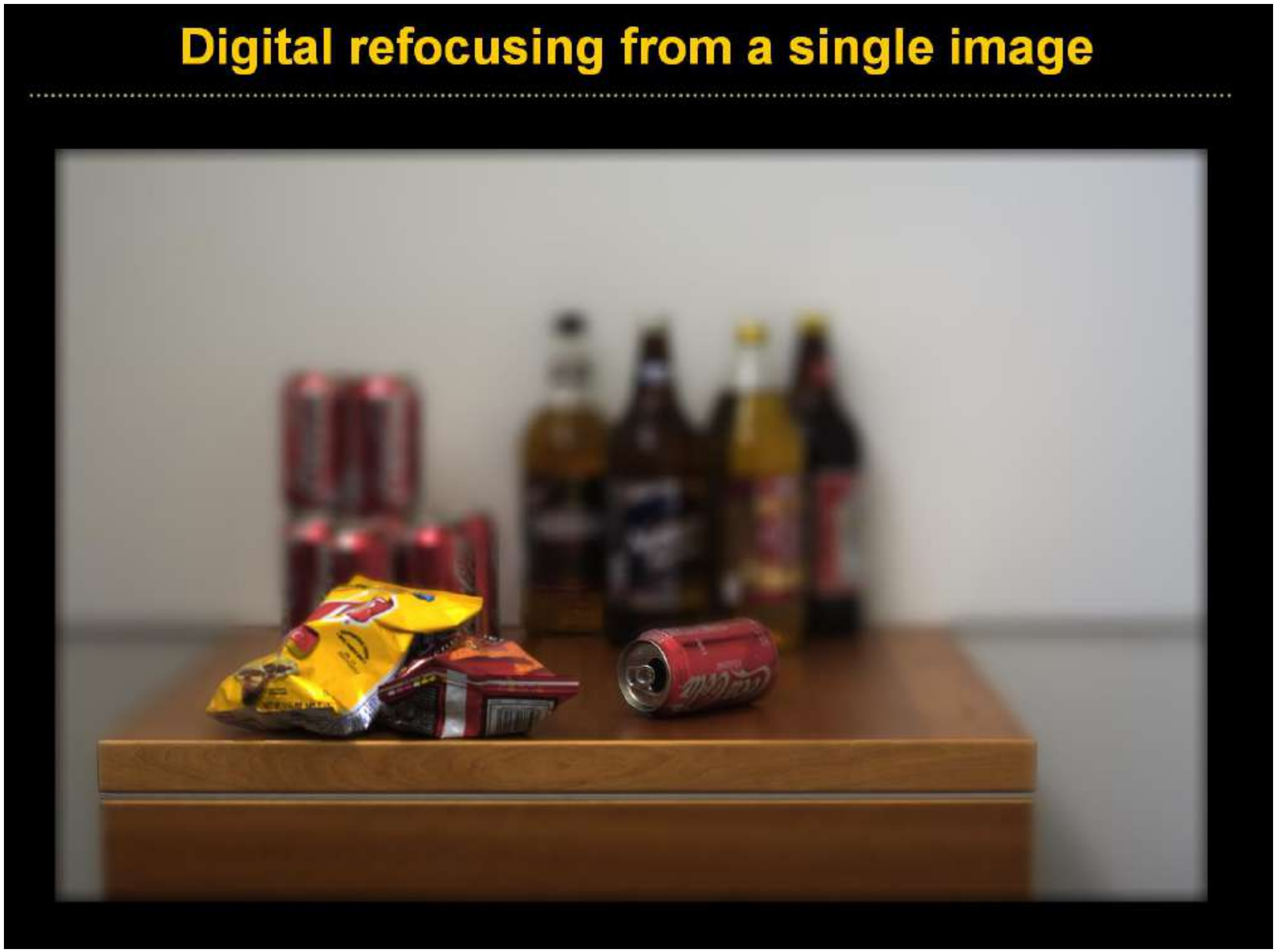


Conventional camera with coded aperture

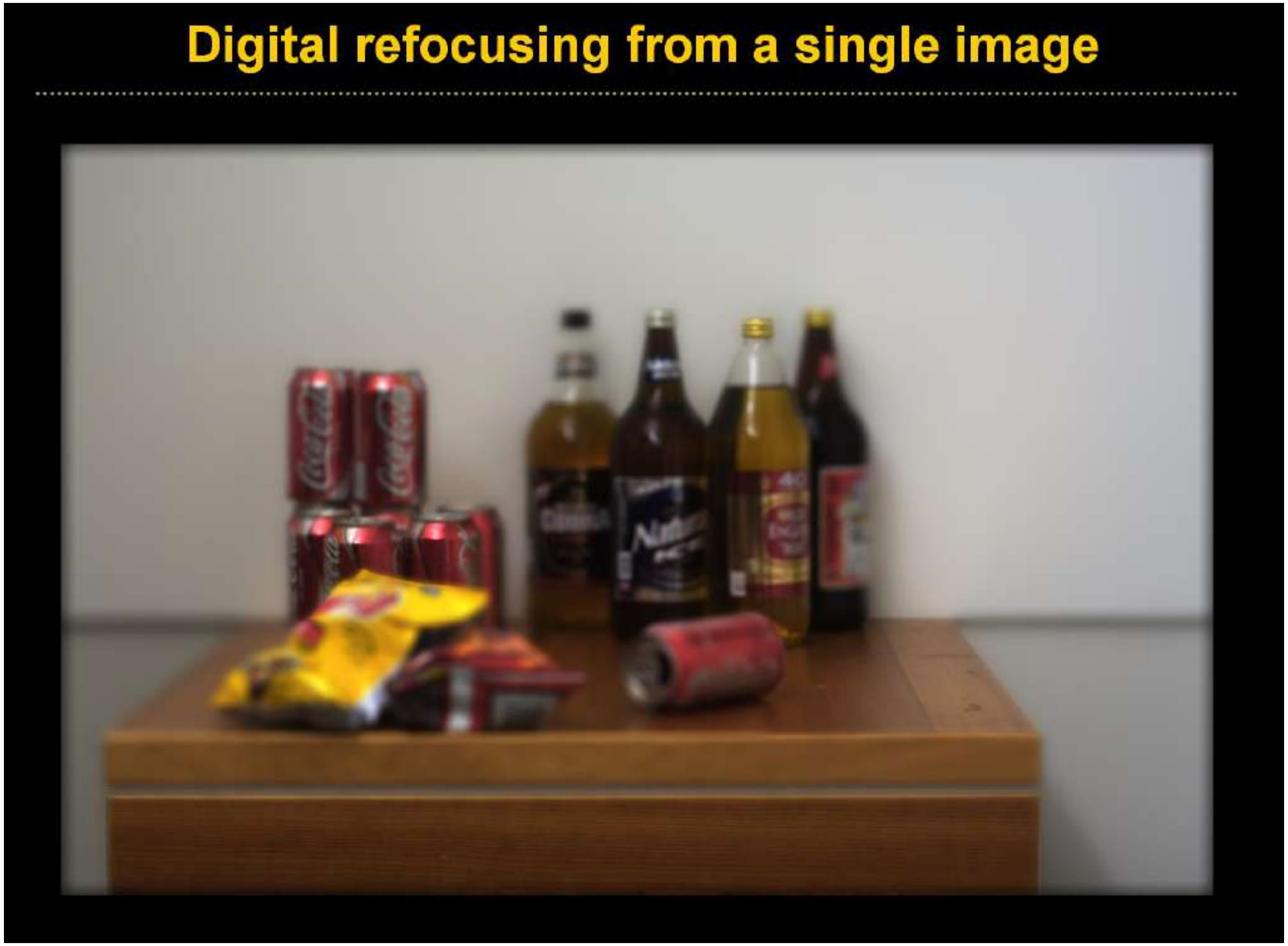
Conventional camera with coded aperture



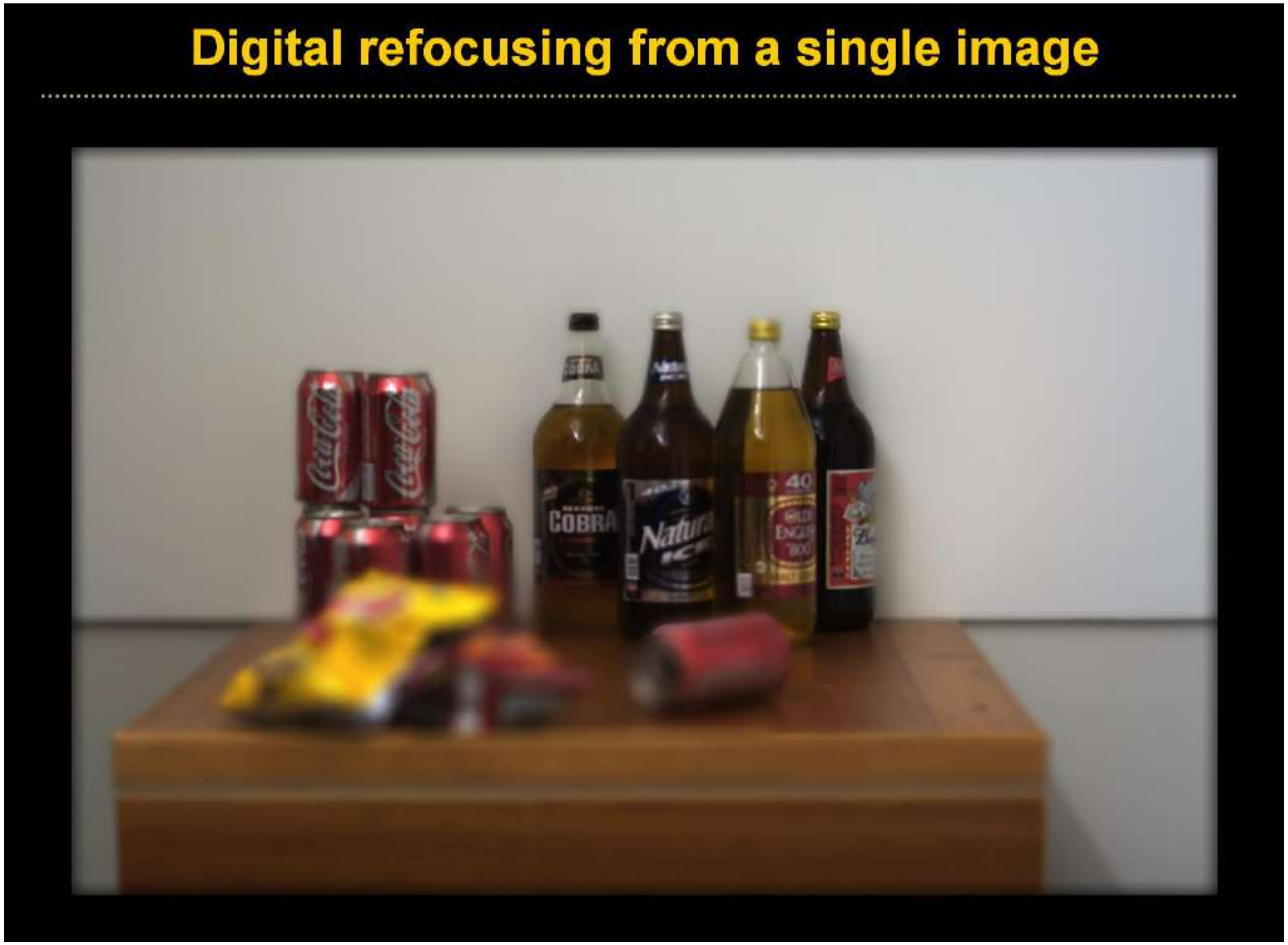
Conventional camera with coded aperture



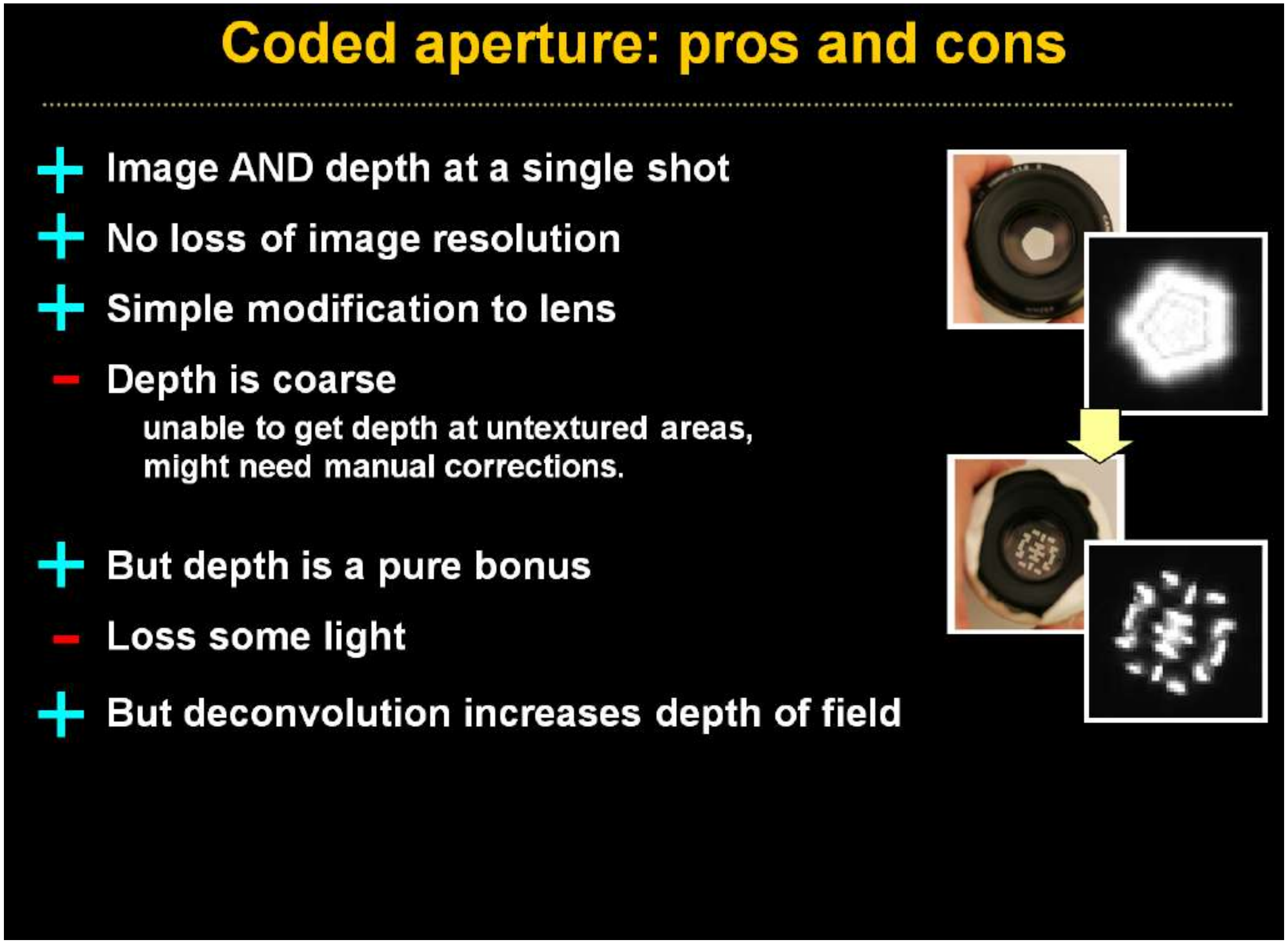
Conventional camera with coded aperture



Conventional camera with coded aperture

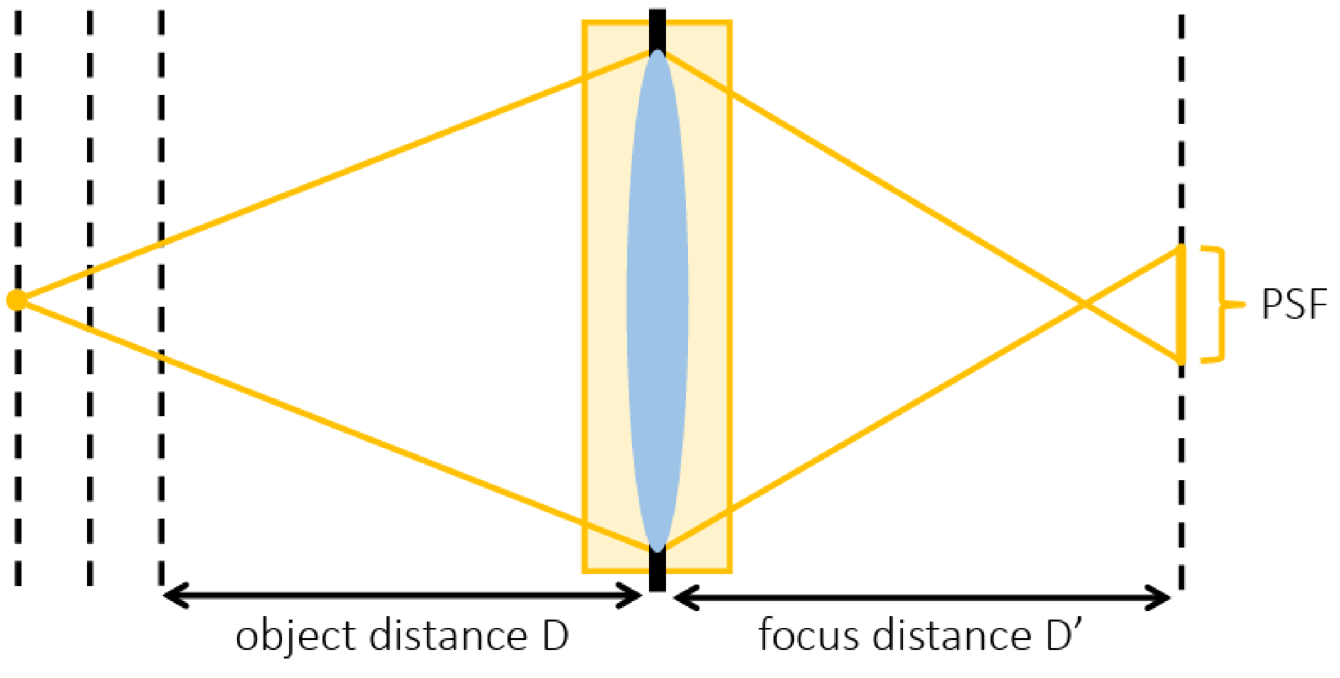


Conventional camera with coded aperture



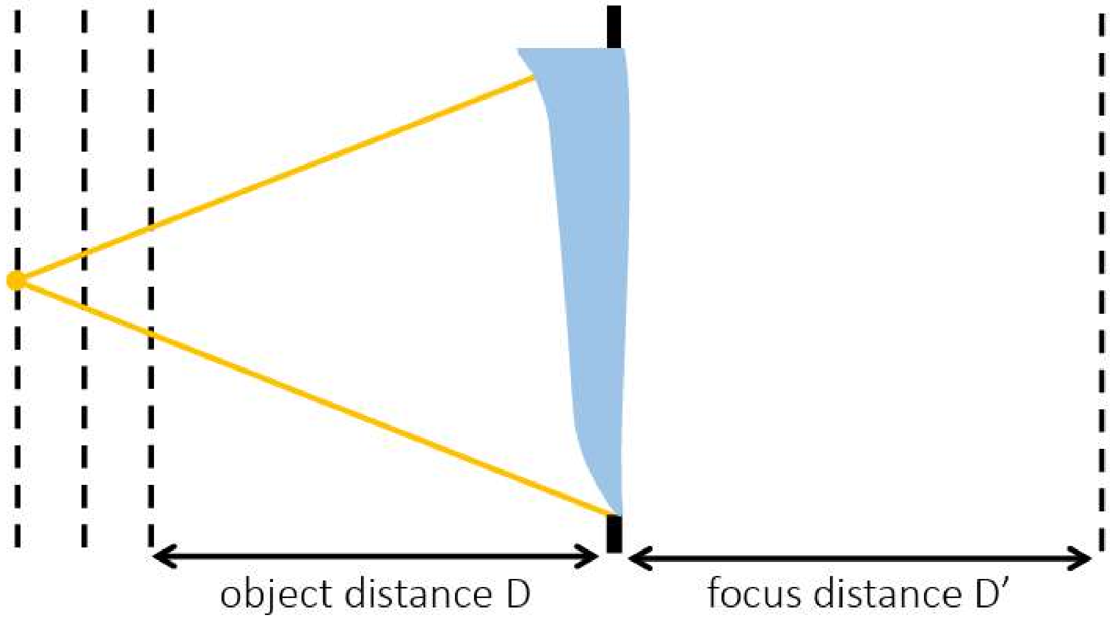
Camera with generalized optics

Change optics, not aperture



Camera with generalized optics

Wavefront coding

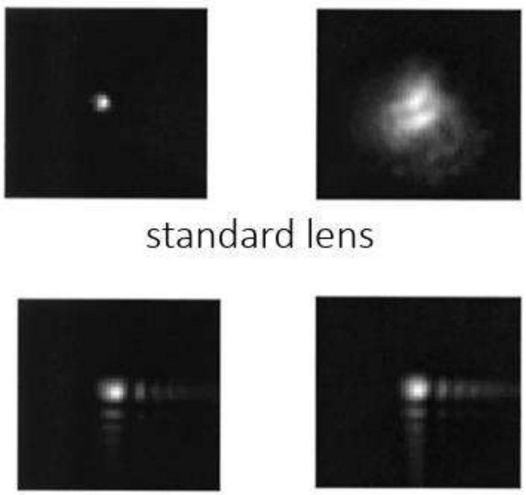


Replace lens with a cubic phase plate

Camera with generalized optics

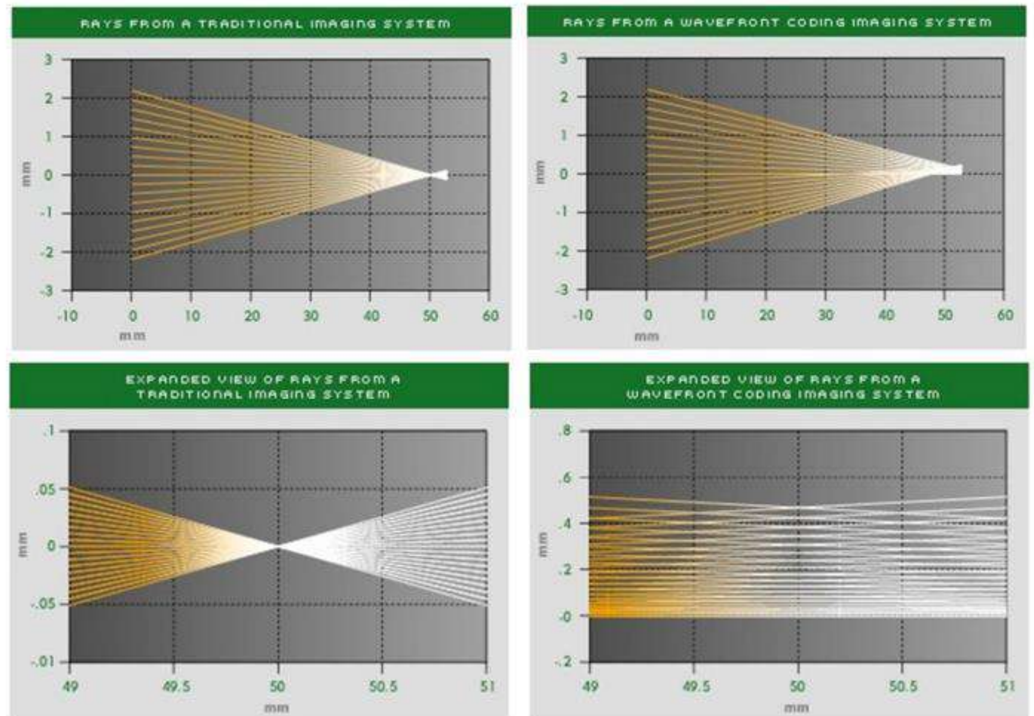
Wavefront coding

In focus Out of focus



standard lens

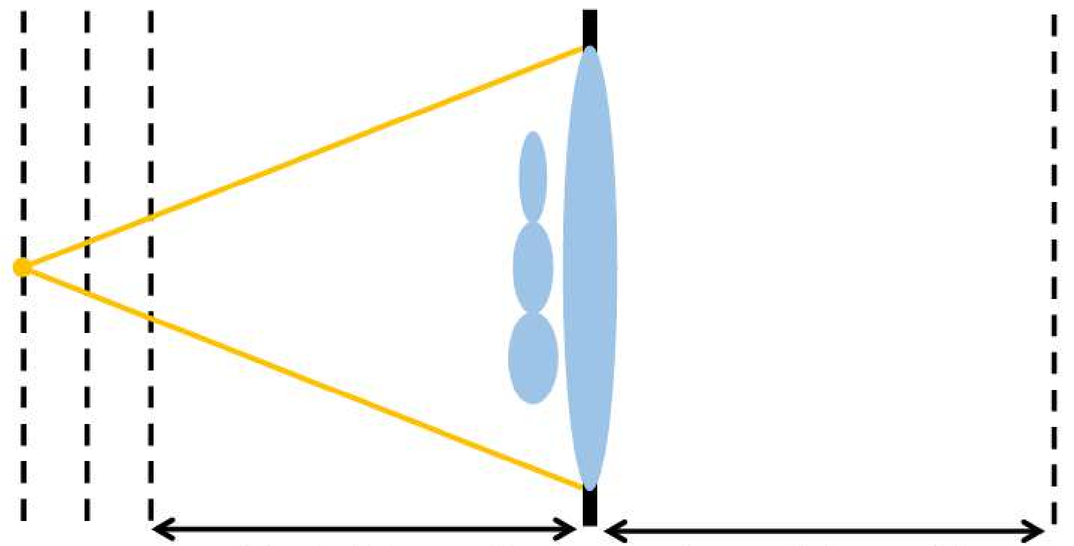
wavefront coding



- Rays no longer converge.
- Approximately depth-invariant PSF for certain range of depths.

Camera with generalized optics

Lattice lens

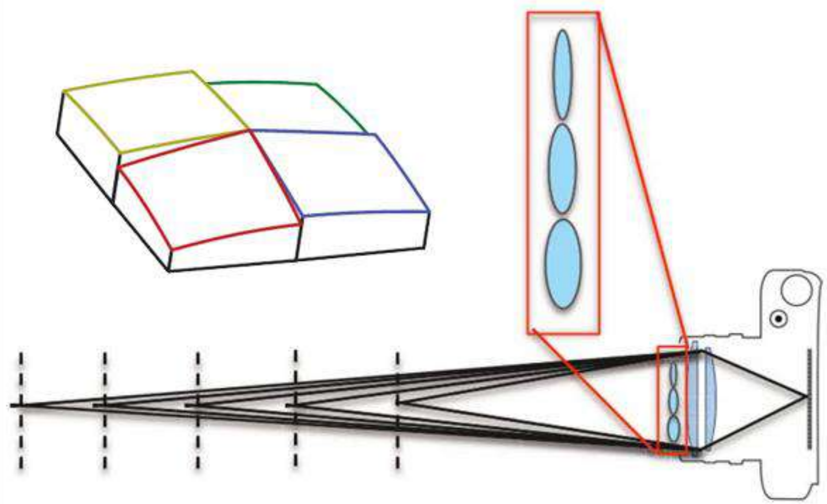


Add lenslet array with varying focal length in front of lens

Camera with generalized optics

Lattice lens

Does this remind you of something?



Camera with generalized optics

Lattice lens

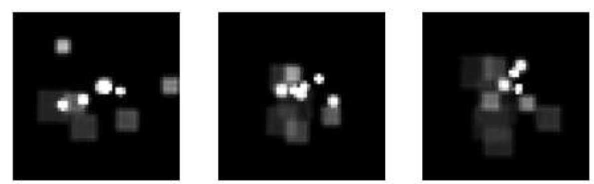
- Effectively captures only the “useful” subset of the 4D lightfield.

Light field spectrum: 4D
Image spectrum: 2D
Depth: 1D
$\Big\}$ 3D
→ Dimensionality gap (Ng 05)

Only the 3D manifold corresponding to physical focusing distance is useful

- PSF is not depth-invariant, so local deconvolution as in coded aperture.

PSFs at different depths



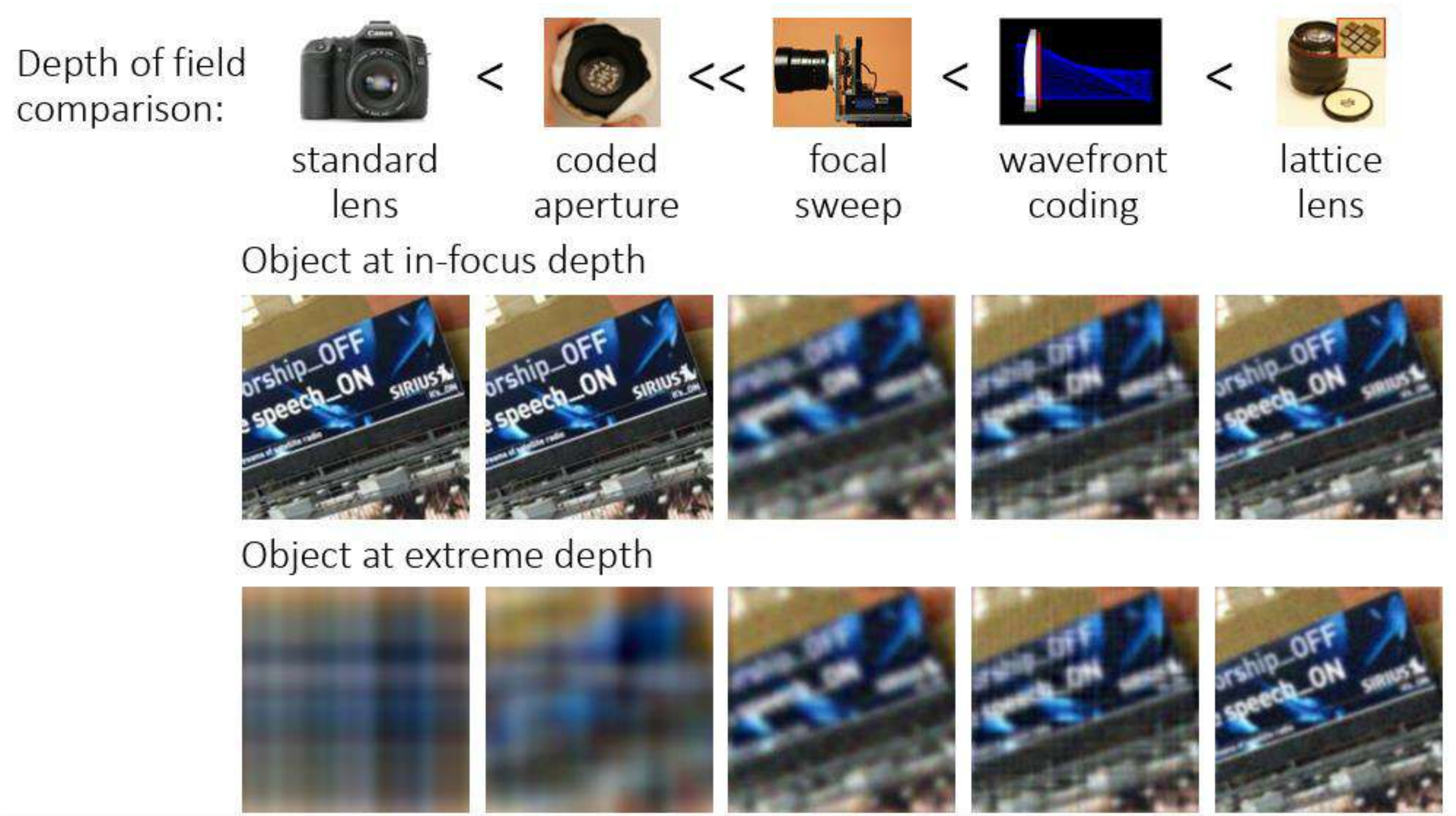
Camera with generalized optics

Results

Camera with generalized optics

Refocusing example

Camera with generalized optics



References

- Computational Photography (Levoy, Adams, Pulli, Stanford)
- Computational Photography SIGGRAPH Course (Raskar & Tumblin)
- Digital and Computational Photography (Durand & Freeman, MIT)
- Computational Photography (Efros, CMU)
- Computational Photography (Gkioulekas, CMU)
- Computational Imaging (Wetzstein, Stanford)
- Computational Photography (Fergus, NYU)
- Computational Imaging (Dragotti, Imperial College)
- Computer Vision (Seitz & Szeliski, UWashington)
- Introduction to Visual Computing and Visual Modeling (Kutulakos, UToronto)